ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE

STUDY MATERIAL

Function & Inverse Trigonometric Functions

English Medium

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FUNCTION

1. CARTESIAN PRODUCT OF TWO SETS :

Given two non-empty sets A and B. The cartesian product $A \times B$ is the set of all ordered pairs of the form (a, b) where the first entry comes from set A & second comes from set B.

$A \times B = \{(a, b) \mid a \in A, b \in B\}$

e.g. $A = \{1, 2, 3\}$ $\quad$ $B = \{p, q\}$

$A \times B = \{(1, p),(1, q), (2, p), (2, q), (3, p), (3, q)\}$

Note :

(i) If either A or B is the null set, then $A \times B$ will also be empty set, i.e. $A \times B = \phi$

(ii) If $n(A) = p, n(B) = q$, then $n(A \times B) = pq$, where $n(X)$ denotes the number of elements in set X.

(iii) A **Relation** R from set A to B is any subset of $A \times B$. If a R b or $(a, b) \in R$ then b is image of a under R and a is preimage of b under R.

Note : If $n(A) = m$, $n(B) = n$, then number of relations defined from set A to B are 2^{mn}.

2. FUNCTION :

A relation R from set A to set B is called a function if each element of A is uniquely associated with some element of B. It is denoted by the symbol :

$$f : A \to B \text{ or } A \xrightarrow{f} B$$

which reads 'f' is a function from A to B 'or' f maps A to B,

If an element $a \in A$ is associated with an element $b \in B$, then b is called 'the f image of a' or 'image of a under f' or 'the value of the function f at a'. Also a is called the 'pre-image of b' or 'argument of b under the function f'. We write it as

$b = f(a)$ or $f : a \to b$ or $f : (a, b)$

Thus a function 'f' from set A to set B is subset of $A \times B$ in which each a belonging to A appears in one and only one ordered pair belonging to f.

Representation of Function :

(a) Ordered pair : Every function from $A \to B$ satisfies the following conditions :

(i) $f \subset A \times B$ (ii) $\forall\ a \in A$ there exist $b \in B$ and (iii) $(a, b) \in f$ & $(a, c) \in f \Rightarrow b = c$

(b) Formula based (uniformly/nonuniformly) :

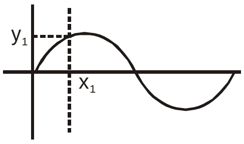
(i) $f : \mathbb{R} \to \mathbb{R}$, $y = f(x) = 4x$, $f(x) = x^2$ (uniformly defined)

(ii) $f(x) = \begin{cases} x+1 & -1 \le x < 4 \\ -x & 4 \le x < 7 \end{cases}$ (non-uniformly defined)

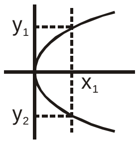
(iii) $f(x) = \begin{cases} x^2 & x \ge 0 \\ -x-1 & x < 0 \end{cases}$ (non-uniformly defined)

(c) Graphical representation :

If a vertical line cuts a given graph at more than one point then it can not be the graph of a function.



Graph (1)



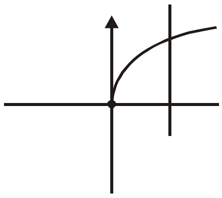
Graph (2)

Graph(1) represent a function but graph(2) does not represent a function.

Every function is a relation but every relation is not necessarily a function.

Illustration 1 : State whether the $f(x)=\sqrt{x}$ is a function or not

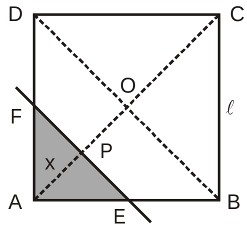
Solution :



it is a function.

Illustration 2 : ABCD is a square of side ℓ. A line parallel to the diagonal BD at a distance 'x' from the vertex A cuts two adjacent sides. Express the area of the segment of the square with A at a vertex, as a function of x. Find this area at $x = 1/\sqrt{2}$ and at x = 2, when ℓ = 2.

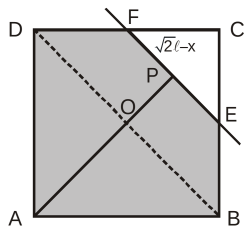
Solution : There are two different situations



Case-I : when $x = AP \le OA$, i.e., $x \le \dfrac{\ell}{\sqrt{2}}$

$$ar(\Delta AEF) = \frac{1}{2}x.2x = x^2 \quad (\because\ PE = PF = AP = x)$$

Case-II : when $x = AP > OA$, i.e., $x > \dfrac{\ell}{\sqrt{2}}$ but $x \le \sqrt{2}\ell$



$$ar(ABEFDA) = ar(ABCD) - ar(\Delta CFE)$$

$$= \ell^2 - \frac{1}{2}\left(\sqrt{2}\ell - x\right).2\left(\sqrt{2}\ell - x\right) \quad [\because\ CP = \sqrt{2}\ell - x]$$

$$= \ell^2 - \left(2\ell^2 + x^2 - 2\sqrt{2}\ell x\right) = 2\sqrt{2}\ell x - x^2 - \ell^2$$

$\therefore$ the required function s(x) is as follows :

$$s(x) = \begin{cases} x^2, & 0 \le x \le \dfrac{\ell}{\sqrt{2}} \\ 2\sqrt{2}\,\ell x - x^2 - \ell^2, & \dfrac{\ell}{\sqrt{2}} < x \le \sqrt{2}\ell \end{cases} \ ; \ \text{area of } s(x) = \begin{cases} \dfrac{1}{2} & \text{at } x = \dfrac{1}{\sqrt{2}} \\ 8(\sqrt{2}-1) & \text{at } x = 2 \end{cases}$$

Ans.

Do yourself - 1 :

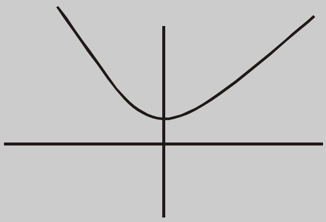
1. Which of the following is (are) function ?

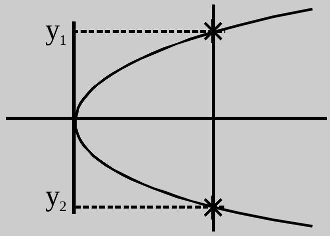
A : {1, 2, 3, 4}　　　　B = {1, 4, 9, 16, 25}

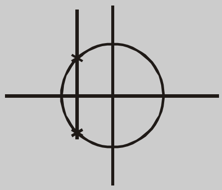
(A) R_1 = {(1, 1) (2, 4) (3, 9) (4, 16) (4, 25)}　　(B) R_2 = {(2, 4) (3, 9) (4, 16)}

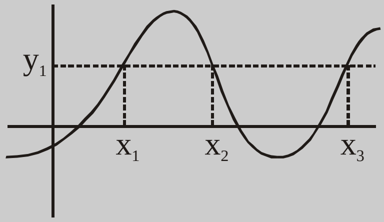
(C) R_3 = {(1, 1) (2, 4) (3, 9) (4, 16)}　　(D) R_4 = {(1, 1) (2, 4) (3, 9) (4, 4)}

2. Which of the following represent graph(s) of a function ?

(A) 

(B) 

(C) 

(D) 

3. Let A = {11, 13, 15, 17} and B = {12, 14, 16, 18} be two sets and 'R' be the relation from A to B defined as $(x, y) \in R \Leftrightarrow x > y$ then write R.

4. If set A = {(1, 2), (3, 4)} & B = {(p, q, r), (s, t)} then A × B is equal to

(A) {(1, 2), (p, q, r), (3, 4), (s, t)}

(B) {(1, 2), (p, q, r)), ((1, 2), (s, t)), ((3, 4), (p,q,r)), ((3, 4), (s, t))}

(C) {(1, p), (1, q), (1, r), (2, p), (2, q), (2, r), (3, p), (3, q), (3, r), (1, s), (2, s), (3, s), (4, s), (1, t), (2, t), (3, t), (4,t)}

(D) {((1, 2), (q, p, r)), ((1, 2), (t, s)), ((3, 4), (p, q, r)), ((3, 4), (t, s))}

5. If n(A) = 3 & n(B) = 2 then number of non-empty relations that can be form set A to set B is

(A) 63　　(B) 31　　(C) 7　　(D) 15

6. Which of the following rule is/are function(s) ?

(A) $f : \left(0, \frac{\pi}{2}\right) \to [0,1]$, f(x) = sinx　　(B) $f : \{-3, -2, -1, 0, 1, 0, 1, 2, 3\} \to \{1, 4, 9\}$, $f(x) = x^2$

(C) $f : \left[0, \frac{\pi}{2}\right) \to [0,1) \to f(x) = \cos x$　　(D) $f : \left(0, \frac{\pi}{2}\right] \to (0,1) \to f(x) = \cos x$

7. Let A and B be two sets such that n(A) = 2 and n(B) = 3 if (1, p) (2, q) (1, r) are in A × B find A and B where p, q, r are distinct elements.

3. DOMAIN, CO-DOMAIN & RANGE OF A FUNCTION :

Let $f : A \to B$, then the set A is known as the domain of f and the set B is known as co-domain of f. The set of f images of all the elements of A is known as the range of f.

Domain of f = $\{a \mid a \in A, (a, f(a)) \in f\}$

Range of f = $\{f(a) \mid a \in A, f(a) \in B\}$

(a) If only the rule of function is given then the domain of the function is the set of those real numbers, where function is defined.

(b) For a continuous function with connected domain, the interval from minimum to maximum value of a function gives the range.

(c) It should be noted that range is a subset of co-domain.

Note :

(i) The complete set of all positive real numbers is denoted by $\mathbb{R}^+$ or $\mathbb{R} > 0$

(ii) The complete set of all negative real numbers is denoted by $\mathbb{R}^-$ or $\mathbb{R} < 0$

(iii) The complete set of all real numbers other then zero is denoted by $\mathbb{R}^*$ or $\mathbb{R} \neq 0$

(iv) The complete set of all integers is denoted by $\mathbb{Z}$.

Basic Problems Based on Domain of a Function :

Illustration 3 : Find the domain of following functions :

(i) $y = \sqrt{5-2x}$ (ii) $y = \dfrac{1}{\sqrt{x - |x|}}$

Solution : (i) $5 - 2x \geq 0 \Rightarrow x \leq \dfrac{5}{2}$ $\therefore$ Domain is $(-\infty, 5/2]$

(ii) $x - |x| > 0 \Rightarrow |x| < x \Rightarrow$ x cannot take any real values $\therefore$ Domain is ϕ

Do yourself - 2 :

1. Find domain of $y = \sqrt{\dfrac{x^2 - 3x + 2}{(x+1)(x+3)}}$

2. Find domain of $f(x) = \sqrt{\dfrac{x(x-1)^3(x-5)}{(x+4)^2(x-2)}}$.

3. $y = \sqrt{(e^x - 1)(x^2 - 4)}$ find domain

4. Find the domain of definition of the given functions :

(i) $y = \sqrt{-px}\,(p > 0)$ (ii) $y = \sqrt{1 - |x|}$ (iii) $y = \dfrac{1}{x^2 + 1}$

(iv) $y = \dfrac{1}{x^3 - x}$ (v) $y = \sqrt{x^2 - 4x + 3}$ (vi) $y = \dfrac{x}{\sqrt{x^2 - 3x + 2}}$

5. Find domain of following functions : $f(x) = \sqrt{\dfrac{(2x+1)}{x^3 - 3x^2 + 2x}}$

6. Find the domain of following functions :

(i) $y = 1 - \log_{10}x$ (ii) $y = \dfrac{1}{\sqrt{x^2 - 4x}}$

7. Find the domain of definition of the given functions :

(i) $y = \log_x 2$ (ii) $y = \dfrac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ (iii) $y = \sqrt{x} + \sqrt[3]{\dfrac{1}{x-2}} - \log_{10}(2x-3)$

8. Find the Domain of

(i) $f(x) = \dfrac{1}{\sqrt{x - |x|}}$ (ii) $f(x) = \sqrt{|x| - x}$

(iii) $f(x) = \sqrt{x - |x|}$ (iv) $f(x) = \dfrac{1}{\sqrt{|x| - x}}$

9. Find the domain of definition of the given functions :

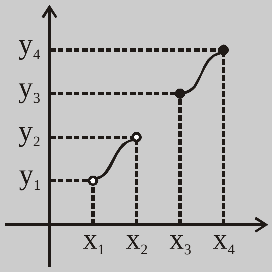
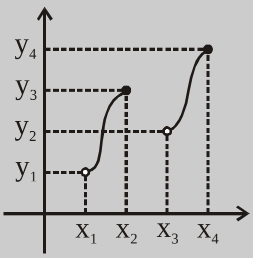
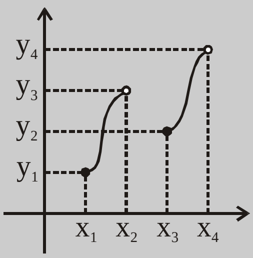
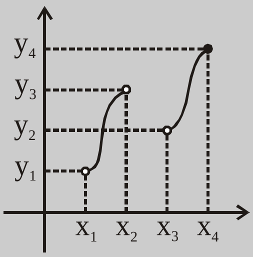
(i) $y = \log_{10}\left(\sqrt{x-4}+\sqrt{6-x}\right)$ (ii) $y = \log_{10}[1 - \log_{10}(x^2 - 5x + 16)]$

10. Find the domain of definition of the given functions :

(i) $y = \dfrac{3}{4-x^2} + \log_{10}\left(x^3 - x\right)$ (ii) $y = \dfrac{1}{\sqrt{\sin x}} + \sqrt[3]{\sin x}$

Comprehension :

Graph of y = f(x) is drawn here, match Domain & Range of the f(x)

y=f(x) Column-I		**(Domain) Column-II**		**(Range) Column-III**	
(P)		(I)	$(x_1, x_2] \cup (x_3, x_4]$	(i)	$(y_1, y_2) \cup [y_3, y_4]$
(Q)		(II)	$(x_1, x_2) \cup [x_3, x_4]$	(ii)	$(y_1, y_4]$
(R)		(III)	$[x_1, x_2) \cup [x_3, x_4)$	(iii)	$[y_1, y_4)$
(S)		(IV)	$(x_1, x_2) \cup (x_3, x_4]$	(iv)	$(y_1, y_4) - \{y_2\}$

11. Select the correct match

(A) P-(II)-(ii) (B) Q-(I)-(ii) (C) R-(III)-(ii) (D) S-(II)-(iv)

12. Select the correct match

(A) P-(II)-(i) (B) Q-(II)-(i) (C) R-(III)-(ii) (D) S-(I)-(ii)

Basic Problems Based on Range of a Function :

Illustration 4 : Find the range of following functions :

(i) $f(x) = \log_{\sqrt{2}}\left((x-1)^2+4\right)$ (ii) $f(x) = 3 - \cos x$

(iii) $f(x) = \dfrac{1}{8-3\sin x}$

Solution : (i) $f(x) = \log_{\sqrt{2}}\left((x-1)^2+4\right)$

$4 \le (x-1)^2 + 4 < \infty$

$\Rightarrow \log_{\sqrt{2}} 4 \le \log_{\sqrt{2}}(x-1)^2 + 4 < \infty$

$\Rightarrow 4 \le \log_{\sqrt{2}}(x-1)^2 + 4 < \infty$

$\therefore$ Range of $f(x) = [4,\infty)$

(ii) $f(x) = 3 - \cos x$

$-1 \le \cos x \le 1$

$2 \le 3 - \cos x \le 4$

$\therefore$ Range of $f(x) = [2,4]$

(iii) $f(x) = \dfrac{1}{8-3\sin x}$

$-1 \le \sin x \le 1$

$\therefore$ Range of f = $\left[\dfrac{1}{11}, \dfrac{1}{5}\right]$

Do yourself - 3 :

1. Find range of the following functions

(i) $f : R \to R$, $f(x) = 3x + 5$ (ii) $f : [-1, 5] \to R$, $f(x) = 3x + 5$

2. Find range of the following functions

(i) $f : R \to R$, $f(x) = x^2 - 8x + 7$ (ii) $f : [-1, 3] \to R$, $f(x) = x^2 - 8x + 7$

(iii) $f : [-1, \infty) \to R$, $f(x) = x^2 - 8x + 7$ (iv) $f : (-\infty, 3] \to R$, $f(x) = x^2 - 8x + 7$

3. Find range of the following functions

(i) $f : R \to R$, $f(x) = 5 - 4\sin x$ (ii) $f : R \to R$; $f(x) = 5 - 4\cos^2 x$

(iii) $f : R \to R$, $f(x) = \sin^2 x - 3\sin x + 8$

4. Find Range :-

(i) $f(x) = \dfrac{1}{x^2 - x + 1}$ (ii) $f(x) = \dfrac{1}{4x^2 - 5x + 1}$

5. If $f(x) = 3x + 4$; $g(x) = \begin{cases} x^2+1; & x \ge 1 \\ 2x+3; & x < 1 \end{cases}$ and $h(x) = \begin{cases} e^x & ; & x \ge 5 \\ 2x^2+3x & ; & 2 < x < 5 \\ 5-x & ; & x < 2 \end{cases}$ then find $h(f(g(-2)))$

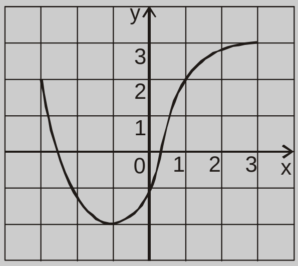
6. Find the range of the following functions :

(i) $f(x) = 2^{x^2} + 1$ (ii) $f(x) = \dfrac{1}{8-3\sin x}$

7. Find range of following functions : $f(x)=\dfrac{1}{2-\cos 3x}$

8. Graph the function $F(x)=\begin{cases}3-x, & x\le 1\\ 2x, & x>1\end{cases}$

9. The graph of a function f is given.



(i) State the value of $f(-1)$.

(ii) For what values of x is $f(x) = 2$

(iii) State the domain and range of f.

(iv) On what interval is f increasing ?

(v) Estimated value of $f(2)$ is -

(A) 2.2 (B) 2.8 (C) 2.5 (D) 3

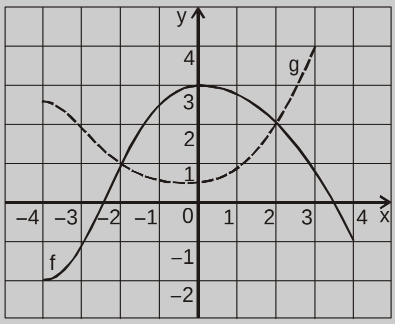
(vi) Estimated value of x such that $f(x) = 0$, is -

(A) –2.5 (B) 0.8 (C) –2.9 (D) 0.3

10. Draw graph of f(x) & find range of f(x)

(i) $f(x)=\begin{cases}x+1; & -1\le x<4\\ -x; & 4\le x<7\end{cases}$ (ii) $f(x)=\begin{cases}x^2; & x\ge 0\\ -x-1; & x<0\end{cases}$

11. The graphs of f and g are given.



(i) State the value of $f(-4)$ and g(3)

(ii) For what value of x is $f(x) = g(x)$?

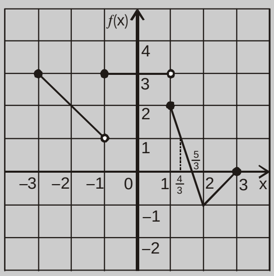
(iii) Estimate the solution of the equation $f(x) = -1$.

(iv) On what interval is f decreasing?

(v) State the domain and range of f.

(vi) State the domain and range of g.

12. Solve the following inequalities using graph of f(x) :



(i) $0 \le f(x) \le 1$

(ii) $-1 \le f(x) \le 2$

(iii) $2 \le f(x) \le 3$

(iv) $f(x) > -1$ & $f(x) < 0$

13. The range of the function $f(x) = e^{-x} + e^{x}$, is -

(A) $f(x) \ge 1$ (B) $f(x) \le 1$ (C) $f(x) \ge 2$ (D) $f(x) \le 2$

14. The range of the function $f : \mathbb{N} \to \mathbb{Z}$; $f(x) = (-1)^{x-1}$, is -

(A) [–1, 1] (B) {– 1, 1} (C) {0, 1} (D) {0, 1, – 1}

4. IMPORTANT TYPES OF FUNCTIONS :

(a) Polynomial Function :

If a function f is defined by $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \ldots\ldots + a_1 x + a_0$ where n is a non negative integer and $a_0, a_1, a_2, \ldots, a_n$ are real numbers. If $a_n \neq 0$, then f is called a polynomial function of degree n.

Note :

(i) Range of odd degree polynomial is always $\mathbb{R}$.

(ii) Range of even degree polynomial is never $\mathbb{R}$.

(iii) A polynomial of degree one with no constant term is called an odd linear function. i.e. $f(x) = ax,\ a \neq 0$

(iv) $f(x) = ax + b,\ a \neq 0$ is a linear polynomial

(v) $f(x) = c$ is non linear polynomial (its degree is zero)

(vi) $f(x) = 0$ is a polynomial. Its degree is **not** defined (or $-\infty$)

(vii) There are four polynomial functions, satisfying the relation ; $f(x) \cdot f(1/x) = f(x) + f(1/x)$. They are : (a) $f(x) = 0$ (b) $f(x) = 2$ (c) $f(x) = x^n + 1$ & (d) $f(x) = 1 - x^n$, where n is a positive integer.

(b) Algebraic Function :

y is an algebraic function of x, if it is a function that satisfies an algebraic equation of the form $P_0(x)\, y^n + P_1(x)\, y^{n-1} + \ldots\ldots + P_{n-1}(x)\, y + P_n(x) = 0$. Where n is a positive integer and $P_0(x)$, $P_1(x)$....... are Polynomials in x. e.g. $x^3 + y^3 - 3xy = 0$.

In other words a function 'f' is called an algebraic function if it can be constructed using finite number of algebraic operations (such as addition, subtraction, multiplication, division, and taking radicals) within polynomials.

Example : $f(x) = \sqrt{x^2+1}$; $g(x) = \dfrac{x^4 - 16x^2}{x + \sqrt{x}} + (x-2)\sqrt[3]{x+1}$

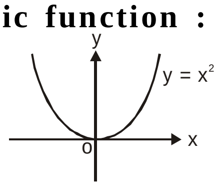
Note :

(i) All polynomial functions are Algebraic but not the converse.

(ii) A function that is not algebraic is called **Transcendental Function.**

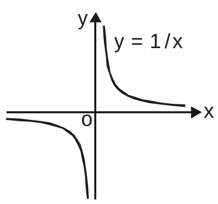
Basic algebraic function :

(i) $y = x^2$



Domain : $\mathbb{R}$
Range : $\mathbb{R}^+ \cup \{0\}$ or $[0, \infty)$

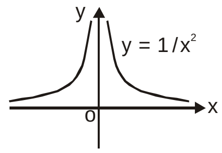
(ii) $y = \dfrac{1}{x}$



Domain : $\mathbb{R} - \{0\}$ or $\mathbb{R}^*$

Range : $\mathbb{R} - \{0\}$

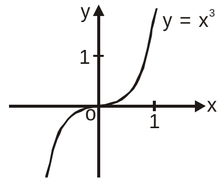
(iii) $y = \dfrac{1}{x^2}$



Domain : $\mathbb{R}^*$

Range : $\mathbb{R}^+$ or $(0, \infty)$

(iv) $y = x^3$



Domain : $\mathbb{R}$

Range : $\mathbb{R}$

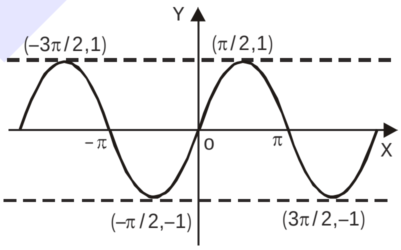
(c) Rational function :

A rational function is a function of the form $y = f(x) = \dfrac{g(x)}{h(x)}$, where g(x) & h(x) are polynomials & $h(x) \neq 0$, **Domain :** $\mathbb{R}-\{x \mid h(x)=0\}$

Any rational function is automatically an algebraic function.

(d) Trigonometric functions :

(i) Sine function

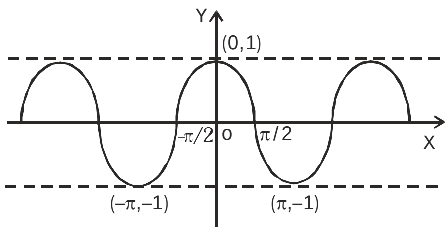


$f(x) = \sin x$

Domain : $\mathbb{R}$

Range : $[-1, 1]$, period 2π

(ii) Cosine function

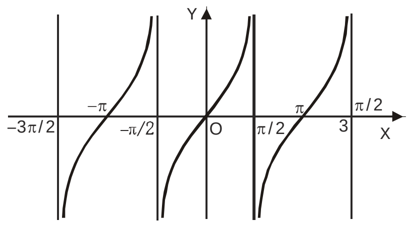


$f(x) = \cos x$

Domain : $\mathbb{R}$

Range : $[-1, 1]$, period 2π

(iii) Tangent function

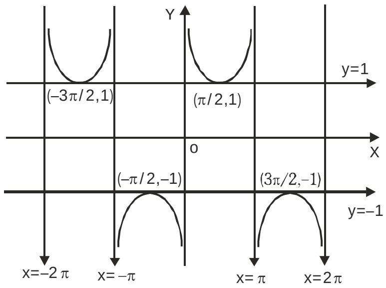


$f(x) = \tan x$

Domain : $\mathbb{R}-\left\{x \mid x = \dfrac{(2n+1)\pi}{2}, n \in \mathbb{Z}\right\}$

Range : $\mathbb{R}$, period π

(iv) Cosecant function

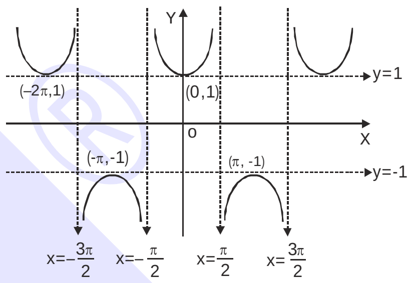


$f(x) = \text{cosec } x$

Domain : $\mathbb{R} - \{x|x = n\pi,\ n \in \mathbb{Z}\}$

Range : $\mathbb{R} - (-1, 1)$, period 2π

(v) Secant function

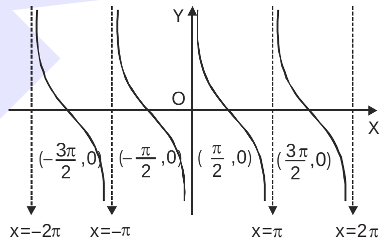


$f(x) = \sec x$

Domain : $\mathbb{R} - \{x|x = (2n + 1)\ \pi/2 : n \in \mathbb{Z}\}$

Range : $\mathbb{R} - (-1, 1)$, period 2π

(vi) Cotangent function



$f(x) = \cot x$

Domain : $\mathbb{R} - \{x|x = n\pi,\ n \in \mathbb{Z}\}$

Range : $\mathbb{R}$, period π

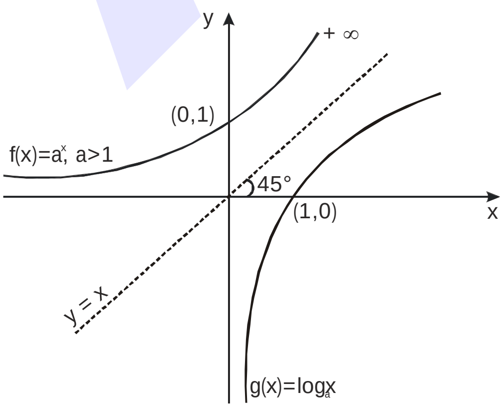
(e) Exponential and Logarithmic Function :

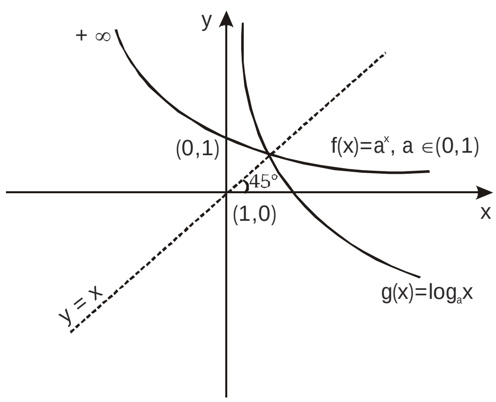
A function $f(x) = a^x (a > 0),\ a \neq 1,\ x \in \mathbb{R}$ is called an exponential function. The inverse of the exponential function is called the logarithmic function, i.e. $g(x) = \log_a x$.

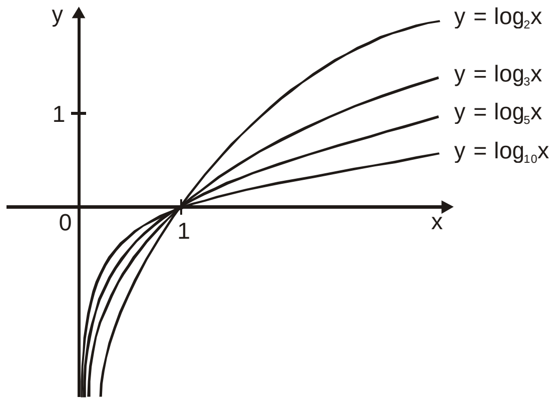
Note that f(x) & g(x) are inverse of each other & their graphs are as shown. (If functions are mirror image of each other about the line y = x)

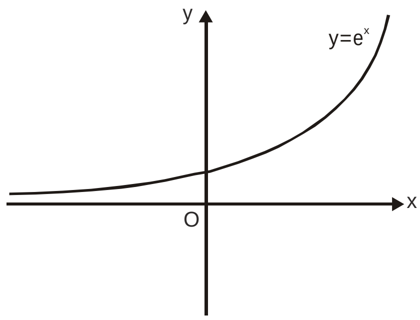
Domain of a^x is $\mathbb{R}$ **Range** $\mathbb{R}^+$

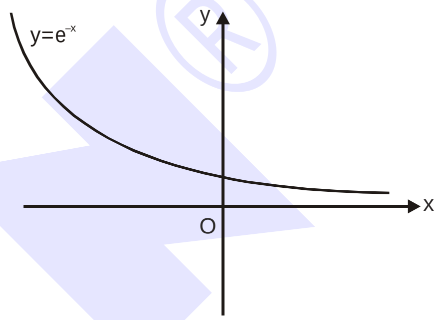
Domain of $\log_a x$ is $\mathbb{R}^+$ **Range** $\mathbb{R}$

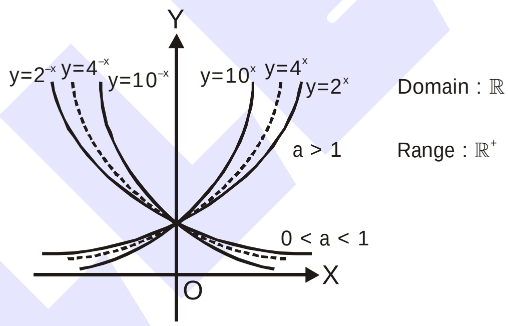












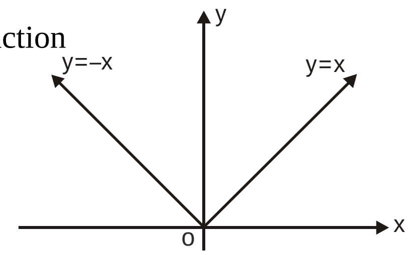
Note-1 : $f(x) = a^{1/x}$, $a > 0$ **Domain** : $\mathbb{R} - \{0\}$ **Range** : $\mathbb{R}^+ - \{1\}$

Note-2 : $f(x) = \log_x a = \dfrac{1}{\log_a x}$ **Domain** : $\mathbb{R}^+ - \{1\}$ **Range** : $\mathbb{R} - \{0\}$

$(a > 0)$ $(a \neq 1)$

(f) Absolute Value Function :

A function $y = f(x) = |x|$ is called the absolute value function or Modulus function. It is defined as :



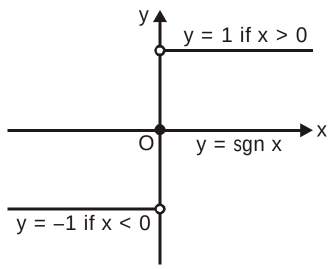
$$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

For $f(x) = |x|$, domain is $\mathbb{R}$ and range is $[0,\infty)$

For $f(x) = \dfrac{1}{|x|}$, domain is $\mathbb{R}^*$ and range is $\mathbb{R}^+$.

(g) Signum Function :

A function y= f (x) = Sgn (x) is defined as follows :



$$y = f(x) = \begin{cases} 1 & \text{for} \quad x > 0 \\ 0 & \text{for} \quad x = 0 \\ -1 & \text{for} \quad x < 0 \end{cases}$$

It is also written as Sgn x = |x|/ x ; $x \neq 0$;
= 0; x = 0

Note : f (x) = (sgn(x))x $\Rightarrow$ f (x) = |x|

Domain : $\mathbb{R}$

Range : {–1, 0, 1}

Do yourself - 4 :

1. Arrange the following in decreasing order $3^x, 5^x, 2^x, 4^x$ if x > 0

(A) $3^x > 5^x > 2^x > 4^x$ (B) $5^x > 4^x > 3^x > 2^x$

(C) $2^x > 3^x > 4^x > 5^x$ (D) $2^x > 5^x > 4^x > 3^x$

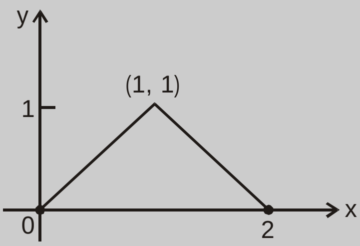
2. Draw the graph of $y = 2^x$, $y = 3^x$, $y = 4^x$, $y = 5^x$ on same cartesion plane.

3. Draw the graph of y = |x + 3| + |x – 1| + |x – 5|

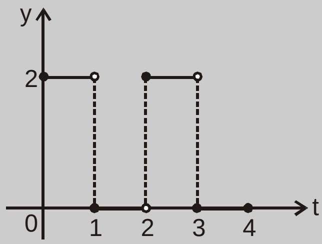
4. Draw the graph of sgn(sinx).

5. Find a formula for each function graphed

(i)

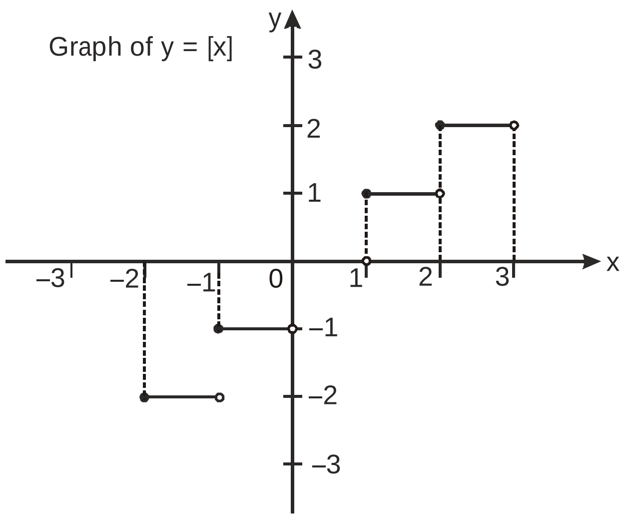


(ii)



(h) Greatest integer or step up function :

The function y = f(x) = [x] is called the greatest integer function where [x] denotes the greatest integer less than or equal to x. Note that for :



x	[x]
[–2,–1)	–2
[–1,0)	–1
[0,1)	0
[1,2)	1

Domain : $\mathbb{R}$

Range : I

Properties of greatest integer function :

(i) $[x] \le x < [x] + 1$ and $x - 1 < [x] \le x$, $0 \le x - [x] < 1$

(ii) $[x + m] = [x] + m$, if m is an integer.

(iii) $[x] + [-x] = \begin{cases} 0, & x \in \mathbb{Z} \\ -1, & x \notin \mathbb{Z} \end{cases}$

Illustration 5 : If y = 2[x] + 3 & y = 3[x – 2] + 5, then find [x + y] where [.] denotes greatest integer function.

Solution : $y = 3[x - 2] + 5 = 3[x] - 1$

so $3[x] - 1 = 2[x] + 3$

$[x] = 4 \Rightarrow 4 \le x < 5$

then y = 11

so x + y will lie in the interval [15, 16)

so [x + y] = 15

Illustration 6 : Find the value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \ldots\ldots \left[\frac{1}{2} + \frac{2946}{1000}\right]$ where [.] denotes greatest integer function ?

Solution :

$$\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{1000}\right] + \ldots\ldots \left[\frac{1}{2} + \frac{499}{1000}\right] + \left[\frac{1}{2} + \frac{500}{1000}\right] + \ldots\ldots \left[\frac{1}{2} + \frac{1499}{1000}\right] + \left[\frac{1}{2} + \frac{1500}{1000}\right] + \ldots\ldots$$

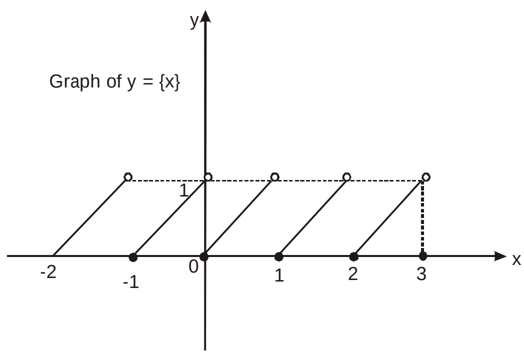
$$+ \left[\frac{1}{2} + \frac{2499}{1000}\right] + \left[\frac{1}{2} + \frac{2500}{1000}\right] + \ldots\ldots \left[\frac{1}{2} + \frac{2946}{1000}\right]$$

$= 0 + 1 \times 1000 + 2 \times 1000 + 3 \times 447 = 3000 + 1341 = 4341$ **Ans.**

(i) Fractional part function :

It is defined as : g(x) = {x} = x – [x]

e.g. the fractional part of the number 2.1 is 2.1–2 = 0.1 and the fractional part of –3.7 is 0.3 The period of this function is 1 and graph of this function is as shown.



x	{x}
[–2,–1)	x+2
[–1,0)	x+1
[0,1)	x
[1,2)	x–1

Domain : $\mathbb{R}$

Range : [0,1)

Properties of fractional part function :

(i) $0 \le \{x\} < 1$ (ii) $\{[x]\} = [\{x\}] = 0$ (iii) $\{\{x\}\} = \{x\}$

(iv) $\{x+m\} = \{x\},\ m \in I$ (v) $\{x\} + \{-x\} = \begin{cases} 1, & x \notin \mathbb{Z} \\ 0, & x \in \mathbb{Z} \end{cases}$

Illustration 7 : Solve the equation $|2x - 1| = 3[x] + 2\{x\}$ where [.] denotes greatest integer and {.} denotes fractional part function.

Solution : We are given that, $|2x - 1| = 3[x] + 2\{x\}$

Let, $2x - 1 \le 0$ i.e. $x \le \frac{1}{2}$. The given equation yields.

$1 - 2x = 3[x] + 2\{x\}$

$\Rightarrow \quad 1 - 2[x] - 2\{x\} = 3[x] + 2\{x\} \Rightarrow \quad 1 - 5[x] = 4\{x\} \Rightarrow \{x\} = \frac{1-5[x]}{4}$

$\Rightarrow \quad 0 \le \frac{1-5[x]}{4} < 1 \Rightarrow 0 \le 1 - 5[x] < 4 \Rightarrow -\frac{3}{5} < [x] \le \frac{1}{5}$

Now, $[x] = 0$ as zero is the only integer lying between $-\frac{3}{5}$ and $\frac{1}{5}$

$\Rightarrow \quad \{x\} = \frac{1}{4} \Rightarrow x = \frac{1}{4}$ which is less than $\frac{1}{2}$, Hence $\frac{1}{4}$ is one solution.

Now, let $2x - 1 > 0$ i.e $x > \frac{1}{2}$

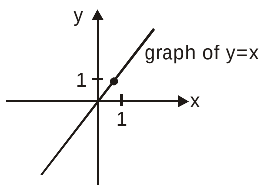
$\Rightarrow \quad 2x - 1 = 3[x] + 2\{x\} \quad \Rightarrow \quad 2[x] + 2\{x\} - 1 = 3[x] + 2\{x\}$

$\Rightarrow \quad [x] = -1 \Rightarrow -1 \le x < 0$ which is not a solution as $x > \frac{1}{2}$

$\Rightarrow \quad x = \frac{1}{4}$ is the only solution.

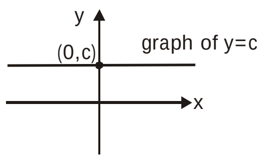
(j) Identity function :

The function $f : A \to A$ defined by $f(x) = x \ \forall \ x \in A$ is called the identity of A and is denoted by I_A. It is easy to observe that identity function defined on R is a bijection.



(k) Constant function :

A function $f : A \to B$ is said to be a constant function if every element of A has the same f image in B. Thus $f : A \to B$; $f(x) = c,\ \forall \ x \in A,\ c \in B$ is a constant function. Note that the range of a constant function is a singleton.



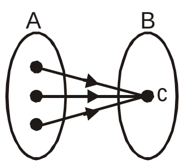


Illustration 8 : Find the domain $f(x) = \dfrac{1}{\sqrt{\|[|x|-5]\|-11}}$ where [.] denotes greatest integer function.

Solution : $\left|[|x|-5]\right| > 11$

so $[|x|-5] > 11$ or $[|x|-5] < -11$

$[|x|] > 16$ $\quad$ $[|x|] < -6$

$|x| \geq 17$ or $|x| < -6$ (Not Possible)

$\Rightarrow$ $x \leq -17$ or $x \geq 17$

so $x \in (-\infty, -17] \cup [17, \infty)$

Do yourself - 5 :

1. Find the range of x if

(i) $|[x]| = 10$ (ii) $[x]^2 - [x] - 2 = 0$ (iii) $[x] \geq 5$ (iv) $[x] < -3$

2. Find the domain of f(x) if $f(x) = \sqrt{[x]^2 - 17[x] + 66}$

3. If $y = 2[x] + 1 = 3[x + 1] - 5$ then find

(i) x (ii) y (iii) $[x + 2y]$

4. Find the value of $\left[\dfrac{1}{3}\right] + \left[\dfrac{1}{3} + \dfrac{1}{100}\right] + \ldots + \left[\dfrac{1}{3} + \dfrac{899}{100}\right]$, where [.] denotes greatest integer function.

5. Find the domain of the function $f(x) = \dfrac{1}{\sqrt{\|[|x|-7]\|-11}}$, where [.] denotes greatest integer function.

6. Find range of $f(x) = \dfrac{3 + x - [x]}{5 + x - [x]}$, where [.] denotes greatest integer function.

7. Draw graph of following functions :

(i) $f(x) = \text{sgn}(x^2 - 9)$ (ii) $f(x) = [x^2]$ (iii) $f(x) = [e^x]$

(iv) $f(x) = [\sin x]$ (v) $f(x) = [\cos x]$ (vi) $f(x) = [\sin x + \cos x]$

(sgn(x) denotes signum function and [.] denotes greatest integer function)

8. Draw graph of following functions :

(i) $f(x) = \{e^x\}$ (ii) $f(x) = \{x^2\}$

({.} denotes fractional part function)

9. Draw the graph of

(i) $y = \left[\sin \dfrac{x}{2}\right]$ (ii) $y = [x^3]$

10. Solve for x : (sgn(x) denotes signum function and [.] denotes greatest integer function and {.} denotes fractional part function)

(i) $3[x] - 2x = \{x\} + \dfrac{1}{2}$ (ii) $|[x - 1]| + 6x = 3\{x\} + 2[x] + 6$

11. Find x if

(i) $2x + \{x\} = 5[x]$ (ii) $x - \{x\} = 2[x]$ (iii) $[x] - 2x = 5\{x\} + 5$

12. Let $\{x\}$ & $[x]$ denote the fractional and integral part of a real number x respectively. Solve $4\{x\} = x + [x]$.

13. If [x] and {x} denotes the greatest integer function less than or equal to x and fractional part function respectively, then the number of real x, satisfying the equation $(x-2)[x] = \{x\} - 1$, is-

(A) 0 (B) 1 (C) 2 (D) infinite

14. If [a] denotes the greatest integer less than or equal to a and $-1 \le x < 0$, $0 \le y < 1$, $1 \le z < 2$,

then $\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is equal to -

(A) [x] (B) [y] (C) [z] (D) none of these

15. Let {x} & [x] denotes the fraction and integral part of a real number x respectively, then match the column.

	Column-I		**Column-II**
(A)	$[x^2] \ge 4$	(p)	$x \in [2, 4)$
(B)	$[x]^2 - 5[x] + 6 = 0$	(q)	$x \in (-\infty, -2] \cup [2, \infty)$
(C)	$x = \{x\}$	(r)	$x \in (-\infty, -5)$
(D)	$[x] < -5$	(s)	$x \in \{-2\}$
		(t)	$x \in [0, 1)$

5. ALGEBRAIC OPERATIONS ON FUNCTIONS :

If f & g are real valued functions of x with domain set A, B respectively, f + g, f – g, (f. g) & (f/g) as follows :

(a) $(f \pm g)(x) = f(x) \pm g(x)$ domain in each case is $A \cap B$

(b) $(f.g)(x) = f(x).g(x)$ domain is $A \cap B$

(c) $\left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)}$ domain $A \cap B - \{x|g(x) = 0\}$

Some more Problems Based on Domain of a Function :

Illustration 9 : Find the domain of the following function :

(i) $y = \log_{(x-4)} (x^2 - 11x + 24)$ (ii) $f(x) = \log_2 \left(-\log_{1/2}\left(1+\dfrac{1}{\sqrt[4]{x}}\right)-1\right)$

Solution : (i) $y = \log_{(x-4)}(x^2 - 11x + 24)$

Here 'y' would assume real value if,

$x - 4 > 0$ and $\ne 1$, $x^2 - 11x + 24 > 0 \Rightarrow x > 4$ and $\ne 5$, $(x - 3)(x - 8) > 0$

$\Rightarrow x > 4$ and $\ne 5$, $x < 3$ or $x > 8 \Rightarrow x > 8 \Rightarrow$ Domain (y) = $(8, \infty)$

(ii) We have $f(x)=\log_2\left(-\log_{1/2}\left(1+\frac{1}{\sqrt[4]{x}}\right)-1\right)$

f(x) is defined if $-\log_{1/2}\left(1+\frac{1}{\sqrt[4]{x}}\right)-1>0$

or if $\log_{1/2}\left(1+\frac{1}{\sqrt[4]{x}}\right)<-1$ or if $\left(1+\frac{1}{\sqrt[4]{x}}\right)>(1/2)^{-1}$

or if $1+\frac{1}{\sqrt[4]{x}}>2$ or if $\frac{1}{\sqrt[4]{x}}>1$ or if $x^{1/4}<1$ or if $0<x<1$

$\therefore$ D(f) = (0, 1)

Do yourself - 6 :

1. Find the Domain of

(i) $f(x)=\log_{3-2\{x\}}\left(x^2-5x+13\right)$ (ii) $f(x)=\sqrt{\frac{9-3^x}{5^{-x}-125}}$

2. Find domain of following functions : $f(x)=\sin\left(\sqrt{1-x^2}\right)+\sqrt{x+2}+\frac{1}{\log_{10}^{(x+1)}}$

3. Find the domains of definitions of the following functions :
(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)

(i) $f(x)=\sqrt{\cos 2x}+\sqrt{16-x^2}$ (ii) $f(x)=\log_7\log_5\log_3\log_2(2x^3+5x^2-14x)$

(iii) $f(x)=\ln\left(\sqrt{x^2-5x-24}-x-2\right)$ (iv) $f(x)=\sqrt{\frac{1-5^x}{7^{-x}-7}}$

4. Find the domains of definitions of the following functions :
(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)

(i) $y=\log_{10}\sin(x-3)+\sqrt{16-x^2}$ (ii) $f(x)=\log_{100x}\left(\frac{2\log_{10}x+1}{-x}\right)$

5. Find the domains of definitions of the following functions :
(Read the symbols [*] and {*} as greatest integers and fractional part functions respectively.)

(i) $f(x)=\sqrt{x^2-|x|}+\frac{1}{\sqrt{9-x^2}}$ (ii) $f(x)=\sqrt{(x^2-3x-10)\ln^2(x-3)}$

(iii) $f(x)=\sqrt{(5x-6-x^2)\left[\{\ell n\{x\}\}\right]}+\sqrt{(7x-5-2x^2)}+\left(\ln\left(\frac{7}{2}-x\right)\right)^{-1}$

6. Find the domains of definitions of the following functions :
(Read the symbols [*] as greatest integers function)

(i) $f(x)=\sqrt{\log_x(\cos 2\pi x)}$ (ii) $f(x)=\frac{[x]}{2x-[x]}$

7. Find domain

(i) $y = \frac{3}{9-x^2} + \log_2 x$ **(ii)** $y = \log_{10}(1 - \log_{10}(x^2 - x + 4))$

8. Find domain

(i) $y = \frac{1}{\sqrt{\log_{x+1}(x-1)}}$ **(ii)** $y = \sqrt{\log_x x^2 - 1}$ **(iii)** $y = \log_{x-2} \frac{x^2-4}{x+1}$

9. The domain of definition of f(x) = $\frac{\log_2(x+3)}{x^2+3x+2}$ is :

(A) R \ {–1, –2} (B) (–2, ∞)
(C) R \ {–1, –2, –3} (D) (–3, ∞) \ {–1, –2}

10. The number of integers lying in the domain of the function $f(x) = \sqrt{\log_{0.5}\left(\frac{5-2x}{x}\right)}$ is -

(A) 3 (B) 2 (C) 1 (D) 0

Illustration 10 : Find the range of following functions :

$f(x) = \log_{\sqrt{2}}\left(2 - \log_2(16\sin^2 x + 1)\right)$

Solution : $f(x) = \log_{\sqrt{2}}\left(2 - \log_2(16\sin^2 x + 1)\right)$

$1 \le 16 \sin^2 x + 1 \le 17$

$\therefore \quad 0 \le \log_2 (16 \sin^2 x + 1) \le \log_2 17$

$\therefore \quad 2 - \log_2 17 \le 2 - \log_2 (16 \sin^2 x + 1) \le 2$

Now consider $0 < 2 - \log_2 (16 \sin^2 x + 1) \le 2$

$\therefore \quad -\infty < \log_{\sqrt{2}}[2 - \log_2(16\sin^2 x + 1)] \le \log_{\sqrt{2}} 2 = 2$

$\therefore$ the range is $(-\infty, 2]$

Illustration 11 : Find the range of f(x) = $\frac{x-[x]}{1+x-[x]}$, where [.] denotes greatest integer function.

Solution : $y = \frac{x-[x]}{1+x-[x]} = \frac{\{x\}}{1+\{x\}}$

$$\therefore \quad \frac{1}{y} = \frac{1}{\{x\}} + 1 \Rightarrow \frac{1}{\{x\}} = \frac{1-y}{y} \Rightarrow \{x\} = \frac{y}{1-y}$$

$0 \le \{x\} < 1 \Rightarrow 0 \le \frac{y}{1-y} < 1$

Range = [0, 1/2)

Do yourself - 7 :

1. The function f(x) is defined on the interval [0, 1]. Find the domain of definition of the functions.
(i) f(sinx) (ii) f(2x + 3)

2. Given that $y = f(x)$ is a function whose domain is [4,7] and range is [–1, 9]. Find the range and domain of
(i) $g(x) = \frac{1}{3}f(x)$ (ii) $h(x) = f(x-7)$

3. Find range of
(i) $f(x) = \ell n(3x^2 - 8x + 1)$ (ii) $f(x) = \log_2\left(\frac{3\cos x + 4\sin x + 10}{3\cos x + 4\sin x + 15}\right)$

4. Find the range of the following functions :
(i) $f(x) = \frac{2}{x}$ (ii) $f(x) = \frac{1}{x^2 - x + 1}$ (iii) $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$

5. Find the range of the following functions :
(i) $f(x) = e^{(x-1)^2}$ (ii) $f(x) = x^3 - x^2 + x + 1$ (iii) $f(x) = \log(x^8 + x^4 + x^2 + 1)$

6. Find Range :-
(i) $f(x) = \frac{5x - 3}{4x + 7}$ (ii) $f(x) = \frac{7x + 9}{3x - 5}$

7. Find range of the following functions
(i) $f : R \to R$, $f(x) = 2\sin^2 x - 3\sin x + 8$ (ii) f(x) = 8cotx.sinx
(iii) $f(x) = \sin(\log_2(x^2 + 1))$

8. Find Range :-
(i) $f(x) = \frac{x^2 - 5x + 6}{4x^2 - 5x - 6}$ (ii) $f(x) = \frac{3x - 6}{x^3 - 4x^2 + 2x + 4}$ (iii) $f(x) = \frac{1}{x^2 - 7x + 6}$

9. Find the range of the following function :
(i) $\log_4\left|x + \frac{1}{x}\right|$ (ii) $f(x) = \sin(3x^2 + 1)$
(iii) $f(x) = 2\sin\left(2x + \frac{\pi}{4}\right)$ (iv) $f(x) = \cos\left(2x + \frac{\pi}{4}\right)$

10. Find range of
(i) $f(x) = \frac{3 - \sin^2 x}{4\cos^2 x + 2}$ (ii) $f(x) = \ell n(3x^2 - 8x + 9)$

11. Find range of
(i) $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ (ii) $f(x) = \frac{(\ell n\, x)^2 - 5\ell n\, x + 4}{(\ell n\, x)^2 - 2\ell n\, x - 3}$

12. Find the domain & range of the following functions.

(i) $y = \log_{\sqrt{5}}\left(\sqrt{2}(\sin x - \cos x) + 3\right)$ (ii) $y = \frac{2x}{1+x^2}$ (iii) $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$

13. Find the domain & range of the following functions.

(i) $f(x) = \frac{x}{1+|x|}$ (ii) $y = \sqrt{2-x} + \sqrt{1+x}$ (iii) $f(x) = \frac{\sqrt{x+4} - 3}{x-5}$

14. Find the range of the following functions :

(i) $f(x) = \frac{x-1}{x+2}$ (ii) $f(x) = \sin^2 x - 2\sin x + 4$

(iii) $f(x) = \sin(\log_2 x)$ (iv) $f(x) = \frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1}$

15. Find range of following functions : $f(x) = \log_2\left(\log_{1/2}(x^2 + 4x + 4)\right)$

16. A function f has domain [– 1, 2] and range [0, 1]. The domain and range respectively of the function g defined by $g(x) = 1 - f(x+1)$ is

(A) [–1, 1] ; [–1, 0] (B) [–2, 1]; [0, 1] (C) [0, 2] ; [–1, 0] (D) [1, 3] ; [–1, 0]

17. The range of the function $f(x) = \operatorname{sgn}\left(\frac{\sin^2 x + 2\sin x + 4}{\sin^2 x + 2\sin x + 3}\right)$ is (where sgn(.) denotes signum function)-

(A) {–1,0,1} (B) {–1,0} (C) {1} (D) {0,1}

18. The range of the function $f(\theta) = \sqrt{8\sin^2\theta + 4\cos^2\theta - 8\sin\theta\cos\theta}$ is -

(A) $\left[\sqrt{5}-1, \sqrt{5}+1\right]$ (B) $\left[0, \sqrt{5}+1\right]$

(C) $\left[\sqrt{6-\sqrt{20}}, \sqrt{6+\sqrt{20}}\right]$ (D) none of these

19. The range of the function $f(x) = \sqrt{4-x^2} + \sqrt{x^2-1}$ is

(A) $[\sqrt{3}, \sqrt{7}]$ (B) $[\sqrt{3}, \sqrt{5}]$ (C) $[\sqrt{2}, \sqrt{3}]$ (D) $[\sqrt{3}, \sqrt{6}]$

20. Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$ is -

(A) [1, 2] (B) $[1, \infty)$ (C) $\left[2, \frac{7}{3}\right]$ (D) $\left(1, \frac{7}{3}\right]$

21. Range of the function $f(x) = \frac{\{x\}}{1+\{x\}}$, where {x} denotes the fractional part function, is

(A) [0 , 1) (B) $\left[0, \frac{1}{2}\right]$ (C) $\left[0, \frac{1}{2}\right)$ (D) $\left(0, \frac{1}{2}\right)$

22. Let $f(x) = (x + 1)(x + 2)(x + 3)(x + 4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in N$ then find the value of $(a + b)$.

6. EQUAL OR IDENTICAL FUNCTION :

Two function f & g are said to be equal if :

(a) The domain of f = the domain of g

(b) The co-domain of f = co-domain of g and

(c) f(x) = g (x), for every x belonging to their common domain (i.e. should have the same graph)

Illustration 12 : The functions $f(x) = \log(x-1) - \log(x-2)$ and $g(x) = \log\left(\frac{x-1}{x-2}\right)$ are identical when x lies in the interval

(A) [1, 2] (B) $[2, \infty)$ (C) $(2, \infty)$ (D) $(-\infty, \infty)$

Solution : Since $f(x) = \log(x-1) - \log(x-2)$.

Domain of f(x) is $x > 2$ or $x \in (2, \infty)$ .....(i)

$g(x) = \log\left(\frac{x-1}{x-2}\right)$ is defined if $\frac{x-1}{x-2} > 0 \Rightarrow x \in (-\infty, 1) \cup (2, \infty)$ .....(ii)

From (i) and (ii), $x \in (2, \infty)$. **Ans. (C)**

7. HOMOGENEOUS FUNCTIONS :

A function is said to be homogeneous with respect to any set of variables when each of its terms is of the same degree with respect to those variables.

For examples $5x^2 + 3y^2 - xy$ is homogenous in x & y. Symbolically if, $f(tx, ty) = t^n f(x, y)$ then f(x, y) is homogeneous function of degree n.

Illustration 13 : Which of the following function is not homogeneous ?

(A) $x^3 + 8x^2y + 7y^3$ (B) $y^2 + x^2 + 5xy$ (C) $\frac{xy}{x^2+y^2}$ (D) $\frac{2x-y+1}{2y-x+1}$

Solution : It is clear that (D) does not have the same degree in each term. **Ans. (D)**

8. BOUNDED FUNCTION :

A function is said to be bounded if there exists a finite M such that $|f(x)| \le M, \forall x \in D_f$.

9. IMPLICIT & EXPLICIT FUNCTION :

A function defined by an equation not solved for the dependent variable is called an **implicit function**. e.g. the equations $x^3 + y^3 = 1$ & $x^y = y^x$, defines y as an implicit function. If y has been expressed in terms of x alone then it is called an **Explicit function.**

Illustration 14 : Which of the following function is implicit function ?

(A) $y = \frac{x^2 + e^x + 5}{\sqrt{1 - \cos^{-1} x}}$ (B) $y = x^2$ (C) $xy - \sin(x+y) = 0$ (D) $y = \frac{x^2 \log x}{\sin x}$

Solution : It is clear that in (C) y is not clearly expressed in x. **Ans. (C)**

Do yourself - 8 :

1. Which of the following function is implicit function ?

(A) $xy - \cos(x + y) = 0$ (B) $y = x^3$

(C) $y = \log(x^2 + x + 1)$ (D) $y = |x|$

2. For the function $f(x) = \dfrac{e^x + 1}{e^x - 1}$, if n(d) denotes the number of integers which are not in its domain and n(r) denotes the number of integers which are not in its range, then n(d) + n(r) is equal to -

(A) 2 (B) 3 (C) 4 (D) Infinite

3. If a function is defined by an implicit equation $2^{|x|+|y|} + 2^{|x|-|y|} = 2$, then -

(A) Domain of function is singleton (B) Range of function is singleton

(C) graph of the function intersects the line y = x (D) maximum value of function is 2

4. Are the following functions identical ?

(i) $f(x) = \dfrac{x}{x^2}$ & $\phi(x) = \dfrac{x^2}{x}$ (ii) $f(x) = x$ & $\phi(x) = \sqrt{x^2}$

(iii) $f(x) = \log_{10}x^2$ & $\phi(x) = 2\log_{10}|x|$

5. Find the boundness of the function $f(x) = \dfrac{x^2}{x^4 + 1}$

6. Convert the implicit form into the explicit function :

(i) $xy = 1$ (ii) $x^2y = 1$.

7. Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit functions :

(i) $10^x + 10^y = 10$ (ii) $x + |y| = 2y$

8. Identify the pair(s) of functions which are identical ?

(where [x] denotes greatest integer and {x} denotes fractional part function)

(i) $f(x) = \text{sgn}(x^2 - 3x + 4)$ and $g(x) = e^{[\{x\}]}$

(ii) $f(x) = \sqrt{\dfrac{1 - \cos 2x}{1 + \cos 2x}}$ and $g(x) = \tan x$

9. Identify the pair(s) of functions which are identical ?

(where [x] denotes greatest integer and {x} denotes fractional part function)

(i) $f(x) = \ln(1 + x) + \ln(1 - x)$ and $g(x) = \ln(1 - x^2)$

(ii) $f(x) = \dfrac{\cos x}{1 - \sin x}$ and $g(x) = \dfrac{1 + \sin x}{\cos x}$

10. APPLICATIONS OF FUNCTIONAL RULE :

Illustration 15 : Determine all functions f satisfying the functional relation.

$$f(x)+f\left(\frac{1}{1-x}\right)=\frac{2(1-2x)}{x(1-x)} \text{ where } x\in\mathbb{R}-\{0,1\}$$

Solution : Given $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2(1-2x)}{x(1-x)}=\frac{2}{x}-\frac{2}{1-x}$ ...(i)

Replacing x by $\frac{1}{1-x}$ we obtain

$$f\left(\frac{1}{1-x}\right)+f\left(\frac{1}{1-\frac{1}{1-x}}\right)=2(1-x)-\frac{2}{1-\frac{1}{1-x}}$$

$$\Rightarrow \quad f\left(\frac{1}{1-x}\right)+f\left(1-\frac{1}{x}\right)=-2x+\frac{2}{x} \quad \text{...(ii)}$$

Again replacing x by $\left(1-\frac{1}{x}\right)$ in (i) we obtain

$$\Rightarrow \quad f\left(1-\frac{1}{x}\right)+f\left(\frac{1}{1-\left(1-\frac{1}{x}\right)}\right)=\frac{2}{1-\frac{1}{x}}-\frac{2}{1-\left(1-\frac{1}{x}\right)}$$

$$\Rightarrow \quad f\left(1-\frac{1}{x}\right)+f(x)=\frac{2x}{x-1}-2x \quad \text{....(iii)}$$

subtracting (ii) from (i) then

$$f(x)-f\left(1-\frac{1}{x}\right)=2x-\frac{2}{1-x}$$

Now adding (iii) and (iv) we get

$$2f(x)=\frac{2x}{x-1}-\frac{2}{1-x}$$

$$\Rightarrow \quad f(x)=\frac{x+1}{x-1}$$

Do yourself - 9 :

1. If $2f(x) - 3f\left(\frac{1}{x}\right) = x^2$, x is not equal to zero, then f(2) is equal to-

(A) $-\frac{7}{4}$ (B) $\frac{5}{2}$ (C) –1 (D) none of these

2. If $f(x) = \frac{4^x}{4^x + 2}$, then f(x) + f(1 – x) is equal to-

(A) 0 (B) –1 (C) 1 (D) 4

3. If $x^4 f(x) - \sqrt{1-\sin 2\pi x} = |f(x)| - 2f(x)$, then f(–2) equals

(A) $\frac{1}{17}$ (B) $\frac{1}{11}$ (C) $\frac{1}{19}$ (D) 0

4. Solve the following problems on functional equation :

(i) The function f(x) defined on the real numbers has the property that f(f(x)).(1+f(x))=–f(x) for all x in the domain of f. If the number 3 is in the domain and range of f, compute the value of f(3).

(ii) Suppose f is a real function satisfying f(x + f(x)) = 4f(x) and f(1) = 4. Find the value of f(21).

5. Number of integers in range of $f(x) = x(x + 2)(x + 4)(x + 6) + 7$, $x \in [-4,2]$ is

6. (i) Let f be function defined from $\mathbb{R}^+ \to \mathbb{R}^+$. If $[f(xy)]^2 = x(f(y))^2$ for all positive numbers x and y and f(2) = 6, find the value of f(50).

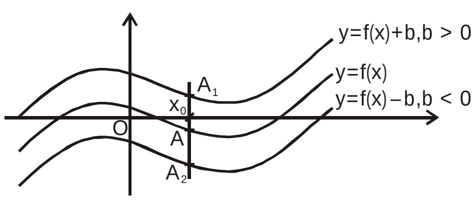
(ii) Let f be a function such that f(3) = 1 and f(3x) = x + f(3x – 3) for all x. Then find the value of f(300).

7. A function f, defined for all x, y $\in \mathbb{R}$ is such that f(1) = 2 & $f(x + y) - kxy = f(x) + 2y^2$, where k is some constant. Find f(x) & show that :

$f(x+y)f\left(\frac{1}{x+y}\right) = k$ for $x + y \neq 0$.

11. BASIC TRANSFORMATIONS ON GRAPHS :

(i) Drawing the graph of y = f(x) + b, b ∈ ℝ, from the known graph of y = f(x)



It is obvious that domain of f(x) and f(x) + b are the same. Let us take any point x_0 in the domain of f(x). $y|_{x=x_0} = f(x_0)$.

The corresponding point on f(x) + b would be $f(x_0) + b$.

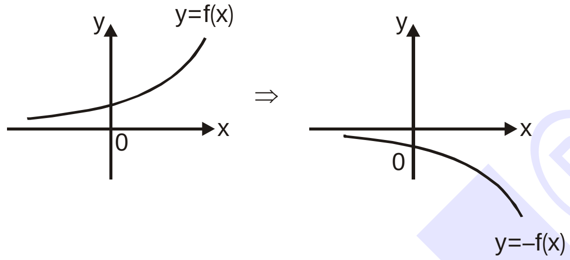
For $b > 0 \Rightarrow f(x_0) + b > f(x_0)$ it means that the corresponding point on f(x) + b would be lying at a distance 'b' units above the point on f(x).

For $b < 0 \Rightarrow f(x_0) + b < f(x_0)$ it means that the corresponding point on $f(x) + b$ would be lying at a distance 'b' units below the point on f(x).

Accordingly the graph of $f(x) + b$ can be obtained by translating the graph of f(x) either in the positive y-axis direction (if $b > 0$) or in the negative y-axis direction (if $b < 0$), through a distance $|b|$ units.

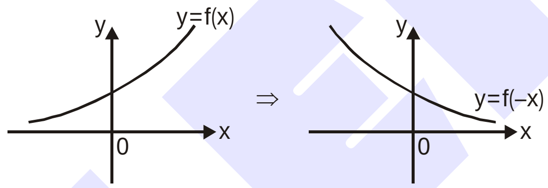
(ii) Drawing the graph of y = –f(x) from the known graph of y = f(x)

To draw $y = -f(x)$, take the image of the curve $y = f(x)$ in the x-axis as plane mirror.



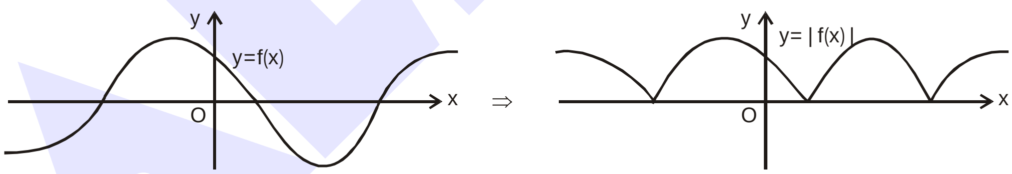
(iii) Drawing the graph of y = f(–x) from the known graph of y = f(x)

To draw $y = f(-x)$, take the image of the curve $y = f(x)$ in the y-axis as plane mirror.



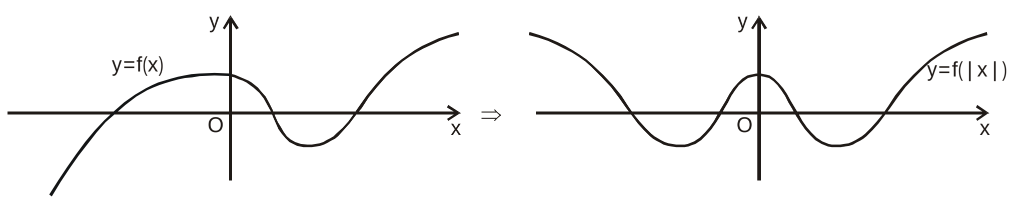
(iv) Drawing the graph of y = |f(x)| from the known graph of y = f(x)

$|f(x)| = f(x)$ if $f(x) \geq 0$ and $|f(x)| = -f(x)$ if $f(x) < 0$. It means that the graph of f(x) and |f(x)| would coincide if $f(x) \geq 0$ and for the portions where $f(x) < 0$ graph of |f(x)| would be image of $y = f(x)$ in x-axis.



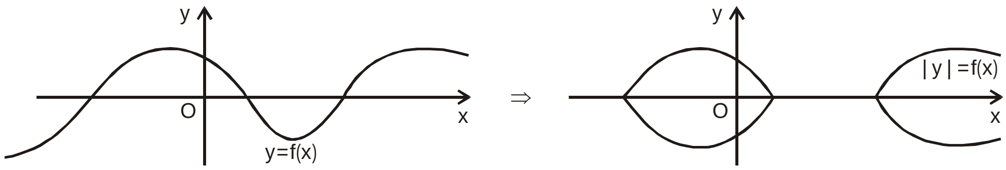
(v) Drawing the graph of y = f(|x|) from the known graph of y = f(x)

It is clear that, $f(|x|) = \begin{cases} f(x), & x \geq 0 \\ f(-x), & x < 0 \end{cases}$. Thus f(|x|) would be a even function, graph of f(|x|) and f(x) would be identical in the first and the fourth quadrants (as $x \geq 0$) and as such the graph of f(|x|) would be symmetric about the y-axis (as (|x|) is even).

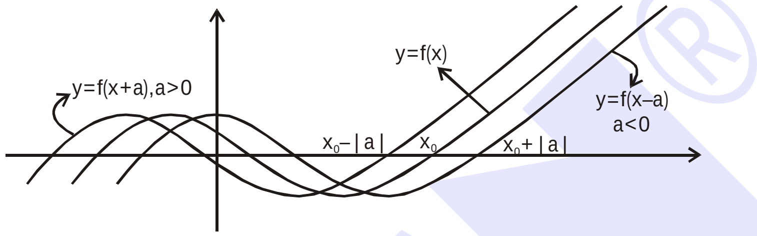


(vi) Drawing the graph of |y| = f(x) from the known graph of y = f(x)

Clearly $|y| \geq 0$. If $f(x) < 0$, graph of $|y| = f(x)$ would not exist. And if $f(x) \geq 0$, $|y| = f(x)$ would give $y = \pm f(x)$. Hence graph of $|y| = f(x)$ would exist only in the regions where f(x) is non-negative and will be reflected about the x-axis only in those regions.



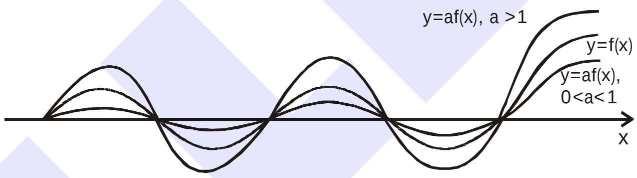
(vii) Drawing the graph of y = f(x + a), a ∈ ℝ from the known graph of y = f(x)



(i) If a > 0, shift the graph of f(x) through 'a' units towards left of f(x).

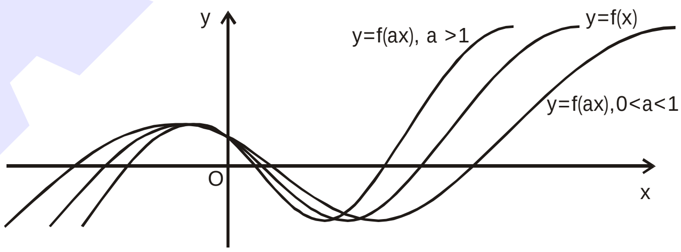
(ii) If a < 0, shift the graph of f(x) through 'a' units towards right of f(x).

(viii) Drawing the graph of y = af(x) from the known graph of y = f(x)



It is clear that the corresponding points (points with same x co-ordinates) would have their ordinates in the ratio of 1 : a.

(ix) Drawing the graph of y = f(ax) from the known graph of y = f(x).



Let us take any point $x_0 \in$ domain of f(x). Let $ax = x_0$ or $x = \dfrac{x_0}{a}$.

Clearly if $0 < a < 1$, then $x > x_0$ and f (x) will stretch by $\dfrac{1}{a}$ units along the y-axis and if $a > 1$, $x < x_0$, then f(x) will compress by 'a' units along the y-axis.

Note :

(i) A function h(x) is defined as

h(x) = max. {f(x), g(x)} then

$$h(x)=\begin{cases} f(x) & f(x)\ge g(x) \\ g(x) & g(x)>f(x) \end{cases}$$

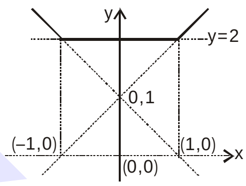
(ii) A function h(x) is defined as

h(x) = min. {f(x), g(x)} then

$$h(x)=\begin{cases} f(x) & f(x)\le g(x) \\ g(x) & g(x)<f(x) \end{cases}$$

Illustration 16 : Find f(x) = max {1 + x, 1 – x, 2}.

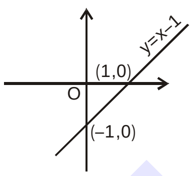
Solution : From the graph it is clear that



$$f(x)\ 3=\begin{cases} 1-x\,; & x<-1 \\ 2\quad ; & -1\le x\le 1 \\ 1+x\,; & x>1 \end{cases}$$

Illustration 17 : Draw the graph of y = |2 –|x –1||.

Solution :



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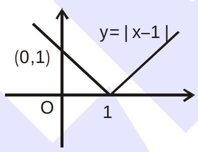

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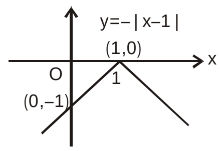
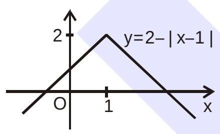




$\Rightarrow$

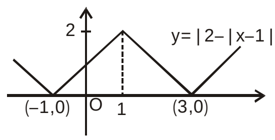
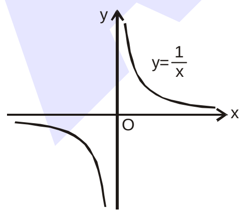
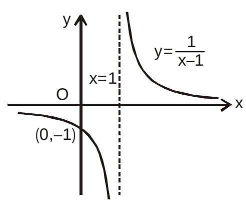


Illustration 18 : Draw the graph of y = $2-\dfrac{4}{|x-1|}$

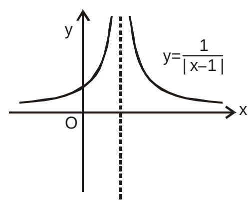
Solution :

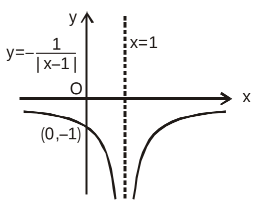


$\Rightarrow$

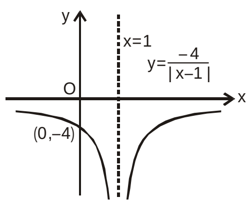


$\Rightarrow$





$\Rightarrow$



$\Rightarrow$

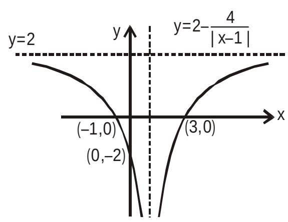


Illustration 19 : Draw the graph of $y = |e^{|x|} - 2|$

Solution :

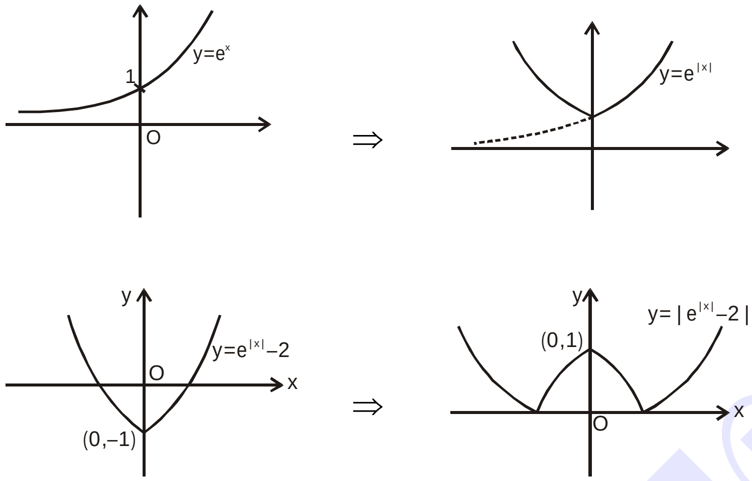
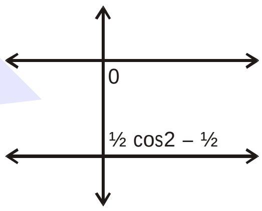


Illustration 20 : Draw the graph of $f(x) = \cos x \cos(x + 2) - \cos^2(x + 1)$.

Solution : $f(x) = \cos x \cos(x + 2) - \cos^2(x + 1)$

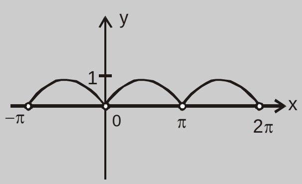
$$= \frac{1}{2}[\cos(2x+2)+\cos 2] - \frac{1}{2}[\cos(2x+2)+1]$$

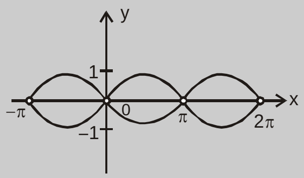
$$= \frac{1}{2}\cos 2 - \frac{1}{2} < 0.$$

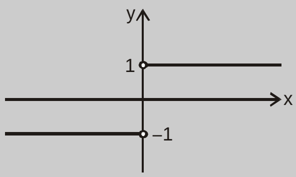


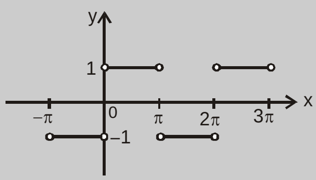
Do yourself - 10 :

1. Which of the following is the graph of $y = \frac{|\sin x|}{\sin x}$?

(A) 

(B) 

(C) 

(D) 

2. Draw graph of following functions :

(i) $y = \left|\ell n|x|\right| + 1$ (ii) $y = \min.\{x^2 + 1, 3-x\}$ (iii) $y = |x - 1| + |x - 3|$

3. A lion moves in the region given by the graph $y - |y| - x + |x| = 0$. Then on which of the following curve a person can move so that he does not encounter lion -

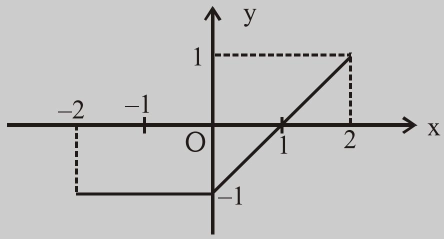
(A) $y = e^{-|x|}$ (B) $y = \frac{1}{x}$ (C) $y = \text{signum}(x)$ (D) $y = -|4 + |x||$

4. Draw graphs of the following function, where [ ] denotes the greatest integer function.

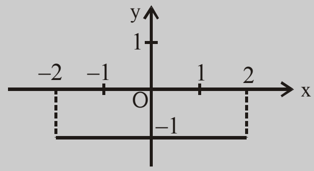
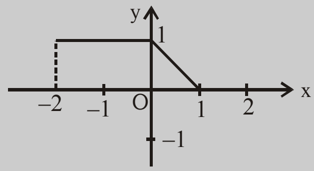
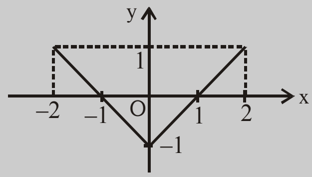
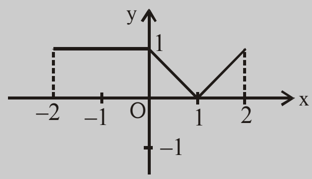
(i) $f(x) = x + [x]$ (ii) $y = (x)^{[x]}$ where $x = [x] + (x)$ & $x > 0$ & $x \le 3$

(iii) $y = \text{sgn}[x]$ (iv) $\text{sgn}(x - |x|)$

5. The graph of the function $y = f(x)$ is as follows :



Match the function mentioned in **Column-I** with the respective graph given in **Column-II**.

Column-I	**Column-II**
(A) $y = \lvert f(x)\rvert$	(P) 
(B) $y = f(\lvert x\rvert)$	(Q) 
(C) $y = f(-\lvert x\rvert)$	(R) 
(D) $y = \frac{1}{2}(\lvert f(x)\rvert - f(x))$	(S) 

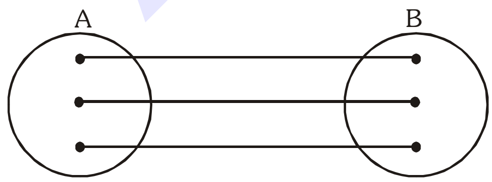
12. CLASSIFICATION OF FUNCTIONS :

One-One Function (Injective mapping) :

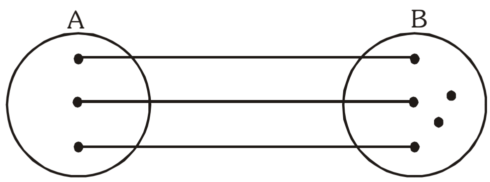
A function $f : A \to B$ is said to be a one-one function or injective mapping if different elements of A have different f images in B.

Thus there exist $x_1, x_2 \in A$ & $f(x_1), f(x_2) \in B$, $\boldsymbol{f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2}$ **or** $\boldsymbol{x_1 \neq x_2 \Leftrightarrow f(x_1) \neq f(x_2)}$.

Diagramatically an injective mapping can be shown as



or

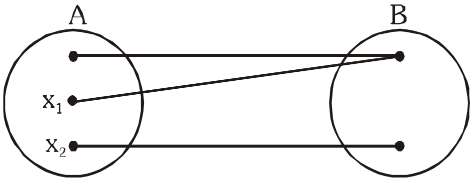


Many-one function (not injective) :

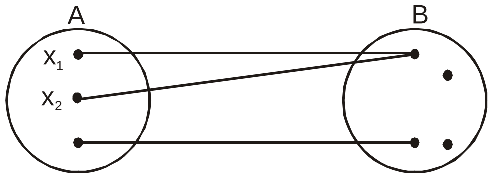
A function $f : A \to B$ is said to be a many one function if two or more elements of A have the same f image in B.

Thus $f : A \to B$ is many one there exist $x_1, x_2 \in A$, $\boldsymbol{f(x_1) = f(x_2)}$ **but** $\boldsymbol{x_1 \neq x_2}$.

Diagramatically a many one mapping can be shown as



or



Note :

(i) If a line parallel to x-axis cuts the graph of the function atmost at one point, then the function is one-one.

(ii) If any line parallel to x-axis cuts the graph of the function atleast at two points, then f is many-one.

(iii) If continuous function $f(x)$ is always increasing or decreasing in whole domain, then $f(x)$ is one-one.

(iv) All linear functions are one-one.

(v) All trigonometric functions in their domain are many one

(vi) All even degree polynomials are many one

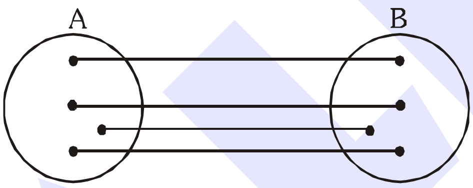
(vii) Linear by Linear is one-one

(viii) Quadratic by quadratic with no common factor is many one.

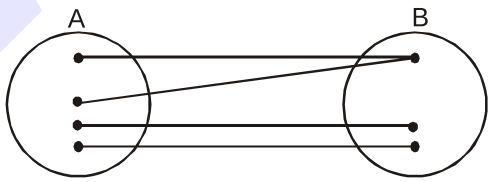
Onto function (Surjective mapping) :

If the function $f : A \to B$ is such that each element in B (co-domain) is the f image of atleast one element in A, then we say that f is a function from A 'onto' B. Thus $f : A \to B$ is surjective iff $\forall\ b \in B$, $\exists$ some $a \in A$ such that $f(a) = b$.

Diagramatically surjective mapping can be shown as



or

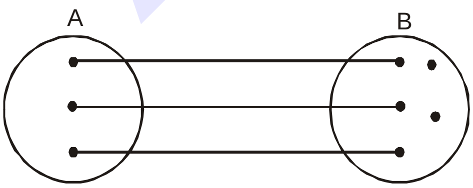


Note that : if range is same as co-domain, then $\boldsymbol{f(x)}$ is onto.

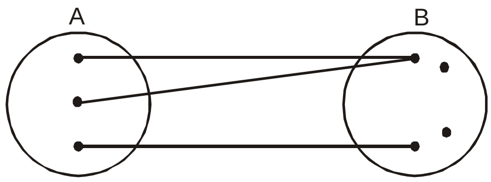
Into function :

If $f : A \to B$ is such that there exists atleast one element in co-domain which is not the image of any element in domain, then $f(x)$ is into.

Diagramatically into function can be shown as



or



Note :

(i) A polynomial function of degree even define from $\mathbb{R} \to \mathbb{R}$ will always be into.

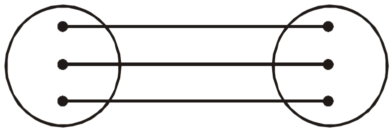
(ii) A polynomial function of degree odd defined from $\mathbb{R} \to \mathbb{R}$ will always be onto.

(iii) Quadratic by quadratic without any common factor define from $\mathbb{R} \to \mathbb{R}$ is always an into function.

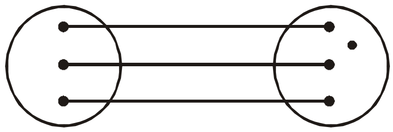
Thus a function can be one of these four types :

(i) one-one onto (injective & surjective)

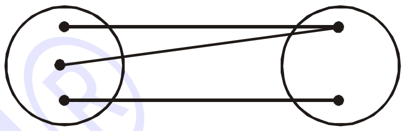
(also known as **Bijective** mapping)



(ii) one-one into (injective but not surjective)



(iii) many-one onto (surjective but not injective)



(iv) many-one into (neither surjective nor injective)

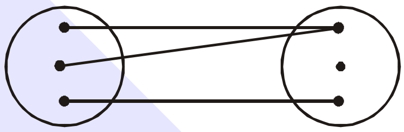
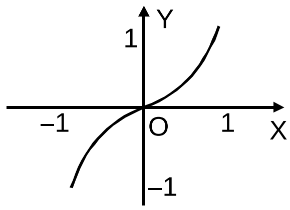


Illustration 21 : Let $A = \{x : -1 \le x \le 1\} = B$ be a mapping $f : A \to B$. For each of the following functions from A to B, find whether it is bijective or non-bijective.

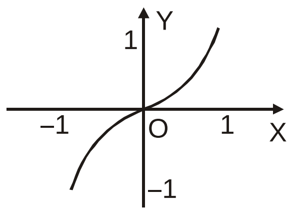
(a) $f(x) = x|x|$ (b) $f(x) = x^3$ (c) $f(x) = \sin\frac{\pi x}{2}$

Solution : (a) $f(x) = x|x| = \begin{cases} -x^2 & , -1 < x < 0 \\ x^2 & , 0 \le x < 1 \end{cases}$,



Graphically,
The graph shows f(x) is one-one, as the straight line parallel to x-axis cuts only at one point.
Here, range
$f(x) \in [-1, 1]$
Thus, range = co-domain
Hence, onto.
Therefore, f(x) is one-one onto or (Bijective).

(b) $f(x) = x^3$,
Graphically;
Graph shows f(x) is one-one onto
(i.e. Bijective)
[as explained in above example]



(c) $f(x) = \sin\frac{\pi x}{2}$
Graphically;
Which shows f(x) is one-one and onto as range

= co-domain.
Therefore, f(x) is bijective.

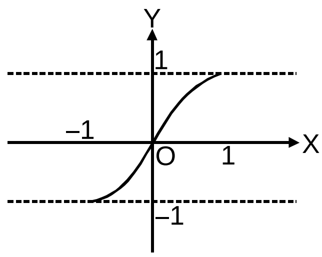


Illustration 22 : Let $f : \mathbb{N} \to \mathbb{Z}$ be a function defined as $f(x) = x - 1000$. Show that $f(x)$ is an into function.

Solution : Let $f(x) = y = x - 1000 \quad \Rightarrow \quad x = y + 1000 = g(y)$ (say)

here $g(y)$ is defined for each $y \in \mathbb{Z}$, but $g(y) \notin \mathbb{N}$ for $y \leq -1000$.

Hence $f(x)$ is into.

Illustration 23 : Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = x + \sqrt{x^2}$, then f is

(A) injective (B) surjective (C) bijective (D) None of these

Solution : We have, $f(x) = x + \sqrt{x^2} = x + |x|$

Clearly, f is not one-one as $f(-1) = f(-2) = 0$ and $-1 \neq -2$

Also, f is not onto as $f(x) \geq 0 \ \forall \ x \in \mathbb{R}$

$\therefore$ range of $f = (0, \infty) \subset \mathbb{R}$ **Ans.(D)**

Illustration 24 : Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as $f(x) = 2x^3 + 6x^2 + 12x + 3\cos x - 4\sin x$; then f is -

(A) Injective (B) Surjective (C) Bijective (D) Not Surjective

Solution : We have $f(x) = 2x^3 + 6x^2 + 12x + 3\cos x - 4\sin x$

$\Rightarrow \ f'(x) = 6x^2 - 12x + 12 - 3\sin x - 4\cos x$

$$f'(x) = \underbrace{6(x-1)^2 + 6}_{g(x)} - \underbrace{(3\sin x + 4\cos x)}_{h(x)}$$

range of $g(x) = [6, \infty)$

range of $h(x) = [-5, 5]$

hence $f'(x)$ always lies in the interval $[1, \infty)$

$\Rightarrow \ f'(x) > 0$

Hence $f(x)$ is increasing i.e. one-one function

Now $x \to \infty \Rightarrow f \to \infty$ & $x \to -\infty \Rightarrow f \to -\infty$ & $f(x)$ is continuous

hence its range is $\mathbb{R} \Rightarrow f$ is onto so f is bijective. **Ans. (C)**

Illustration 25 : Let $f(x) = \dfrac{x^2 + 3x + a}{x^2 + x + 1}$, where $f : \mathbb{R} \to \mathbb{R}$. Find the value of parameter 'a' so that the given function is one-one.

Solution : $$f(x) = \frac{x^2 + 3x + a}{x^2 + x + 1}$$

$$f'(x) = \frac{(x^2 + x + 1)(2x + 3) - (x^2 + 3x + a)(2x + 1)}{(x^2 + x + 1)^2} = \frac{-2x^2 + 2x(1 - a) + (3 - a)}{(x^2 + x + 1)^2}$$

Let, $g(x) = -2x^2 + 2x(1 - a) + (3 - a)$

$g(x)$ will be negative if $4(1 - a)^2 + 8(3 - a) < 0$

$\Rightarrow \ 1 + a^2 - 2a + 6 - 2a < 0 \Rightarrow (a - 2)^2 + 3 < 0$

which is not possible. Therefore function is not monotonic.

Hence, no value of a is possible.

Do yourself - 11 :

1. Let function $f : R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then f is -

(A) one to one and onto (B) one to one but not onto

(C) onto but not one to one (D) neither one to one nor onto

2. Let $f(x) = \frac{x}{1+x}$ defined from $(0, \infty) \to [0, \infty)$ then by f(x) is -

(A) one-one but not onto (B) one-one and onto

(C) many one but not onto (D) many one and onto

3. Let $f : N \to N$ be defined as $f(n) = \begin{cases} \frac{n+1}{2}, & \text{if n is odd} \\ \frac{n}{2}, & \text{if n is even} \end{cases}$ for all $n \in N$. Then f is-

(A) injective but not surjective (B) surjective but not injective

(C) both injective as well as surjective (D) neither injective nor surjective

4. Let E = {1, 2, 3, 4} & F = {1, 2}. Then the number of onto functions from E to F is -

(A) 14 (B) 16 (C) 12 (D) 8

5. **(i)** Is the function $f : \mathbb{N} \to \mathbb{N}$ (the set of natural numbers) defined by $f(x) = 2x + 3$ surjective ?

(ii) Let $A = R - \{3\}$, $\mathbb{R} = \mathbb{R} - \{1\}$ and let $f : A \to B$ defined by $f(x) = \frac{x-2}{x-3}$. Check whether the function f(x) is bijective or not.

6. Which of the following function is surjective but not injective

(A) $f : \mathbb{R} \to \mathbb{R}$ $f(x) = x^4 + 2x^3 - x^2 + 1$ (B) $f : \mathbb{R} \to \mathbb{R}$ $f(x) = x^3 + x + 1$

(C) $f : \mathbb{R} \to \mathbb{R}^+$ $f(x) = \sqrt{1+x^2}$ (D) $f : \mathbb{R} \to \mathbb{R}$ $f(x) = x^3 + 2x^2 - x + 1$

7. A mapping $f : A \to [-1,1]$ defined by $f(x) = \sin x$, $\forall x \in \mathbb{R}$, where A is a subset of $\mathbb{R}$ (the set of all real numbers) is one-one and onto if A is the interval, then A is belongs to

(A) $[0,2\pi]$ (B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (C) $[-\pi,\pi]$ (D) $[0,\pi]$

8. Classify the following functions f(x) defined in $\mathbb{R} \to \mathbb{R}$ as injective, surjective, both or none.

(i) $f(x) = \frac{x^2+4x+30}{x^2-8x+18}$ (ii) $f(x) = x^3 - 6x^2 + 11x - 6$

(iii) $f(x) = (x^2 + x + 5)(x^2 + x - 3)$

9. Match the functions given in column-I correctly with mappings given in column-II.

Column-I	Column-II
(A) $f:\left[-\frac{1}{2},\frac{1}{2}\right]\to\left[\frac{4}{7},\frac{4}{3}\right]$	(P) Injective mapping
$f(x)=\frac{1}{x^2+x+1}$	(Q) Non-injective mapping
(B) $f:[-2, 2]\to[-1, 1]$	(R) Surjective mapping
$f(x) = \sin x$	(S) Non-surjective mapping
(C) $f:\mathbb{R}-I\to\mathbb{R}$	(T) Bijective mapping
$f(x) = \ell n\{x\}$, (where $\{.\}$ represents fractional part function)	
(D) $f:(-\infty, 0]\to[1, \infty)$, $f(x)=(1+\sqrt{-x})+(\sqrt{-x}-x)$	

13. COMPOSITE OF UNIFORMLY & NON-UNIFORMLY DEFINED FUNCTION:

Let $f: A\to B$ & $g: B\to C$ be two functions. Then the function $gof: A\to C$ defined by $(gof)(x) = g(f(x))$ $\forall\ x\in A$ is called the composite of the two functions f & g.

Diagramatically $\xrightarrow{x}\boxed{f}\xrightarrow{f(x)}\boxed{g}\longrightarrow g(f(x))$

Thus the image of every $x\in A$ under the function gof is the g-image of f-image of x.

Note that gof is defined only if $\forall\ x\in A$, f(x) is an element of the domain of 'g' so that we can take its g-image.

Properties of composite functions:

(a) In general composite of functions is not commutative i.e. $gof \neq fog$.

(b) The composition of functions is associative i.e. if f, g, h are three functions such that fo(goh) & (fog)oh are defined, then fo(goh) = (fog)oh.

(c) The composition of two bijections is a bijection i.e. if f & g are two bijections such that gof is defined, then gof is also a bijection.

Illustration 26 : If $f(x) = x^2 + 1$, $g(x)=\frac{1}{x-1}$, then find (fog) (x) and (gof) (x).

Solution : Given, $f(x) = x^2 + 1$ .....(1) $\quad g(x)=\frac{1}{x-1}$ ...(2)

Now $(fog)(x) = f(g(x)) = f\left(\frac{1}{x-1}\right) = f(z)$, where $z=\frac{1}{x-1}$

$= z^2 + 1$ $\quad [\because f(x) = x^2 + 1]$

$$=\left(\frac{1}{x-1}\right)^2+1=\frac{1}{(x-1)^2}+1$$

Note : Domain of fog(x) is $x\in\mathbb{R}-\{1\}$

$(gof)(x) = g(f(x)) = g(x^2 + 1) = g(u)$, where $u = x^2+1$

$$=\frac{1}{u-1}=\frac{1}{x^2+1-1}=\frac{1}{x^2}$$

Note : Domain of gof(x) is $x\in\mathbb{R}-\{0\}$

Illustration 27 : If f be the greatest integer function and g be the modulus function, then

$$(gof)\left(-\frac{5}{3}\right)-(fog)\left(-\frac{5}{3}\right)=$$

(A) 1 (B) –1 (C) 2 (D) 4

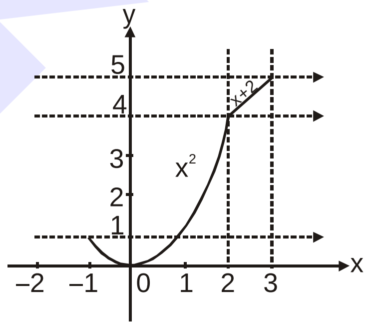
Solution : Given $(gof)\left(\frac{-5}{3}\right)-(fog)\left(\frac{-5}{3}\right) = g\left\{f\left(\frac{-5}{3}\right)\right\}-f\left\{g\left(\frac{-5}{3}\right)\right\} = g(-2)-f\left(\frac{5}{3}\right) = 2-1=1$

Ans.(A)

Illustration 28 : Let $f(x)=\begin{cases} x+1, & x\le 1 \\ 2x+1, & 1<x\le 2 \end{cases}$ and $g(x)=\begin{cases} x^2, & -1\le x<2 \\ x+2, & 2\le x\le 3 \end{cases}$, find (fog)

Solution : $f(g(x))=\begin{cases} g(x)+1, & g(x)\le 1 \\ 2g(x)+1, & 1<g(x)\le 2 \end{cases}$

Here, g(x) becomes the variable that means we should draw the graph.



It is clear that $g(x)\le 1$; $\forall\ x\in[-1,1]$

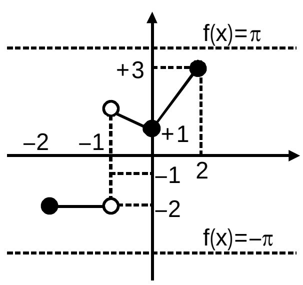
and $1<g(x)\le 2$; $\forall\ x\in(1,\sqrt{2}\,]$

$$\Rightarrow\quad f(g(x))=\begin{cases} x^2+1, & -1\le x\le 1 \\ 2x^2+1, & 1<x\le\sqrt{2} \end{cases}$$

Illustration 29: Find the domain and range of h(x) = g(f(x)), where

$f(x)=\begin{cases} [x], & -2\le x\le -1 \\ |x|+1, & -1<x\le 2 \end{cases}$ and $g(x)=\begin{cases} [x], & -\pi\le x\le 0 \\ \sin x, & 0\le x\le\pi \end{cases}$, [.] denotes the greatest integer function.

Solution : $h(x)=g(f(x))=\begin{cases} [f(x)], & -\pi\le f(x)<0 \\ \sin(f(x)), & 0\le f(x)\le\pi \end{cases}$



From graph of f(x), we get

$$h(x)=\begin{cases} [[x]], & -2\le x\le -1 \\ \sin(|x|+1), & -1<x\le 2 \end{cases}$$

$\Rightarrow$ Domain of h(x) is [–2, 2]

and Range of h(x) is $\{-2,-1\}\cup[\sin 3, 1]$

Do yourself - 12 :

1. Let f(x) = $\frac{\alpha x}{x+1}$, x ≠ – 1. Then for what value of α is f(f (x)) = x ?

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) 1 (D) –1

2. Let g (x) = 1 + x – [x] & f (x) = $\begin{cases} -1, & x<0 \\ 0, & x=0 \\ 1, & x>0 \end{cases}$. Then for all x, f(g (x)) is equal to, where [ ] denotes greatest integer function.

(A) x (B) 1 (C) f(x) (D) g(x)

3. $f:\mathbb{R}\to\mathbb{R}$ such that $f(x) = \ell n\left(x+\sqrt{x^2+1}\right)$. Another function g(x) is defined such that go$f$(x) = x ∀ x ∈ R. Then g(2) is -

(A) $\frac{e^2+e^{-2}}{2}$ (B) e^2 (C) $\frac{e^2-e^{-2}}{2}$ (D) e^{-2}

4. (i) $f(x) = x^3 - x$ & g(x) = sin 2x, find

(a) f(f(1)) (b) f(f(–1)) (c) $f\left(g\left(\frac{\pi}{2}\right)\right)$

(d) $f\left(g\left(\frac{\pi}{4}\right)\right)$ (e) g(f(1)) (f) $g\left(g\left(\frac{\pi}{2}\right)\right)$

(ii) If $f(x) = \begin{cases} x+1; & 0\le x<2 \\ |x|; & 2\le x<3 \end{cases}$, then find fof(x).

5. Suppose f(x) = sinx and g(x) = 1 – $\sqrt{x}$. Then find the domain and range of the following functions.

(i) fog (ii) gof (iii) fof (iv) gog

6. Find the formula for the function fogoh, given $f(x)=\frac{x}{x+1}$; $g(x)=x^{10}$ and $h(x)=x+3$. Find also the domain of this function. Also compute (fogoh)(–1).

7. If f(x) = max(x, 1/x) for x > 0 where max (a, b) denotes the greater of the two real numbers a and b. Define the function g(x) = f(x) f(1/x) and plot its graph.

8. If f(x) = –1 + |x – 2|, 0 ≤ x ≤ 4

g(x) = 2 – |x|, –1 ≤ x ≤ 3

Then find fog(x) & gof(x). Draw rough sketch of the graphs of fog(x) & gof(x).

9. A function $f:\mathbb{R}\to\mathbb{R}$ is such that $f\left(\frac{1-x}{1+x}\right)=x$ for all x ≠ –1. Prove the following.

(i) $f(f(x)) = x$ (ii) $f(1/x) = -f(x)$, x ≠ 0 (iii) $f(-x-2) = -f(x) - 2$.

10. $f(x)=\begin{bmatrix} 1-x & \text{if } x\le 0 \\ x^2 & \text{if } x>0 \end{bmatrix}$ and $g(x)=\begin{bmatrix} -x & \text{if } x<1 \\ 1-x & \text{if } x\ge 1 \end{bmatrix}$ find (fog)(x) and (gof)(x).

11. If $f(x)=\begin{cases} 0 & x<1 \\ 2x-2 & x\ge 1 \end{cases}$; then the number of solutions of the equation $f(f(f(x))) = x$ is

14. INVERSE OF A FUNCTION :

Let $f : A \to B$ be a one-one & onto function, then there exists a unique function $g : B \to A$ such that $f(x) = y \Leftrightarrow g(y) = x$, $\forall\ x \in A$ & $y \in B$. Then g is said to be inverse of f.

Thus $g = f^{-1} : B \to A = \{(f(x), x)) | (x, f(x)) \in f\}$.

Properties of inverse function :

(a) The inverse of a bijection is unique.

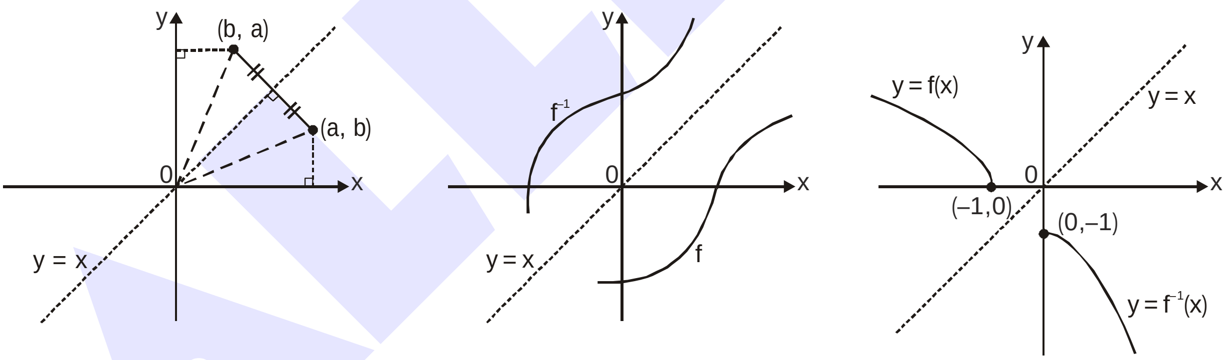
(b) If $f : A \to B$ is a bijection & $g : B \to A$ is the inverse of f, then $fog = I_B$ and $gof = I_A$, where I_A & I_B are identity functions on the sets A & B respectively. If $fof = I$, then f is inverse of itself.

(c) The inverse of a bijection is also a bijection.

(d) If f & g are two bijections $f : A \to B$, $g : B \to C$ then the inverse of gof exists and

$(gof)^{-1} = f^{-1} \text{ o } g^{-1}$.

(e) Since $f(a) = b$ if and only if $f^{-1}(b) = a$, the point (a, b) is on the graph of 'f' if and only if the point (b, a) is on the graph of f^{-1}. But we get the point (b, a) from (a, b) by reflecting about the line $y = x$.



The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

Drawing the graph of $y = f^{-1}(x)$ from the known graph of $y = f(x)$

For drawing the graph of $y = f^{-1}(x)$ take the reflection of $y = f(x)$ about the line $y = x$. The reflected part would give us the graph of $y = f^{-1}(x)$.

e.g. let us draw the graph of $y = \sin^{-1}x$.

We know that $y = f(x) = \sin x$ is invertible if $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$

$\Rightarrow$ the inverse mapping would be $f^{-1} : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

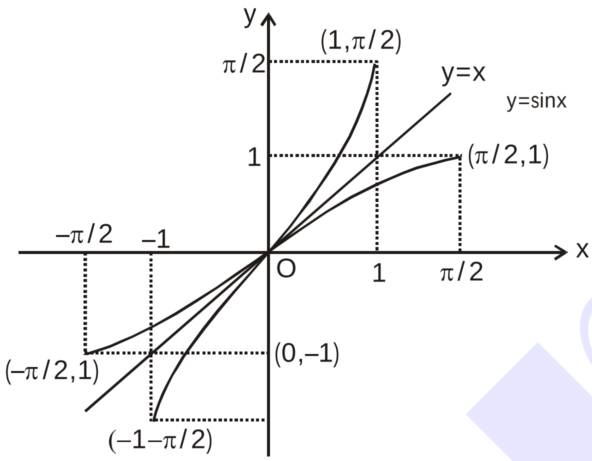


Illustration 30 : Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = (e^x - e^{-x})/2$. Is f(x) invertible ? If so, find its inverse.

Solution : Let us check for invertibility of f(x) :

(a) One-One :

$$f(x) = \frac{1}{2}\left(e^x - e^{-x}\right) \Rightarrow f'(x) = \frac{1}{2}\left(e^x + e^{-x}\right)$$

$\Rightarrow f'(x) > 0$, $f(x)$ is increasing function

$\therefore$ $f(x)$ is one-one function.

(b) Onto :

As x tends to larger and larger values so does f(x) and when $x \rightarrow \infty$, $f(x) \rightarrow \infty$.

Similarly as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ i.e. $-\infty < f(x) < \infty$ so long as $x \in (-\infty, \infty)$

Hence the range of f is same as the set R. Therefore f(x) is onto.

Since f(x) is both one-one and onto, f(x) is invertible.

(c) To find $f^{-1}(x)$: Interchange x & y

$$\frac{e^y - e^y}{2} = x \Rightarrow e^{2y} - 2xe^y - 1 = 0$$

$$\Rightarrow \quad e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} \Rightarrow e^y = x \pm \sqrt{1 + x^2}$$

Since $e^y > 0$, hence negative sign is ruled out and

Hence $e^y = x + \sqrt{1 + x^2}$

Taking logarithm, we have $y = \ell n(x + \sqrt{1 + x^2})$ or $f^{-1}(x) = \ell n(x + \sqrt{1 + x^2})$

Illustration 31 : Find the inverse of the function $f(x) = \log_a\left(x+\sqrt{(x^2+1)}\right)$; $a > 1$ and assuming it to be an onto function.

Solution : Given $f(x) = \log_a\left(x+\sqrt{x^2+1}\right)$

$$\therefore \quad f'(x) = \frac{\log_a e}{\sqrt{(1+x^2)}} > 0$$

which is a strictly increasing functions.

Thus, f(x) is injective, given that f(x) is onto. Hence the given function f(x) is invertible.

Interchanging x & y

$$\Rightarrow \log_a\left(y+\sqrt{(y)^2+1}\right) = x$$

$$\Rightarrow y+\sqrt{(y)^2+1} = a^x \qquad ...(i)$$

$$\text{and } \sqrt{(y)^2+1} - y = a^{-x} \qquad ...(ii)$$

From (i) and (ii), we get $y = \frac{1}{2}(a^x - a^{-x})$ or $f^{-1}(x) = \frac{1}{2}(a^x - a^{-x})$

Illustration 32 : Find the inverse of the function $f(x) = \ell n(x^2+3x+1)$; $x \in [1,3]$ and assuming it to be an onto function.

Solution : Given $f(x) = \ell n(x^2 + 3x + 1)$

$$\therefore \quad f'(x) = \frac{2x+3}{(x^2+3x+1)} > 0 \,\forall x \in [1,3]$$

Which is a strictly increasing function. Thus f(x) is injective, given that f(x) is onto. Hence the given function f(x) is invertible.

Interchanging x & y

$\Rightarrow (y)^2 + 3(y) + 1 - e^x = 0$

$$\therefore \quad y = \frac{-3 \pm \sqrt{9-4.(1-e^x)}}{2} = \frac{-3 \pm \sqrt{(5+4e^x)}}{2}$$

$$\Rightarrow y = \frac{-3 + \sqrt{(5+4e^x)}}{2} \qquad (\text{as } y \in [1,3])$$

$$\text{Hence } f^{-1}(x) = \frac{-3 + \sqrt{(5+4e^x)}}{2}$$

Do yourself - 13 :

1. If $f(x) = x|x|$ then $f^{-1}(x)$ equals-

(A) $\sqrt{|x|}$ (B) $(\text{sgn } x).\sqrt{|x|}$ (C) $-\sqrt{|x|}$ (D) Does not exist

(where sgn(x) denotes signum function of x)

2. Let f be a function defined by $f(x) = (x - 1)^2 + 1, (x \geq 1)$

Statement - 1 : The set $\{x : f(x) = f^{-1}(x)\} = \{1, 2\}$

Statement - 2 : f is bijection and $f^{-1}(x) = 1 + \sqrt{x-1}$, $x \geq 1$.

(1) Statement–1 is true, Statement–2 is false.

(2) Statement–1 is false, Statement–2 is true.

(3) Statement–1 is true, Statement–2 is true ; Statement–2 is a correct explanation for Statement–1.

(4) Statement–1 is true, Statement–2 is true ; Statement–2 is not a correct explanation for statement–1.

3. If $f(x) = 3x - 5$, then $f^{-1}(x)$ -

(A) is given by $\frac{1}{3x-5}$ (B) is given by $\frac{x+5}{3}$

(C) does not exist because f is not one–one (D) does not exist because f is not onto

4. If the function $f : [1, \infty) \rightarrow [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is -

(A) $\left(\frac{1}{2}\right)^{x(x-1)}$ (B) $\frac{1}{2}\left(1+\sqrt{1+4\log_2 x}\right)$ (C) $\frac{1}{2}\left(1-\sqrt{1+4\log_2 x}\right)$ (D) Not defined

5. If $f : (-\infty, 3] \rightarrow [7, \infty)$; $f(x) = x^2 - 6x + 16$, then which of the following is true -

(A) $f^{-1}(x) = 3 + \sqrt{x-7}$ (B) $f^{-1}(x) = 3 - \sqrt{x-7}$

(C) $f^{-1}(x) = \frac{1}{x^2 - 6x + 16}$ (D) f is many-one

6. If $f : [1, \infty) \rightarrow [2, \infty)$ is given by, $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals -

(A) $\frac{x+\sqrt{x^2-4}}{2}$ (B) $\frac{x}{1+x^2}$ (C) $\frac{x-\sqrt{x^2-4}}{2}$ (D) $1 - \sqrt{x^2-4}$

7. Suppose $f(x) = (x + 1)^2$ for $x \geq -1$. If g(x) is the function whose graph is the reflection of the graph of f(x) with respect to the line y = x, then g(x) equals -

(A) $-\sqrt{x} - 1, x \geq 0$ (B) $\frac{1}{(1+x)^2}, x \geq -1$

(C) $\sqrt{x+1}, x \geq -1$ (D) $\sqrt{x} - 1, x \geq 0$

8. (i) Let $f : [-1, 1] \rightarrow [-1, 1]$ defined by $f(x) = x|x|$, find $f^{-1}(x)$.

(ii) $f(x) = 1 + \ell n(x + 2)$, find $f^{-1}(x)$.

9. Compute the inverse of the functions :

(i) $f(x) = \ell n\left(x + \sqrt{x^2+1}\right)$ (ii) $f(x) = 2^{\frac{x}{x-1}}$ (iii) $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$

10. Find the inverse of $f(x) = 2^{\log_{10} x} + 8$ and hence solve the equation $f(x) = f^{-1}(x)$.

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15. ODD & EVEN FUNCTIONS :

Consider a function f (x) such that both x and –x are in its domain then

$$\text{If } \begin{cases} f(-x) = f(x) & \text{then } f \text{ is said to be an even function} \\ f(-x) = -f(x) & \text{then } f \text{ is said to be an odd function} \end{cases}$$

Note :

(i) $f(x) - f(-x) = 0 \Rightarrow f(x)$ is even & $f(x) + f(-x) = 0 \Rightarrow f(x)$ is odd.

(ii) A function may neither be odd nor even.

(iii) The only function which is defined on the entire number line & is even and odd at the same time is $f(x) = 0$.

(iv) Every constant function is even function.

(v) Inverse of an even function is not defined.

(vi) Every even function is symmetric about the y-axis & every odd function is symmetric about the origin.

Special Note : If a function f(x) is defined as $f(a + x) = f(a - x)$ then this function is symmetric about line x = a

(vii) Every function which has '–x' in it's domain whenever 'x' is in it's domain, can be expressed as the sum of an even & an odd function .

$$\text{i.e. } f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{EVEN}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{ODD}}$$

(viii) If f (x) is odd and defined at x = 0, then $f(0) = 0$.

f(x)	g(x)	f(x) + g(x)	f(x) – g(x)	f(x) . g(x)	f(x)/g(x)	(go$f$)(x)	($f$og)(x)
odd	odd	odd	odd	even	even	odd	odd
even	even	even	even	even	even	even	even
odd	even	neither odd nor even	neither odd nor even	odd	odd	even	even
even	odd	neither odd nor even	neither odd nor even	odd	odd	even	even

Illustration 33 : Which of the following functions is (are) even, odd or neither :

(i) $f(x) = x^2 \sin x$ (ii) $f(x) = \sin x - \cos x$ (iii) $f(x) = \dfrac{e^x + e^{-x}}{2}$

Solution :

(i) $f(-x) = (-x)^2 \sin(-x) = -x^2 \sin x = -f(x)$. Hence f(x) is odd.

(ii) $f(-x) = \sin(-x) - \cos(-x) = -\sin x - \cos x$.

Hence f(x) is neither even nor odd.

(iii) $f(-x) = \dfrac{e^{-x} + e^{-(-x)}}{2} = \dfrac{e^{-x} + e^{x}}{2} = f(x)$. Hence f(x) is even

Illustration 34: Identify the given functions as odd, even or neither :

(i) $f(x) = \dfrac{x}{e^x - 1} + \dfrac{x}{2} + 1$ (ii) $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$

Solution : (i) $f(x) = \dfrac{x}{e^x - 1} + \dfrac{x}{2} + 1$

Clearly domain of f(x) is $\mathbb{R} \sim \{0\}$. We have,

$$f(-x) = \frac{-x}{e^{-x}-1} - \frac{x}{2} + 1 = \frac{-e^x.x}{1-e^x} - \frac{x}{2} + 1 = \frac{(e^x - 1 + 1)x}{(e^x - 1)} - \frac{x}{2} + 1$$

$$= x + \frac{x}{e^x - 1} - \frac{x}{2} + 1 = \frac{x}{e^x - 1} + \frac{x}{2} + 1 = f(x)$$

Hence f(x) is an even function.

(ii) $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$

Replacing x, y by zero, we get $f(0) = 2f(0) \Rightarrow f(0) = 0$

Replacing y by –x, we get $f(x) + f(-x) = f(0) = 0 \Rightarrow f(x) = -f(-x)$

Hence f(x) is an odd function.

Do yourself - 14 :

1. Which of the following functions is (are) even, odd or neither :

(i) $f(x) = x^3 \sin 3x$ (ii) $f(x) = \dfrac{e^{x^2} + e^{-x^2}}{2x}$

(iii) $f(x) = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$ (iv) $f(x) = x^2 + 2^x$

2. Find whether the following functions are even or odd or none :

(i) $f(x) = \log\left(x + \sqrt{1+x^2}\right)$ (ii) $f(x) = \dfrac{x(a^x + 1)}{a^x - 1}$

(iii) $f(x) = \sin x + \cos x$ (iv) $f(x) = x\sin^2 x - x^3$

(v) $f(x) = \sin x - \cos x$ (vi) $f(x) = \dfrac{(1+2^x)^2}{2^x}$

(vii) $f(x) = \dfrac{x}{e^x - 1} + \dfrac{x}{2} + 1$ (viii) $f(x) = [(x + 1)^2]^{1/3} + [(x - 1)^2]^{1/3}$

3. Let a and b be real numbers and let $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in R$.

If $f(\log_{10}(\log_3 10)) = 5$ then find the value of $f(\log_{10}(\log_{10} 3))$.

16. PERIODIC FUNCTION :

A function $f(x)$ is called periodic if there exists a positive number T such that $f(x+T) = f(x) = f(x-T)$, for all values of x within the domain of f. Smallest positive T (if exists) is called fundamental period of function $f(x)$.

Note :

(i) Odd powers of sinx, cosx, secx, cosecx are periodic with period 2π.

(ii) None zero integral powers of tanx, cotx are periodic with period π.

(iii) Non zero even powers or modulus of sinx, cosx, secx, cosecx are periodic with period π.

(iv) $f(T) = f(0) = f(-T)$, where 'T' is the period.

(v) if $f(x)$ has a period T then **$f(ax + b)$ has a period T/|a|**$(a \neq 0)$.

Proof : Let $f(x + T) = f(x)$ and $f[a(x + T') + b] = f(ax + b)$

$f(ax + b + aT') = f(ax + b)$

$f(y + aT') = f(y) = f(y + T) \Rightarrow T = aT' \Rightarrow T' = \dfrac{T}{a}$

(vi) If $f(x)$ & $g(x)$ are periodic with period T_1 & T_2 respectively, then a period (need not be fundamental) of $f(x) \pm g(x)$ is L.C.M. of (T_1, T_2).

(a) LCM of T_1 & T_2 is defined when T_1/T_2 is rational.

(b) LCM of $\left\{\dfrac{a}{b}, \dfrac{p}{q}\right\} = \dfrac{\text{LCM of }(a, p)}{\text{HCF of }(b, q)}$

In case if there exists a positive K such that K < LCM of T_1 and T_2 and overall function repeats itself after every K, then fundamental period of the function will be K.

(vii) Every constant function is always periodic, whose fundamental period is undefined.

(viii) Periodic functions are non invertible.

Illustration 35 : Find the fundamental periods (if periodic) of the following functions, where [.] denotes the greatest integer function

(i) $f(x) = e^{\ell n(\sin x)} + \tan^3 x - \text{cosec}(3x - 5)$ (ii) $f(x) = x - [x - b],\ b \in \mathbb{R}$

(iii) $f(x) = \dfrac{|\sin x + \cos x|}{|\sin x| + |\cos x|}$ (iv) $f(x) = \tan\dfrac{\pi}{2}[x]$

(v) $f(x) = \cos(\sin x) + \cos(\cos x)$ (vi) $f(x) = \dfrac{(1+\sin x)(1+\sec x)}{(1+\cos x)(1+\text{cosec}\, x)}$

(vii) $f(x) = e^{x-[x]+|\cos \pi x|+|\cos 2\pi x|+........+|\cos n\pi x|}$

Solution : (i) $f(x) = e^{\ell n(\sin x)} + \tan^3 x - \text{cosec}(3x - 5)$

Period of $e^{\ell n \sin x} = 2\pi$, $\tan^3 x = \pi$

$\text{cosec}\,(3x - 5) = \dfrac{2\pi}{3}$

$\therefore$ Period $= 2\pi$

(ii) $f(x) = x - [x - b] = b + \{x - b\}$

$\therefore$ Period = 1

(iii) $f(x) = \dfrac{|\sin x + \cos x|}{|\sin x| + |\cos x|}$

Since period of $|\sin x + \cos x| = \pi$ and period of $|\sin x| + |\cos x|$ is $\dfrac{\pi}{2}$. Hence f(x) is periodic with π as its period

(iv) $f(x) = \tan\dfrac{\pi}{2}[x]$

$\tan\dfrac{\pi}{2}[x + T] = \tan\dfrac{\pi}{2}[x] \Rightarrow \dfrac{\pi}{2}[x + T] = n\pi + \dfrac{\pi}{2}[x]$

$\therefore$ T = 2

$\therefore$ Period = 2

(v) Let f(x) is periodic then $f(x + T) = f(x)$

$\Rightarrow$ $\cos(\sin(x + T)) + \cos(\cos(x + T)) = \cos(\sin x) + \cos(\cos x)$

If x = 0 then $\cos(\sin T) + \cos(\cos T)$

$= \cos(0) + \cos(1) = \cos\left(\cos\dfrac{\pi}{2}\right) + \cos\left(\sin\dfrac{\pi}{2}\right)$

On comparing $T = \dfrac{\pi}{2}$

(vi) $f(x) = \dfrac{(1+\sin x)(1+\sec x)}{(1+\cos x)(1+\operatorname{cosec} x)} = \dfrac{(1+\sin x)(1+\cos x)\sin x}{\cos x(1+\sin x)(1+\cos x)}$

$\Rightarrow$ $f(x) = \tan x$

Hence f(x) has period π.

(vii) $f(x) = e^{x-[x]+|\cos\pi x|+|\cos 2\pi x|+........+|\cos n\pi|}$

Period of $x - [x] = 1$

Period of $|\cos\pi x| = 1$

Period of $|\cos 2\pi x| = \dfrac{1}{2}$

....

Period of $|\cos n\pi x| = \dfrac{1}{n}$

So period of f(x) will be L.C.M. of all period = 1

Illustration 36 : Find the fundamental periods (if periodic) of the following functions, where [.] denotes the greatest integer function

(i) $f(x) = e^{x-[x]} + \sin x$ (ii) $f(x) = \sin\frac{\pi x}{\sqrt{2}} + \cos\frac{\pi x}{\sqrt{3}}$ (iii) $f(x) = \sin\frac{\pi x}{\sqrt{3}} + \cos\frac{\pi x}{2\sqrt{3}}$

Solution : (i) Period of $e^{x-[x]} = 1$

period of $\sin x = 2\pi$

$\because$ L.C.M. of rational and an irrational number does not exist.

$\therefore$ not periodic.

(ii) Period of $\sin\frac{\pi x}{\sqrt{2}} = \frac{2\pi}{\pi/\sqrt{2}} = 2\sqrt{2}$

Period of $\cos\frac{\pi x}{\sqrt{3}} = \frac{2\pi}{\pi/\sqrt{3}} = 2\sqrt{3}$

$\because$ L.C.M. of two different kinds of irrational number does not exist.

$\therefore$ not periodic.

(iii) Period of $\sin\frac{\pi x}{\sqrt{3}} = \frac{2\pi}{\pi/\sqrt{3}} = 2\sqrt{3}$

Period of $\cos\frac{\pi x}{2\sqrt{3}} = \frac{2\pi}{\pi/2\sqrt{3}} = 4\sqrt{3}$

$\because$ L.C.M. of two similar irrational number exist.

$\therefore$ Periodic with period = $4\sqrt{3}$ **Ans.**

Do yourself - 15 :

1. Let $f(x) = \sin^6 x + \cos^6 x$, then -

(A) $f(x) \in [0, 1]\ \forall\ x \in \mathbb{R}$ (B) $f(x) = 0$ has no solution

(C) $f(x) \in \left[\frac{1}{4}, 1\right] \forall x \in \mathbb{R}$ (D) $f(x)$ is an injective function

2. The period of the function $\frac{\sin x + \sin 5x}{\cos x + \cos 5x}$ is -

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{2}$ (C) π (D) 2π

3. Let $f(x)$ be a periodic function with period 'p' satisfying

$f(x) + f(x + 3) + f(x + 6) + ...+ f(x + 42) = \text{constant}\ \forall\ x \in \mathbb{R}$, then sum of digits of 'p' is

4. Suppose that f is an even, periodic function with period 2, and that f(x) = x for all x in interval [0,1]. Find the value of f(3.14).

5. Find the fundamental periods (if periodic) of the following functions.

(i) $f(x) = \ell n(\cos x) + \tan^3 x$. (ii) $f(x) = e^{x-[x]}$, [.] denotes greatest integer function

(iii) $f(x) = \left|\sin\frac{x}{2}\right| + \left|\cos\frac{x}{2}\right|$

6. Match the function mentioned in column-I with the respective classification given in column-II. (where [.] and {.} denote greatest integer function and fractional part function respectively)

	Column-I		Column-II
(A)	$f : R \to R^+$ $f(x) = \left(e^{[x]}\right)\left(e^{\{x\}}\right)$	(P)	one-one
(B)	$f : (-\infty,-2) \cup (0,\infty) \to R$ $f(x) = \ell n(x^2 + 2x)$	(Q)	many-one
(C)	$f : [-2,2] \to [-1,1]$ $f(x) = \sin x$	(R)	onto
(D)	$f : R \to R$ $f(x) = x^3 - 3x^2 + 3x - 7$	(S)	periodic
		(T)	aperiodic

17. GENERAL :

If f is a continuous function, then :

(a) $f(xy) = f(x) + f(y)\ \forall\ x,y \in \mathbb{R}^+ \Rightarrow f(x) = k\ell n\ x$

(b) $f(xy) = f(x) \,.\, f(y)\ \forall\ x,y \in \mathbb{R} \Rightarrow f(x) = x^n,\ n \in \mathbb{R}$ or $f(x) = 0$

(c) $f(x + y) = f(x) \,.\, f(y)\ \forall\ x,y \in \mathbb{R} \Rightarrow f(x) = a^{kx}$ or $f(x) = 0$

(d) $f(x + y) = f(x) + f(y)\ \forall\ x,y \in \mathbb{R} \Rightarrow f(x) = kx$, where k is a constant.

Illustration 37 : If the function f(x) satisfies the functional rule, $f(x + y) = f(x) + f(y)\ \forall\ x,y \in \mathbb{R}$ & $f(1) = 5$, then find $\sum_{n=1}^{m} f(n)$.

Solution : Here, $f(x + y) = f(x) + f(y)$; put $x = t - 1,\ y = 1$

$f(t) = f(t-1) + f(1)$ ....(1)

$\therefore\ f(t) = f(t-1) + 5$

$\Rightarrow\ f(t) = \{f(t-2) + 5\} + 5$

$\Rightarrow\ f(t) = f(t-2) + 2(5)$

$\Rightarrow\ f(t) = f(t-3) + 3(5)$

....

....

$\Rightarrow\ f(t) = f\{t-(t-1)\} + (t-1)5$

$\Rightarrow\ f(t) = f(1) + (t-1)5$

$\Rightarrow\ f(t) = 5 + (t-1)5$

$\Rightarrow\ f(t) = 5t$

$$\therefore\ \sum_{n=1}^{m} f(n) = \sum_{n=1}^{m}(5n) = 5[1+2+3+.....+m] = \frac{5m(m+1)}{2}$$

Hence, $\sum_{n=1}^{m} f(n) = \frac{5m(m+1)}{2}$.

INVERSE TRIGONOMETRIC FUNCTION

1. INTRODUCTION :

The inverse trigonometric functions, denoted by $\sin^{-1}x$ or (arc sinx), $\cos^{-1}x$ etc., denote the angles whose sine, cosine etc, is equal to x. The angles are usually the numerically smallest angles, except in the case of $\cot^{-1}x$ and if positive & negative angles have same numerical value, the positive angle has been chosen.

It is worthwhile noting that the functions sinx, cosx etc are in general not invertible. Their inverse is defined by choosing an appropriate domain & co-domain so that they become invertible. For this reason the chosen value is usually the simplest and easy to remember.

2. DOMAIN & RANGE OF INVERSE TRIGONOMETRIC FUNCTIONS :

S.No	*f* (x)	Domain	Range
(1)	$\sin^{-1}x$	$\lvert x\rvert \le 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(2)	$\cos^{-1}x$	$\lvert x\rvert \le 1$	$[0,\ \pi]$
(3)	$\tan^{-1}x$	$x \in \mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(4)	$\sec^{-1}x$	$\lvert x\rvert \ge 1$	$[0,\ \pi]-\left\{\frac{\pi}{2}\right\}$ or $\left[0, \frac{\pi}{2}\right)\cup\left(\frac{\pi}{2}, \pi\right]$
(5)	$\text{cosec}^{-1}x$	$\lvert x\rvert \ge 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]-\{0\}$
(6)	$\cot^{-1}x$	$x \in \mathbb{R}$	$(0,\ \pi)$

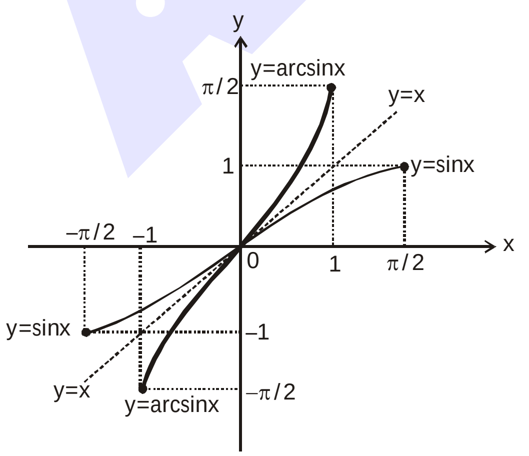
3. GRAPH OF INVERSE TRIGONOMETRIC FUNCTIONS :

(a) $f : \left[\frac{-\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$

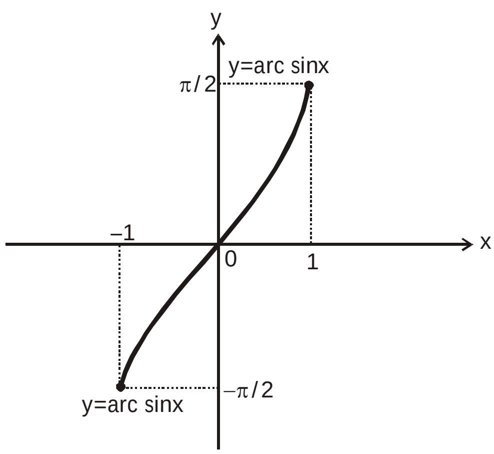
$f(x) = \sin x$

$f^{-1} : [-1,\ 1] \to [-\pi/2, \pi/2]$

$f^{-1}(x) = \sin^{-1}(x)$



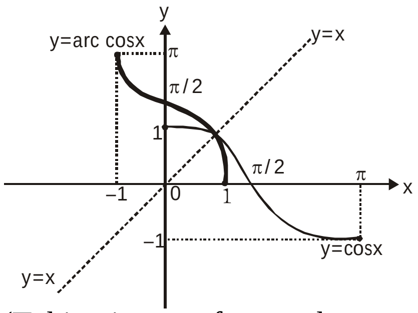
(Taking image of sin x about y = x to get $\sin^{-1}x$)



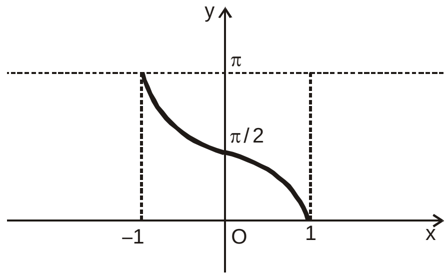
($y = \sin^{-1}x$)

(b) $f : [0, \pi] \to [-1, 1]$ $\qquad$ $f^{-1} : [-1, 1] \to [0, \pi]$

$f(x) = \cos x$ $\qquad$ $f^{-1}(x) = \cos^{-1} x$



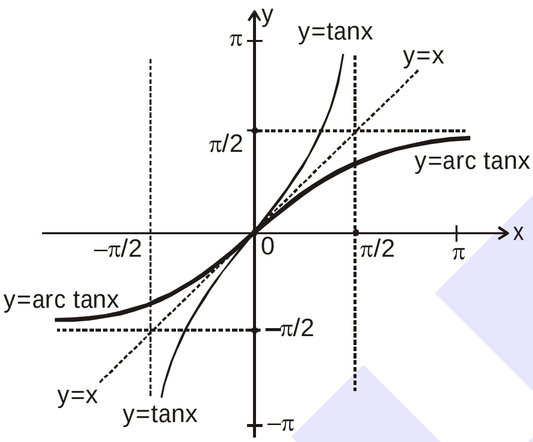
(Taking image of cos x about y = x)



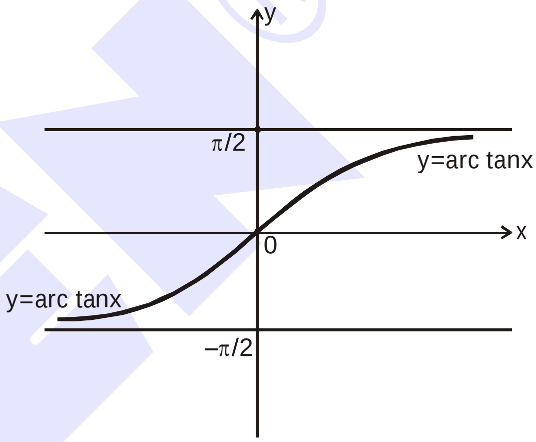
($y = \cos^{-1}x$)

(c) $f : (-\pi/2, \pi/2) \to \mathbb{R}$ $\qquad$ $f^{-1} : \mathbb{R} \to (-\pi/2, \pi/2)$

$f(x) = \tan x$ $\qquad$ $f^{-1}(x) = \tan^{-1} x$



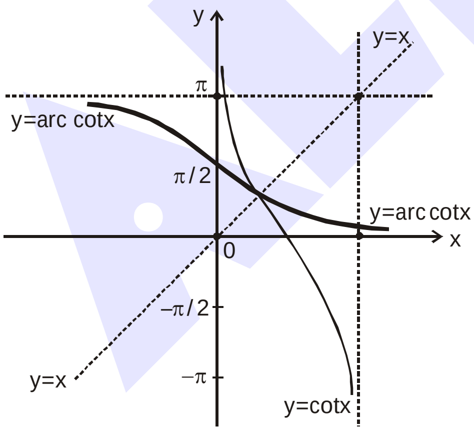
(Taking image of tan x about y = x)



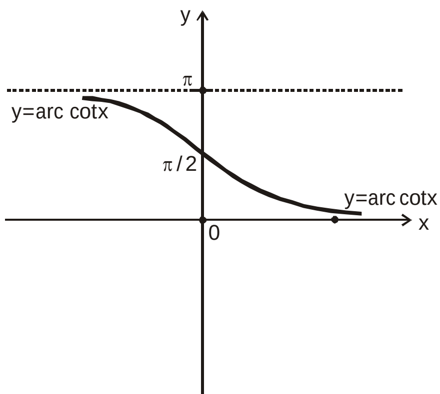
($y = \tan^{-1}x$)

(d) $f : (0, \pi) \to \mathbb{R}$ $\qquad$ $f^{-1} : \mathbb{R} \to (0, \pi)$

$f(x) = \cot x$ $\qquad$ $f^{-1}(x) = \cot^{-1} x$



(Taking image of cot x about y = x)



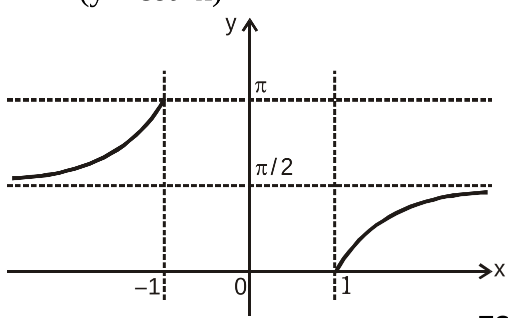
($y = \cot^{-1}x$)

(e) $f : [0, \pi/2) \cup (\pi/2, \pi] \to (-\infty, -1] \cup [1, \infty)$

$f(x) = \sec x$

$f^{-1} : (-\infty, -1] \cup [1, \infty) \to [0, \pi/2) \cup (\pi/2, \pi]$

$f^{-1}(x) = \sec^{-1} x$

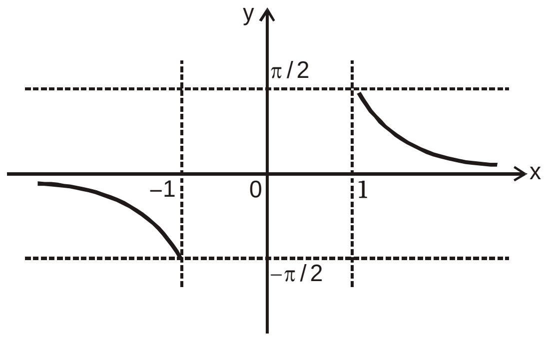


(f) $f : [-\pi/2, 0) \cup (0, \pi/2] \to (-\infty, -1] \cup [1, \infty)$

$f(x) = \text{cosec } x$

$f^{-1} : (-\infty, -1] \cup [1, \infty) \to [-\pi/2, 0) \cup (0, \pi/2]$

$f^{-1}(x) = \text{cosec}^{-1} x$



From the above discussions following IMPORTANT points can be concluded :

(i) All the inverse trigonometric functions represent an angle.

(ii) If $x > 0$, then all six inverse trigonometric functions viz $\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$, $\sec^{-1}x$, $\text{cosec}^{-1}x$, $\cot^{-1}x$ represent an acute angle.

(iii) If $x < 0$, then $\sin^{-1}x$, $\tan^{-1}x$ & $\text{cosec}^{-1}x$ represent an angle from $-\pi/2$ to 0 (IV^th quadrant)

(iv) If $x < 0$, then $\cos^{-1}x$, $\cot^{-1}x$ & $\sec^{-1}x$ represent an obtuse angle. (II^nd quadrant)

(v) III^rd quadrant is never used in range of inverse trigonometric function.

Illustration 1 : The value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ is equal to

(A) $\frac{\pi}{4}$ (B) $\frac{5\pi}{12}$ (C) $\frac{3\pi}{4}$ (D) $\frac{13\pi}{12}$

Solution : $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$ **Ans.(C)**

Illustration 2 : If $\sum_{i=1}^{2n} \cos^{-1} x_i = 0$ then find the value of $\sum_{i=1}^{2n} x_i$

Solution : We know, $0 \le \cos^{-1} x \le \pi$

Hence, each value $\cos^{-1}x_1, \cos^{-1}x_2, \cos^{-1}x_3, \ldots\ldots, \cos^{-1}x_{2n}$ are non-negative their sum is zero only when each value is zero.

i.e., $\cos^{-1}x_i = 0$ for all i

$\Rightarrow$ $x_i = 1$ for all i

$$\therefore \quad \sum_{i=1}^{2n} x_i = x_1 + x_2 + x_3 + \ldots\ldots + x_{2n} = \underbrace{\{1+1+1\ldots\ldots\ldots+1\}}_{2n \text{ times}} = 2n \qquad \{\text{using (i)}\}$$

$$\Rightarrow \quad \sum_{i=1}^{2n} x_i = 2n$$ **Ans.**

Do yourself - 1 :

1. (i) If α, β are roots of the equation $6x^2 + 11x + 3 = 0$, then

(A) both $\cos^{-1}\alpha$ and $\cos^{-1}\beta$ exists (B) both $\text{cosec}^{-1}\alpha$ and $\text{cosec}^{-1}\beta$ exists

(C) both $\cot^{-1}\alpha$ and $\cot^{-1}\beta$ exists (D) none of these

(ii) If $\sin^{-1}x + \sin^{-1}y = \pi$ and $x = ky$, then find the value of $39^{2k} + 5^k$.

2. Find the domain of each of the following functions :

(i) $f(x) = \dfrac{\sin^{-1}x}{x}$ (ii) $f(x) = \sqrt{1-2x} + 3\sin^{-1}\left(\dfrac{3x-1}{2}\right)$ (iii) $f(x) = 2^{\sin^{-1}x} + \dfrac{1}{\sqrt{x-2}}$

3. Find the domain of definition the following functions.

(i) $f(x) = \arccos\dfrac{2x}{1+x}$ (ii) $f(x) = \sqrt{\cos(\sin x)} + \sin^{-1}\dfrac{1+x^2}{2x}$

4. Find the domain of definition the following functions.

(i) $f(x) = \sin^{-1}\left(\dfrac{x-3}{2}\right) - \log_{10}(4-x)$ (ii) $f(x) = \sin^{-1}(2x + x^2)$

5. Find the domain of definition the following function.

$f(x) = \dfrac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$, where $\{x\}$ is the fractional part of x.

6. Find the domain of definition the following functions.

(Read the symbols [*] and { * } as greatest integers and fractional part functions respectively)

(i) $f(x) = \sqrt{3-x} + \cos^{-1}\left(\dfrac{3-2x}{5}\right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$

(ii) $f(x) = e^{\sin^{-1}\left(\frac{x}{2}\right)} + \tan^{-1}\left[\dfrac{x}{2} - 1\right] + \ell n\left(\sqrt{x-[x]}\right)$

7. Find the range of each of the following functions :

(i) $f(x) = \ell n(\sin^{-1}x)$ (ii) $f(x) = \sin^{-1}\left(\dfrac{\sqrt{3x^2+1}}{5x^2+1}\right)$

8. Find the range of each of the following function :

$f(x) = \cos^{-1}\left(\dfrac{(x-1)(x+5)}{x(x-2)(x-3)}\right)$

9. A function g(x) satisfies the following conditions

(i) Domain of g is $(-\infty, \infty)$ (ii) Range of g is $[-1, 7]$

(iii) g has a period π and (iv) $g(2) = 3$

Then which of the following may be possible.

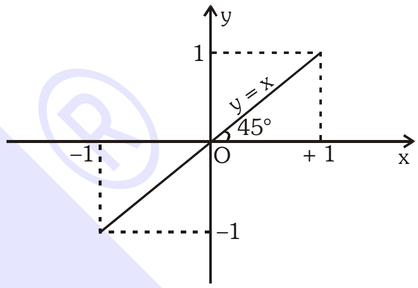
(A) $g(x) = 3 + 4\sin(n\pi + 2x - 4),\ n \in I$ (B) $g(x) = \begin{cases} 3 & ;\ x = n\pi \\ 3 + 4\sin x & ;\ x \neq n\pi \end{cases}$

(C) $g(x) = 3 + 4\cos(n\pi + 2x - 4),\ n \in I$ (D) $g(x) = 3 - 8\sin(n\pi + 2x - 4),\ n \in I$

4. PROPERTIES OF INVERSE CIRCULAR FUNCTIONS :

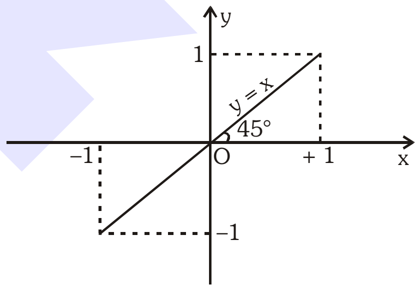
P-1 (i) $y = \sin(\sin^{-1}x) = x$

$x \in [-1,1],\ y \in [-1,1]$



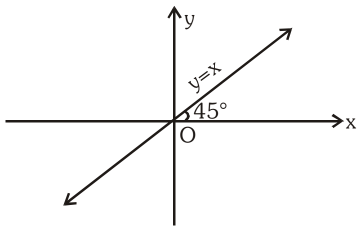
(ii) $y = \cos(\cos^{-1}x) = x$

$x \in [-1,1],\ y \in [-1,1]$



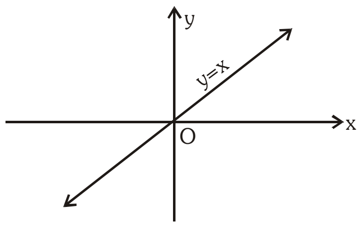
(iii) $y = \tan(\tan^{-1}x) = x$

$x \in \mathbb{R},\ y \in \mathbb{R}$



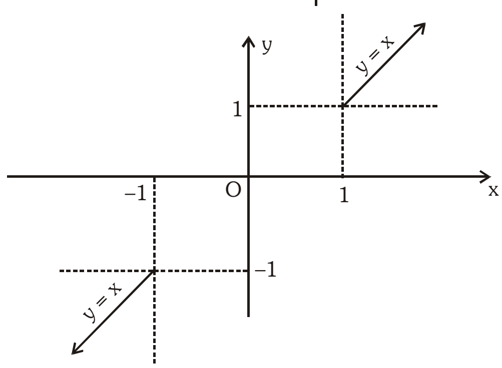
(iv) $y = \cot(\cot^{-1}x) = x,$

$x \in \mathbb{R};\ y \in \mathbb{R}$



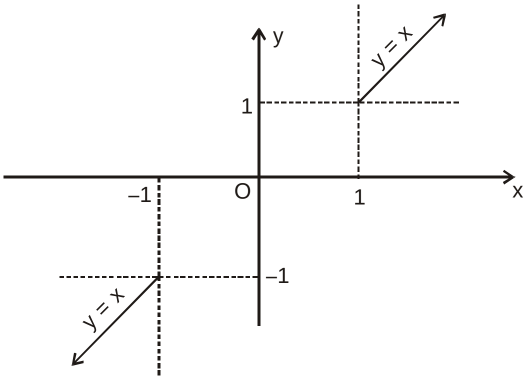
(v) $y = \text{cosec}(\text{cosec}^{-1}x) = x,$

$|x| \geq 1,\ |y| \geq 1$



(vi) $y = \sec(\sec^{-1} x) = x$

$|x| \geq 1$; $|y| \geq 1$



Note : All the above functions are aperiodic.

Illustration 3 : Evaluate the following :

(i) $\sin(\cos^{-1}3/5)$ (ii) $\cos(\tan^{-1} 3/4)$ (iii) $\sin\left(\frac{\pi}{2}-\sin^{-1}\left(-\frac{1}{2}\right)\right)$

(iv) $\tan\left\{2\tan^{-1}\frac{1}{5}-\frac{\pi}{4}\right\}$

Solution : (i) Let $\cos^{-1} 3/5 = \theta$. Then,

$\cos\theta = 3/5 \Rightarrow \sin\theta = 4/5$

$\therefore \sin(\cos^{-1} 3/5) = \sin \theta = 4/5$

(ii) Let $\tan^{-1} 3/4 = \theta$. Then,

$\tan\theta = 3/4$

$\Rightarrow \cos\theta = \frac{4}{5}$ $\left\{\because \text{ as } \cos^2\theta = \frac{1}{1+\tan^2\theta}\right\}$

$\therefore \cos(\tan^{-1} 3/4) = \cos\theta = 4/5$

(iii) $\sin\left(\frac{\pi}{2}-\sin^{-1}\left(\frac{-1}{2}\right)\right) = \sin\left(\frac{\pi}{2}-\left(-\frac{\pi}{6}\right)\right) = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$

(iv) Let $\tan^{-1}\left(\frac{1}{5}\right) = \theta \Rightarrow \tan\theta = \frac{1}{5}$

$\tan\left(2\theta-\frac{\pi}{4}\right) = \frac{\tan(2\theta)-1}{1+\tan 2\theta}$ and $\tan 2\theta = \frac{2\times\frac{1}{5}}{1-\frac{1}{25}} = \frac{5}{12}$ $\left(\because \tan\theta = \frac{1}{5}\right)$

$\Rightarrow \tan\left(2\theta-\frac{\pi}{4}\right) = \frac{\frac{5}{12}-1}{1+\frac{5}{12}} = -\frac{7}{17}$ **Ans.**

Illustration 4 : Prove that $\sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1} 3) = 15$

Solution : We have,

$\sec^2 (\tan^{-1}2) + \text{cosec}^2 (\cot^{-1}3)$

$$=\left\{\sec\left(\tan^{-1} 2\right)\right\}^2+\left\{\text{cosec}\left(\cot^{-1} 3\right)\right\}^2=\left\{\sec\left(\tan^{-1}\frac{2}{1}\right)\right\}^2+\left\{\text{cosec}\left(\cot^{-1}\frac{3}{1}\right)\right\}^2$$

$$=\left\{\sec\left(\sec^{-1}\sqrt{5}\right)\right\}^2+\left\{\text{cosec}\left(\text{cosec}^{-1}\sqrt{10}\right)\right\}^2=\left(\sqrt{5}\right)^2+\left(\sqrt{10}\right)^2=15$$

Do yourself - 2 :

1. Evaluate the following :

(i) $\tan\left(\cos^{-1}\left(\frac{8}{17}\right)\right)$ (ii) $\cos\left(\tan^{-1}\frac{3}{4}\right)$ (iii) $\sin\left(\cos^{-1}\frac{3}{5}\right)$

(iv) $\tan\left(\cos^{-1}\frac{1}{3}\right)$ (v) $\text{cosec}\left(\sec^{-1}\frac{\sqrt{41}}{5}\right)$ (vi) $\tan\left(\text{cosec}^{-1}\frac{65}{63}\right)$

2. Find the following :

(i) $\tan\left[\cos^{-1}\frac{1}{2}+\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right]$ (ii) $\tan\left(\sin^{-1}\frac{3}{5}+\cot^{-1}\frac{3}{2}\right)$

(iii) $\sin\left[\frac{\pi}{2}-\sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)\right]$ (iv) $\cos\left[\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)+\frac{\pi}{6}\right]$

(v) $\sin\left(\frac{\pi}{3}-\sin^{-1}\left(\frac{-1}{2}\right)\right)$ (vi) $\sin\left(\frac{\pi}{6}+\cos^{-1}\frac{1}{4}\right)$

(vii) $\cos\left(\sin^{-1}\frac{4}{5}+\cos^{-1}\frac{2}{3}\right)$

3. Find the following :

(i) $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ (ii) $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

4. Find the simplified value of the following expressions :

(i) $\sin\left(\frac{1}{2}\cos^{-1}\left(\frac{4}{5}\right)\right)$ (ii) $\sin^{-1}\left[\cos\left\{\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right\}\right]$ (iii) $\sin\left(\frac{1}{4}\text{arc}\sin\frac{\sqrt{63}}{8}\right)$

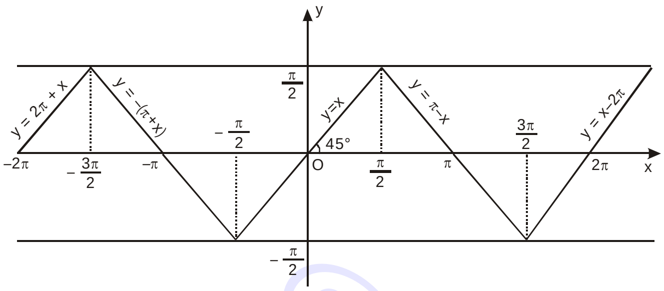
5. Evaluate the following expressions :

(i) $\sec\left(\tan\left\{\tan^{-1}\left(-\frac{\pi}{3}\right)\right\}\right)$ (ii) $\cos\tan^{-1}\sin\cot^{-1}\left(\frac{1}{2}\right)$

(iii) $\tan\left[\cos^{-1}\left(\frac{3}{4}\right)+\sin^{-1}\left(\frac{3}{4}\right)-\sec^{-1}3\right]$

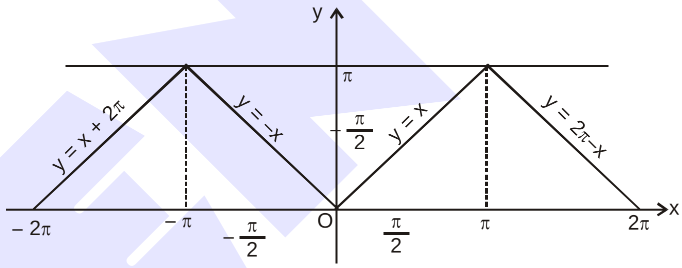
P-2 (i) $y = \sin^{-1}(\sin x),\ x \in \mathbb{R},\ y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ periodic with period 2π and it is an odd function.

$$\sin^{-1}(\sin x) = \begin{cases} -\pi - x & , -\pi \le x \le -\dfrac{\pi}{2} \\ x & , -\dfrac{\pi}{2} < x < \dfrac{\pi}{2} \\ \pi - x, & \dfrac{\pi}{2} \le x \le \pi \end{cases}$$



(ii) $y = \cos^{-1}(\cos x),\ x \in \mathbb{R},\ y \in [0,\pi]$, periodic with period 2π and it is an even function.

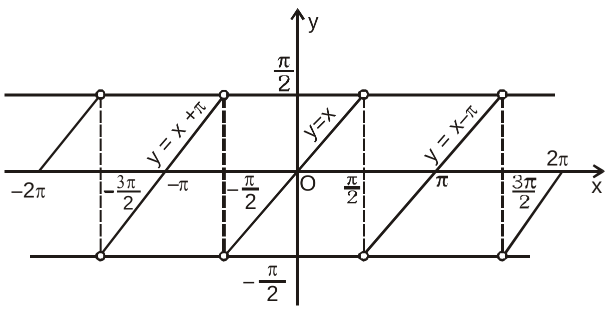
$$\cos^{-1}(\cos x) = \begin{cases} -x & , -\pi \le x \le 0 \\ x & , 0 < x \le \pi \end{cases}$$



(iii) $y = \tan^{-1}(\tan x)$

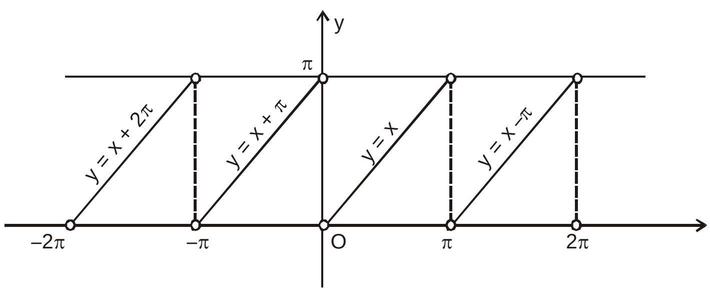
$x \in \mathbb{R} - \left\{(2n-1)\frac{\pi}{2}, n \in I\right\};\quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, periodic with period π and it is an odd function.

$$\tan^{-1}(\tan x) = \begin{cases} x + \pi\ , & -\dfrac{3\pi}{2} < x < -\dfrac{\pi}{2} \\ x\ , & -\dfrac{\pi}{2} < x < \dfrac{\pi}{2} \\ x - \pi\ , & \dfrac{\pi}{2} < x < \dfrac{3\pi}{2} \end{cases}$$

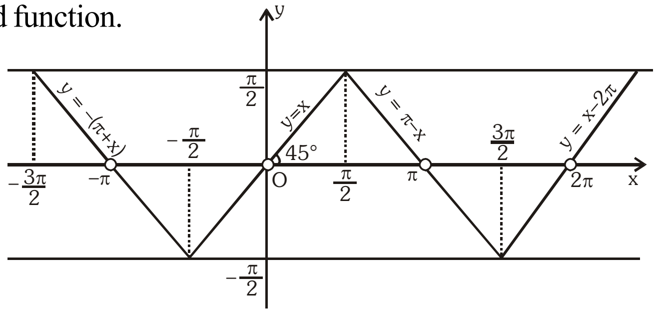


(iv) $y = \cot^{-1}(\cot x),\ x \in \mathbb{R} - \{n\pi, n \in I\},\ y \in (0, \pi)$, periodic with period π and neither even nor odd function.

$$\cot^{-1}(\cot x) = \begin{cases} x + \pi\ , & -\pi < x < 0 \\ x\ , & 0 < x < \pi \\ x - \pi\ , & \pi < x < 2\pi \end{cases}$$



(v) $y = \text{cosec}^{-1}(\text{cosec } x), x \in \mathbb{R} - \{n\pi, n \in I\}$ $y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$, is periodic with period 2π and it is an odd function.



(vi) $y = \sec^{-1}(\sec x)$, y is periodic with period 2π and it is an even function.

$x \in \mathbb{R} - \left\{(2n-1)\frac{\pi}{2} n \in I\right\}$, $y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

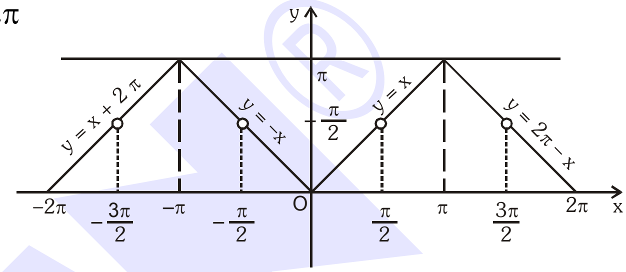


Illustration 5 : The value of $\sin^{-1}(-\sqrt{3}/2) + \cos^{-1}(\cos(7\pi/6))$ is -

(A) $5\pi/6$ (B) $\pi/2$ (C) $3\pi/2$ (D) none of these

Solution : $\sin^{-1}\left(-\sqrt{3}/2\right) = -\sin^{-1}\left(\sqrt{3}/2\right) = -\pi/3$

and $\cos^{-1}(\cos(7\pi/6) = \cos^{-1}\cos(2\pi - 5\pi/6) = \cos^{-1}\cos(5\pi/6) = 5\pi/6$

Hence $\sin^{-1}(-\sqrt{3}/2) + \cos^{-1}(\cos 7\pi/6) = -\frac{\pi}{3} + \frac{5\pi}{6} = \frac{\pi}{2}$ **Ans.(B)**

Illustration 6 : Evaluate the following :

(i) $\sin^{-1}(\sin 10)$ (ii) $\tan^{-1}(\tan(-6))$ (iii) $\cot^{-1}(\cot 4)$

Solution : (i) We know that $\sin^{-1}(\sin\theta) = \theta$, if $-\pi/2 \le \theta \le \pi/2$

Here, $\theta = 10$ radians which does not lie between $-\pi/2$ and $\pi/2$

But, $3\pi - \theta$ i.e., $3\pi - 10$ lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

Also, $\sin(3\pi - 10) = \sin 10$

$\therefore$ $\sin^{-1}(\sin 10) = \sin^{-1}(\sin(3\pi - 10)) = (3\pi - 10)$

(ii) We know that,

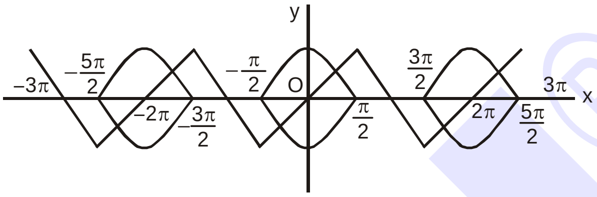
(ii) We know that,
$\tan^{-1}(\tan\theta) = \theta$, if $-\pi/2 < \theta < \pi/2$. Here, $\theta = -6$, radians which does not lie between $-\pi/2$ and $\pi/2$. We find that $2\pi - 6$ lies between $-\pi/2$ and $\pi/2$ such that;
$\tan(2\pi - 6) = -\tan 6 = \tan(-6)$
$\therefore \quad \tan^{-1}(\tan(-6)) = \tan^{-1}(\tan(2\pi - 6)) = (2\pi - 6)$

(iii) $\cot^{-1}(\cot 4) = \cot^{-1}(\cot(\pi + (4 - \pi))) = \cot^{-1}(\cot(4 - \pi)) = (4 - \pi)$

Ans.

Illustration 7 : Find the number of solutions of (x, y) which satisfy $|y| = \cos x$ and $y = \sin^{-1}(\sin x)$, where $|x| \le 3\pi$.

Solution : Graphs of $y = \sin^{-1}(\sin x)$ and $|y| = \cos x$ meet exactly six times in $[-3\pi, 3\pi]$.



Do yourself - 3 :

1. Evaluate the following inverse trigonometric expressions :

(i) $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$ (ii) $\tan^{-1}\left(\tan\left(\frac{7\pi}{6}\right)\right)$ (iii) $\sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right)$

(iv) $\sin^{-1}\left(\sin\frac{7\pi}{6}\right)$ (v) $\tan^{-1}\left(\tan\frac{2\pi}{3}\right)$ (vi) $\cos^{-1}\left(\cos\frac{5\pi}{4}\right)$ (vii) $\sec^{-1}\left(\sec\frac{7\pi}{4}\right)$

2. Show that : $\sin^{-1}\left(\sin\frac{33\pi}{7}\right) + \cos^{-1}\left(\cos\frac{46\pi}{7}\right) + \tan^{-1}\left(-\tan\frac{13\pi}{8}\right) + \cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right) = \frac{13\pi}{7}$

3. Prove that the identities :

(i) $\sin^{-1}\cos(\sin^{-1}x) + \cos^{-1}\sin(\cos^{-1}x) = \frac{\pi}{2}, |x| \le 1$

(ii) $\tan(\tan^{-1}x + \tan^{-1}y + \tan^{-1}z) = \cot(\cot^{-1}x + \cot^{-1}y + \cot^{-1}z)$

4. Prove that the equation, $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = \alpha\pi^3$ has no roots for $\alpha < \frac{1}{32}$ and $\alpha > \frac{7}{8}$.

5. Find the set of values of 'a' for which the equation $2\cos^{-1}x = a + a^2(\cos^{-1}x)^{-1}$ posses a solution.

6. If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, then $4x^2 - 4xy\cos\alpha + y^2$ is equal to -

(A) $2\sin 2\alpha$ (B) 4 (C) $4\sin^2\alpha$ (D) $-4\sin^2\alpha$

7. If $\sin^{-1}\left(\frac{x}{5}\right) + \text{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$, then a value of x is-

(A) 1 (B) 3 (C) 4 (D) 5

8. The value of $\cot\left(\text{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right)$ is equal to-

(A) $\frac{6}{17}$ (B) $\frac{3}{17}$ (C) $\frac{4}{17}$ (D) $\frac{5}{17}$

9. If $\pi \le x \le 2\pi$, then $\cos^{-1}(\cos x)$ is equal to

(A) x (B) $\pi - x$ (C) $2\pi + x$ (D) $2\pi - x$

10. Find the value of the following inverse trigonometric expressions :

(i) $\sin^{-1}(\sin 4)$ (ii) $\cos^{-1}(\cos 10)$ (iii) $\tan^{-1}(\tan(-6))$

(iv) $\cot^{-1}(\cot(-10))$ (v) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\left(\cos\frac{9\pi}{10} - \sin\frac{9\pi}{10}\right)\right)$

11. Find the value of $\sin^{-1}(\cos(\sin^{-1}x)) + \cos^{-1}(\sin(\cos^{-1}x))$

P-3 (i) $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ $\qquad |x| \le 1$

(ii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$ $\qquad x \in \mathbb{R}$

(iii) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$ $\qquad |x| \ge 1$

P-4 (i) $\sin^{-1}(-x) = -\sin^{-1}x$, $\qquad |x| \le 1$

(ii) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$, $\qquad |x| \ge 1$

(iii) $\tan^{-1}(-x) = -\tan^{-1}x$, $\qquad x \in \mathbb{R}$

(iv) $\cot^{-1}(-x) = \pi - \cot^{-1}x$, $\qquad x \in \mathbb{R}$

(v) $\cos^{-1}(-x) = \pi - \cos^{-1}x$, $\qquad |x| \le 1$

(vi) $\sec^{-1}(-x) = \pi - \sec^{-1}x$, $\qquad |x| \ge 1$

P-5 (i) $\operatorname{cosec}^{-1}x = \sin^{-1}\frac{1}{x}$; $\qquad |x| \ge 1$

(ii) $\sec^{-1}x = \cos^{-1}\frac{1}{x}$; $\qquad |x| \ge 1$

(iii) $$\cot^{-1}x = \begin{cases} \tan^{-1}\frac{1}{x} & ; \quad x > 0 \\ \pi + \tan^{-1}\frac{1}{x} & ; \quad x < 0 \end{cases}$$

Illustration 8 : Find value of x if $\cos^{-1}(-x) + \tan^{-1}(-x) - 2\sin^{-1}(x) + \sec^{-1}\left(-\frac{1}{x}\right) = \frac{\pi}{4}$ for $|x| \le 1$.

Solution :

$$\pi - \cos^{-1}(x) - \tan^{-1}(x) - 2\sin^{-1}(x) + \cos^{-1}(-x) = \frac{\pi}{4}$$

$$\pi - \cos^{-1}(x) - \tan^{-1}(x) - 2\sin^{-1}(x) + \pi - \cos^{-1}(x) = \frac{\pi}{4}$$

$$2\pi - 2(\sin^{-1}x + \cos^{-1}x) - \frac{\pi}{4} = \tan^{-1}x$$

$$2\pi - \pi - \frac{\pi}{4} = \tan^{-1}x \Rightarrow \tan^{-1}x = \frac{3\pi}{4}$$ Hence no soluton **Ans.**

Do yourself - 4 :

1. Prove the following :

(i) $\cos^{-1}\left(\frac{5}{13}\right)=\tan^{-1}\left(\frac{12}{5}\right)$ (ii) $\sin^{-1}\left(-\frac{4}{5}\right)=\tan^{-1}\left(-\frac{4}{3}\right)=\cos^{-1}\left(-\frac{3}{5}\right)-\pi$

2. Find the value of $\sin(\tan^{-1}a + \tan^{-1}\frac{1}{a})$; $a \neq 0$

3. Identify the pair(s) of functions which are identical. Also plot the graphs in each case.

(i) $y=\tan(\cos^{-1}x);\ y=\frac{\sqrt{1-x^2}}{x}$ (ii) $y=\tan(\cot^{-1}x);y=\frac{1}{x}$

(iii) $y=\sin(\operatorname{arc}\tan x);y=\frac{x}{\sqrt{1+x^2}}$ (iv) $y=\cos(\operatorname{arc}\tan x);y=\sin(\operatorname{arc}\cot x)$

4. Find all values of k for which there is a triangle whose angles have measure $\tan^{-1}\left(\frac{1}{2}\right),\tan^{-1}\left(\frac{1}{2}+k\right)$ and $\tan^{-1}\left(\frac{1}{2}+2k\right)$.

5. If x, y, z are in A.P. and $\tan^{-1}x$, $\tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then

(A) $x = y = z$ (B) $2x = 3y = 6z$ (C) $6x = 3y = 2z$ (D) $6x = 4y = 3z$

6. Find the value of following expressions :

(i) $\cot(\tan^{-1}a + \cot^{-1}a)$ (ii) $\sin(\sin^{-1}x + \cos^{-1}x)$, $|x| \le 1$

7. If $a > 0$, $b > 0$, $\tan^{-1}\left(\frac{a}{x}\right) + \tan^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$ then prove that $x=\sqrt{ab}$.

P-6 (i) (a) $$\tan^{-1}x+\tan^{-1}y=\begin{cases}\tan^{-1}\dfrac{x+y}{1-xy} & \text{where } x>0,\ y>0\ \&\ xy<1\\ \pi+\tan^{-1}\dfrac{x+y}{1-xy} & \text{where } x>0,\ y>0\ \&\ xy>1\\ \dfrac{\pi}{2}, & \text{where } x>0,\ y>0\ \&\ xy=1\end{cases}$$

(b) $$\tan^{-1}x-\tan^{-1}y=\tan^{-1}\frac{x-y}{1+xy} \text{ where } x>0,\ y>0$$

(c) $$\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right] \text{if } x>0, y>0, z>0\ \&\ xy+yz+zx<1$$

(ii) (a) $$\sin^{-1}x+\sin^{-1}y=\begin{cases}\sin^{-1}[x\sqrt{1-y^2}+y\sqrt{1-x^2}] & \text{where } x>0,\ y>0\ \&\ (x^2+y^2)\le 1\\ \pi-\sin^{-1}[x\sqrt{1-y^2}+y\sqrt{1-x^2}] & \text{where } x>0,\ y>0\ \&\ x^2+y^2>1\end{cases}$$

(b) $$\sin^{-1}x-\sin^{-1}y=\sin^{-1}[x\sqrt{1-y^2}-y\sqrt{1-x^2}] \text{ where } x>0,\ y>0$$

(iii) (a) $\cos^{-1} x + \cos^{-1} y = \cos^{-1}\left[xy - \sqrt{1-x^2}\sqrt{1-y^2}\right]$ where $x > 0,\ y > 0$

(b) $$\cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right) \ ; & x < y, \quad x, y > 0 \\ -\cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right); & x > y, \quad x, y > 0 \end{cases}$$

Note : In the above results x & y are taken positive. In case if these are given as negative, we first apply **P-4** and then use above results.

Illustration 9 : Prove that

(i) $\tan^{-1}\dfrac{1}{7} + \tan^{-1}\dfrac{1}{13} = \tan^{-1}\dfrac{2}{9}$ (ii) $\tan^{-1}\dfrac{1}{5} + \tan^{-1}\dfrac{1}{7} + \tan^{-1}\dfrac{1}{3} + \tan^{-1}\dfrac{1}{8} = \dfrac{\pi}{4}$

Solution : (i) L.H.S. $= \tan^{-1}\dfrac{1}{7} + \tan^{-1}\dfrac{1}{13}$

$$= \tan^{-1}\left\{\frac{\frac{1}{7}+\frac{1}{13}}{1-\frac{1}{7}\times\frac{1}{13}}\right\} \qquad \left\{\because \tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right); \text{ if } xy < 1\right\}$$

$$= \tan^{-1}\left(\frac{20}{90}\right) = \tan^{-1}\left(\frac{2}{9}\right) = \text{R.H.S.}$$

(ii) $$\left(\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{7}\right) + \left(\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{8}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{5}+\frac{1}{7}}{1-\frac{1}{5}\times\frac{1}{7}}\right) + \tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{8}}{1-\frac{1}{3}\times\frac{1}{8}}\right) = \tan^{-1}\left(\frac{6}{17}\right) + \tan^{-1}\left(\frac{11}{23}\right)$$

$$= \tan^{-1}\left(\frac{\frac{6}{17}+\frac{11}{23}}{1-\frac{6}{17}\times\frac{11}{23}}\right) = \tan^{-1}\left(\frac{325}{325}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$ **Ans.**

Illustration 10 : Prove that $\sin^{-1}\dfrac{12}{13} + \cot^{-1}\dfrac{4}{3} + \tan^{-1}\dfrac{63}{16} = \pi$

Solution : We have,

$$\sin^{-1}\frac{12}{13} + \cos^{-1}\frac{4}{5} + \tan^{-1}\frac{63}{16}$$

$$= \tan^{-1}\frac{12}{5} + \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{63}{16} \qquad \left[\because \sin^{-1}\frac{12}{13} = \tan^{-1}\frac{12}{5} \text{ and } \cos^{-1}\frac{4}{5} = \tan^{-1}\frac{3}{4}\right]$$

$$= \pi + \tan^{-1}\left\{\frac{\frac{12}{5}+\frac{3}{4}}{1-\frac{12}{5}\times\frac{3}{4}}\right\} + \tan^{-1}\frac{63}{16} \quad \left[\because\ \tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), \text{if } xy > 1\right]$$

$$= \pi + \tan^{-1}\left(\frac{63}{-16}\right) + \tan^{-1}\left(\frac{63}{16}\right)$$

$$= \pi - \tan^{-1}\frac{63}{16} + \tan^{-1}\frac{63}{16} \qquad \left[\because \tan^{-1}(-x) = -\tan^{-1}x\right]$$

$$= \pi$$

Illustration 11 : Prove that : $\cos^{-1}\frac{12}{13} + \sin^{-1}\frac{3}{5} = \sin^{-1}\frac{56}{65}$

Solution : We have, L.H.S. = $\cos^{-1}\left(\frac{12}{13}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{3}{4}\right)$

$$\left[\because \cos^{-1}\left(\frac{12}{13}\right) = \tan^{-1}\left(\frac{5}{12}\right) \ \&\ \sin^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left(\frac{3}{4}\right)\right]$$

$$\text{L.H.S.} = \tan^{-1}\left(\frac{\frac{5}{12}+\frac{3}{4}}{1-\frac{5}{12}\times\frac{3}{4}}\right) = \tan^{-1}\left(\frac{56}{33}\right)$$

$$\text{R.H.S.} = \sin^{-1}\left(\frac{56}{65}\right) = \tan^{-1}\left(\frac{56}{33}\right)$$

L.H.S. = R.H.S. Hence Proved

Do yourself - 5 :

1. Prove the following :

(i) $\sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{8}{17}\right) = \cos^{-1}\left(\frac{36}{85}\right)$ (ii) $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right) = \frac{\pi}{4}$

(iii) $\tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{7}{24}\right) = \tan^{-1}\left(\frac{1}{2}\right)$

2. Prove that :

(i) $2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25} = \pi$ (ii) $\text{arc}\cos\sqrt{\frac{2}{3}} - \text{arc}\cos\frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$

3. Find the simplest value of

(i) $f(x) = \text{arc cos}x + \text{arc cos}\left(\frac{x}{2}+\frac{1}{2}\sqrt{3-3x^2}\right),\ x \in \left(\frac{1}{2}, 1\right)$

(ii) $f(x) = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right),\ x \in R - \{0\}$

4. Let $f:(-1, 1) \to B$, be a function defined by $f(x) = \tan^{-1}\frac{2x}{1-x^2}$, then f is both one-one and onto when B is the interval-

(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (C) $\left(0, \frac{\pi}{2}\right)$ (D) $\left[0, \frac{\pi}{2}\right)$

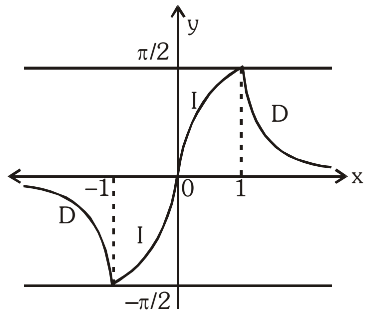
5. Let $\tan^{-1} y = \tan^{-1} x + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, where $|x| < \frac{1}{\sqrt{3}}$. Then a value of y is :

(A) $\frac{3x-x^3}{1+3x^2}$ (B) $\frac{3x+x^3}{1+3x^2}$ (C) $\frac{3x-x^3}{1-3x^2}$ (D) $\frac{3x+x^3}{1-3x^2}$

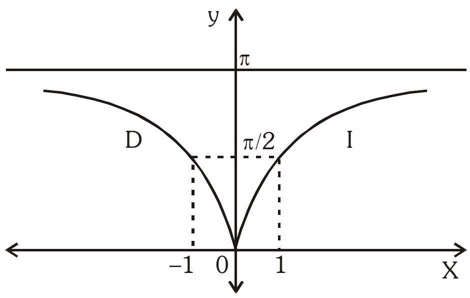
6. If $\tan^{-1}x + \cot^{-1}\frac{1}{y} + 2\tan^{-1}z = \pi$, then prove that $x + y + 2z = xz^2 + yz^2 + 2xyz$

5. SIMPLIFIED INVERSE TRIGONOMETRIC FUNCTIONS :

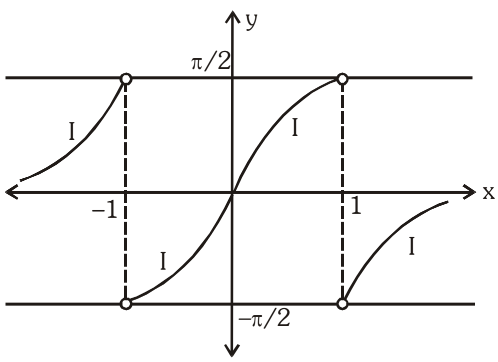
(a) $$y = f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & \text{if } |x| \le 1 \\ \pi - 2\tan^{-1}x & \text{if } x > 1 \\ -(\pi + 2\tan^{-1}x) & \text{if } x < -1 \end{cases}$$



(b) $$y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & \text{if } x \ge 0 \\ -2\tan^{-1}x & \text{if } x < 0 \end{cases}$$

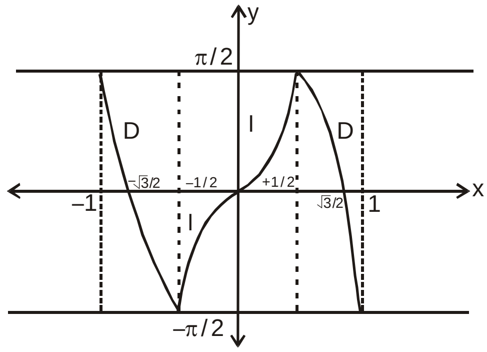


(c) $$y = f(x) = \tan^{-1}\frac{2x}{1-x^2} = \begin{cases} 2\tan^{-1}x & \text{if } |x| < 1 \\ \pi + 2\tan^{-1}x & \text{if } x < -1 \\ -(\pi - 2\tan^{-1}x) & \text{if } x > 1 \end{cases}$$



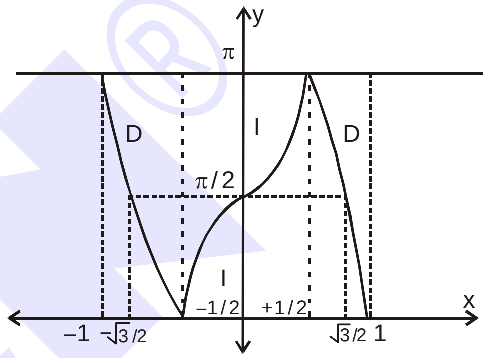
(d) $y = f(x) = \sin^{-1}(3x - 4x^3)$

$$= \begin{cases} -(\pi + 3\sin^{-1} x) & \text{if} \quad -1 \le x \le -\dfrac{1}{2} \\ 3\sin^{-1} x & \text{if} \quad -\dfrac{1}{2} \le x \le \dfrac{1}{2} \\ \pi - 3\sin^{-1} x & \text{if} \quad \dfrac{1}{2} \le x \le 1 \end{cases}$$

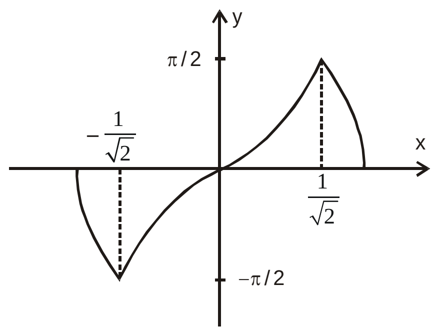


(e) $y = f(x) = \cos^{-1}(4x^3 - 3x)$

$$= \begin{cases} 3\cos^{-1} x - 2\pi & \text{if} \quad -1 \le x \le -\dfrac{1}{2} \\ 2\pi - 3\cos^{-1} x & \text{if} \quad -\dfrac{1}{2} \le x \le \dfrac{1}{2} \\ 3\cos^{-1} x & \text{if} \quad \dfrac{1}{2} \le x \le 1 \end{cases}$$



(f) $$\sin^{-1}\left(2x\sqrt{1-x^2}\right) = \begin{cases} -\left(\pi + 2\sin^{-1} x\right) & -1 \le x \le -\dfrac{1}{\sqrt{2}} \\ 2\sin^{-1} x & -\dfrac{1}{\sqrt{2}} \le x \le \dfrac{1}{\sqrt{2}} \\ \pi - 2\sin^{-1} x & \dfrac{1}{\sqrt{2}} \le x \le 1 \end{cases}$$



(g) $$\cos^{-1}\left(2x^2 - 1\right) = \begin{cases} 2\cos^{-1} x & 0 \le x \le 1 \\ 2\pi - 2\cos^{-1} x & -1 \le x \le 0 \end{cases}$$

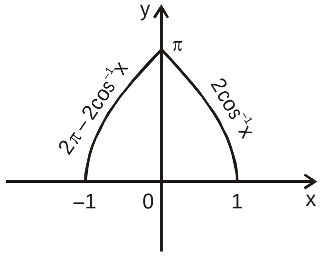


Illustration 12 : Prove that : $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}=\tan^{-1}\frac{31}{17}$

Solution : We have, $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}$

$$=\tan^{-1}\left\{\frac{2\times\frac{1}{2}}{1-\left(\frac{1}{2}\right)^2}\right\}+\tan^{-1}\frac{1}{7} \qquad \left[\because 2\tan^{-1}x=\tan^{-1}\left(\frac{2x}{1-x^2}\right),\text{if } -1<x<1\right]$$

$$=\tan^{-1}\frac{4}{3}+\tan^{-1}\frac{1}{7}=\tan^{-1}\left\{\frac{\frac{4}{3}+\frac{1}{7}}{1-\frac{4}{3}\times\frac{1}{7}}\right\}=\tan^{-1}\frac{31}{17}$$

Illustration 13 : Prove that $\tan^{-1}\sqrt{x}=\frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right), x\in[0,1]$

Solution : We have, $\frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\cos^{-1}\left\{\frac{1-\left(\sqrt{x}\right)^2}{1+\left(\sqrt{x}\right)^2}\right\}=\frac{1}{2}\times 2\tan^{-1}\sqrt{x}=\tan^{-1}\sqrt{x}.$

Alter : Putting $\sqrt{x}=\tan\theta$, we have $\Rightarrow \theta\in\left[0,\frac{\pi}{4}\right]$

$$\text{RHS}=\frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}\cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)=\frac{1}{2}\cos^{-1}(\cos 2\theta)=\theta \quad \because \left(2\theta\in\left[0,\frac{\pi}{2}\right]\right)$$

$$=\tan^{-1}\sqrt{x}=\text{LHS}$$

Illustration 14 : Prove that : (i) $4\tan^{-1}\frac{1}{5}-\tan^{-1}\frac{1}{70}+\tan^{-1}\frac{1}{99}=\frac{\pi}{4}$

(ii) $2\tan^{-1}\frac{1}{5}+\sec^{-1}\frac{5\sqrt{2}}{7}+2\tan^{-1}\frac{1}{8}=\frac{\pi}{4}$

Solution : (i) $4\tan^{-1}\frac{1}{5}-\tan^{-1}\frac{1}{70}+\tan^{-1}\frac{1}{99}=2\left\{2\tan^{-1}\frac{1}{5}\right\}-\tan^{-1}\frac{1}{70}+\tan^{-1}\frac{1}{99}$

$$=2\left\{\tan^{-1}\frac{2\times 1/5}{1-(1/5)^2}\right\}-\tan^{-1}\frac{1}{70}+\tan^{-1}\frac{1}{99} \qquad \left[\begin{array}{c}\because 2\tan^{-1}x\\ =\tan^{-1}\frac{2x}{1-x^2},\text{if } |x|<1\end{array}\right]$$

$$= 2\tan^{-1}\frac{5}{12} - \left\{\tan^{-1}\frac{1}{70} - \tan^{-1}\frac{1}{99}\right\} = \tan^{-1}\left\{\frac{2\times 5/12}{1-(5/12)^2}\right\} - \tan^{-1}.\left\{\frac{\frac{1}{70}-\frac{1}{99}}{1+\frac{1}{70}\times\frac{1}{99}}\right\}$$

$$= \tan^{-1}\frac{120}{119} - \tan^{-1}\frac{29}{6931} = \tan^{-1}\frac{120}{119} - \tan^{-1}\frac{1}{239} = \tan^{-1}\left\{\frac{\frac{120}{119}-\frac{1}{239}}{1+\frac{120}{119}\times\frac{1}{239}}\right\} = \tan^{-1}1 = \frac{\pi}{4}$$

(ii) $$2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2\tan^{-1}\frac{1}{8} = 2\left\{\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right\} + \sec^{-1}\frac{5\sqrt{2}}{7}$$

$$= 2\tan^{-1}\left\{\frac{\frac{1}{5}+\frac{1}{8}}{1-\frac{1}{5}\times\frac{1}{8}}\right\} + \tan^{-1}\sqrt{\left(\frac{5\sqrt{2}}{7}\right)^2 - 1} \qquad \left[\because \sec^{-1}x = \tan^{-1}\sqrt{x^2-1}\right]$$

$$= 2\tan^{-1}\frac{13}{39} + \tan^{-1}\frac{1}{7} = 2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7}$$

$$= \tan^{-1}\left\{\frac{2\times 1/3}{1-(1/3)^2}\right\} + \tan^{-1}\frac{1}{7} \qquad \left[\because 2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2},\ \text{if } |x|<1\right]$$

$$= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{7} = \tan^{-1}\left\{\frac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}\times\frac{1}{7}}\right\} = \tan^{-1}1 = \frac{\pi}{4}$$

Illustration 15 : If $\tan^{-1} y = 4\tan^{-1} x, \left(|x| < \tan\frac{\pi}{8}\right)$, find y as an algebraic function of x and hence prove that $\tan\frac{\pi}{8}$ is a root of the equation $x^4 - 6x^2 + 1 = 0$.

Solution : We have $\tan^{-1} y = 4\tan^{-1} x$

$$\Rightarrow \tan^{-1} y = 2\tan^{-1}\frac{2x}{1-x^2} \qquad (\text{as } |x| < 1)$$

$$= \tan^{-1}\frac{\frac{4x}{1-x^2}}{1-\frac{4x^2}{(1-x^2)^2}} = \tan^{-1}\frac{4x(1-x^2)}{x^4-6x^2+1} \qquad \left(\text{as}\left|\frac{2x}{1-x^2}\right|<1\right)$$

$\Rightarrow \quad y = \dfrac{4x(1-x^2)}{x^4-6x^2+1}$

If $\quad x = \tan\dfrac{\pi}{8}$

$\Rightarrow \quad \tan^{-1}y = 4\tan^{-1}x = \dfrac{\pi}{2} \;\Rightarrow\;$ y is not defined $\;\Rightarrow\; x^4 - 6x^2 + 1 = 0$ **Ans.**

Illustration 16 : If $A = 2\tan^{-1}\left(2\sqrt{2}-1\right)$ and $B = 3\sin^{-1}(1/3) + \sin^{-1}(3/5)$, then show $A > B$.

Solution : We have, $A = 2\tan^{-1}\left(2\sqrt{2}-1\right) = 2\tan^{-1}(1.828)$

$\Rightarrow \quad A > 2\tan^{-1}\left(\sqrt{3}\right) \;\Rightarrow\; A > \dfrac{2\pi}{3}$ .... (i)

also we have, $\sin^{-1}\left(\dfrac{1}{3}\right) < \sin^{-1}\left(\dfrac{1}{2}\right) \Rightarrow \sin^{-1}\left(\dfrac{1}{3}\right) < \dfrac{\pi}{6}$

$\Rightarrow \quad 3\sin^{-1}\left(\dfrac{1}{3}\right) < \dfrac{\pi}{2}$

also, $3\sin^{-1}\left(\dfrac{1}{3}\right) = \sin^{-1}\left(3.\dfrac{1}{3} - 4\left(\dfrac{1}{3}\right)^3\right) = \sin^{-1}\left(\dfrac{23}{27}\right) = \sin^{-1}(0.852)$

$\Rightarrow \quad 3\sin^{-1}(1/3) < \sin^{-1}\left(\sqrt{3}/2\right) \Rightarrow 3\sin^{-1}(1/3) < \pi/3$

also, $\sin^{-1}(3/5) = \sin^{-1}(0.6) < \sin^{-1}\left(\sqrt{3}/2\right) \Rightarrow \sin^{-1}(3/5) < \pi/3$

Hence, $B = 3\sin^{-1}(1/3) + \sin^{-1}(3/5) < \dfrac{2\pi}{3}$ .... (ii)

From (i) and (ii), we have $A > B$.

Illustration 17 : If $\theta = \tan^{-1}(2\tan^2\theta) - \dfrac{1}{2}\sin^{-1}\left(\dfrac{3\sin 2\theta}{5+4\cos 2\theta}\right)$ then find the sum of all possible values of $\tan\theta$.

Solution : $\theta = \tan^{-1}(2\tan^2\theta) - \dfrac{1}{2}\sin^{-1}\left(\dfrac{3\sin 2\theta}{5+4\cos 2\theta}\right) \Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \dfrac{1}{2}\sin^{-1}\left(\dfrac{6\tan\theta}{9+\tan^2\theta}\right)$

$\Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \dfrac{1}{2}\sin^{-1}\left[\dfrac{2\left(\dfrac{1}{3}\tan\theta\right)}{1+\left(\dfrac{1}{3}\tan\theta\right)^2}\right] \Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \dfrac{2}{2}\tan^{-1}\left(\dfrac{1}{3}\tan\theta\right)$

$\Rightarrow \quad \theta = \tan^{-1}(2\tan^2\theta) - \tan^{-1}\left(\dfrac{1}{3}\tan\theta\right)$ .... (i) taking tangent on both sides

$\Rightarrow \quad \tan\theta = \dfrac{6\tan^2\theta - \tan\theta}{3 + 2\tan^3\theta} \Rightarrow \quad 2\tan^4\theta - 6\tan^2\theta + 4\tan\theta = 0$

$\Rightarrow \quad 2\tan\theta(\tan^3\theta - 3\tan\theta + 2) = 0 \Rightarrow \quad 2\tan\theta(\tan\theta - 1)^2(\tan\theta + 2) = 0$

$\Rightarrow \quad \tan\theta = 0, 1, -2$ which satisfy equation (i)

$\therefore \quad$ sum $= 0 + 1 - 2 = -1$ **Ans.**

Do yourself - 6 :

1. Prove the following results :

(i) $2\tan^{-1}\left(\dfrac{1}{5}\right) + \tan^{-1}\left(\dfrac{1}{8}\right) = \tan^{-1}\left(\dfrac{4}{7}\right)$ (ii) $2\sin^{-1}\left(\dfrac{3}{5}\right) - \tan^{-1}\left(\dfrac{17}{31}\right) = \dfrac{\pi}{4}$

2. Prove each of the following relations :

(i) $\tan^{-1}x = -\pi + \cot^{-1}\dfrac{1}{x} = \sin^{-1}\dfrac{x}{\sqrt{1+x^2}} = -\cos^{-1}\dfrac{1}{\sqrt{1+x^2}}$ when $x < 0$.

(ii) $\cos^{-1}x = \sec^{-1}\dfrac{1}{x} = \pi - \sin^{-1}\sqrt{1-x^2} = \pi + \tan^{-1}\dfrac{\sqrt{1-x^2}}{x} = \cot^{-1}\dfrac{x}{\sqrt{1-x^2}}$ when $-1 < x < 0$

3. Express in terms of

(i) $\tan^{-1}\dfrac{2x}{1-x^2}$ to $\tan^{-1}x$ for $x > 1$ (ii) $\sin^{-1}(2x\sqrt{1-x^2})$ to $\sin^{-1}x$ for $1 \ge x > \dfrac{1}{\sqrt{2}}$

(iii) $\cos^{-1}(2x^2 - 1)$ to $\cos^{-1}x$ for $-1 \le x < 0$

4. Simplify $\tan\left\{\dfrac{1}{2}\sin^{-1}\left(\dfrac{2x}{1+x^2}\right) + \dfrac{1}{2}\cos^{-1}\left(\dfrac{1-y^2}{1+y^2}\right)\right\}$, if $x > y > 1$.

5. Prove that

(i) $\sin^{-1}\left(\dfrac{3}{5}\right) + \sin^{-1}\left(\dfrac{8}{17}\right) = \sin^{-1}\dfrac{77}{85}$ (ii) $\tan^{-1}\dfrac{3}{4} + \sin^{-1}\dfrac{5}{13} = \cos^{-1}\dfrac{33}{65}$

(iii) $\sin^{-1}\left(\dfrac{1}{\sqrt{5}}\right) + \cot^{-1}3 = \dfrac{\pi}{4}$ (iv) $\tan^{-1}\left(\dfrac{1}{3}\right) + \tan^{-1}\left(\dfrac{1}{5}\right) + \tan^{-1}\left(\dfrac{1}{7}\right) + \tan^{-1}\left(\dfrac{1}{8}\right) = \dfrac{\pi}{4}$

6. The numerical value of $\cot\left(2\sin^{-1}\dfrac{3}{5} + \cos^{-1}\dfrac{3}{5}\right)$ is

(A) $\dfrac{-4}{3}$ (B) $\dfrac{-3}{4}$ (C) $\dfrac{3}{4}$ (D) $\dfrac{4}{3}$

6. EQUATIONS INVOLVING INVERSE TRIGONOMETRIC FUNCTIONS :

Illustration 18 : The equation $2\cos^{-1}x + \sin^{-1}x = \frac{11\pi}{6}$ has

(A) no solution (B) only one solution (C) two solutions (D) three solutions

Solution : Given equation is $2\cos^{-1}x + \sin^{-1}x = \frac{11\pi}{6}$

$\Rightarrow \cos^{-1}x + (\cos^{-1}x + \sin^{-1}x) = \frac{11\pi}{6} \Rightarrow \cos^{-1}x + \frac{\pi}{2} = \frac{11\pi}{6} \Rightarrow \cos^{-1}x = 4\pi/3$

which is not possible as $\cos^{-1}x \in [0, \pi]$ **Ans.(A)**

Illustration 19 : If $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = 5\pi^2/8$, then x is equal to-

(A) –1 (B) 0 (C) 1 (D) none of these

Solution : The given equation can be written as $(\tan^{-1}x + \cot^{-1}x)^2 - 2\tan^{-1}x\cot^{-1}x = 5\pi^2/8$

Since $\tan^{-1}x + \cot^{-1}x = \pi/2$ we have

$(\pi/2)^2 - 2\tan^{-1}x(\pi/2 - \tan^{-1}x) = 5\pi^2/8$

$\Rightarrow 2(\tan^{-1}x)^2 - 2(\pi/2)\tan^{-1}x - 3\pi^2/8 = 0 \Rightarrow \tan^{-1}x = -\pi/4 \Rightarrow x = -1$ **Ans. (A)**

Illustration 20 : Solve the equation : $\tan^{-1}\frac{x-1}{x-2} + \tan^{-1}\frac{x+1}{x+2} = \frac{\pi}{4}$

Solution : $\tan^{-1}\frac{x-1}{x-2} + \tan^{-1}\frac{x+1}{x+2} = \frac{\pi}{4}$

taking tangent on both sides

$$\Rightarrow \tan\left(\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right)\right) = 1 \Rightarrow \frac{\tan\left(\tan^{-1}\left(\frac{x-1}{x-2}\right)\right) + \tan\left(\tan^{-1}\left(\frac{x+1}{x+2}\right)\right)}{1 - \tan\left(\tan^{-1}\left(\frac{x-1}{x-2}\right)\right)\tan\left(\tan^{-1}\left(\frac{x+1}{x+2}\right)\right)} = 1$$

$$\Rightarrow \frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{x-1}{x-2}\cdot\frac{x+1}{x+2}} = 1 \Rightarrow \frac{(x-1)(x+2) + (x-2)(x+1)}{x^2 - 4 - (x^2 - 1)} = 1 \Rightarrow 2x^2 - 4 = -3 \Rightarrow x = \pm\frac{1}{\sqrt{2}}$$

Now verify $x = \frac{1}{\sqrt{2}}$

$$= \tan^{-1}\left(\frac{\frac{1}{\sqrt{2}} - 1}{\frac{1}{\sqrt{2}} - 2}\right) + \tan^{-1}\left(\frac{\frac{1}{\sqrt{2}} + 1}{\frac{1}{\sqrt{2}} + 2}\right) = \tan^{-1}\left(\frac{\sqrt{2} - 1}{2\sqrt{2} - 1}\right) + \tan^{-1}\left(\frac{\sqrt{2} + 1}{2\sqrt{2} + 1}\right)$$

$$= \tan^{-1}\left(\frac{(2\sqrt{2}+1)(\sqrt{2}-1)+(2\sqrt{2}-1)(\sqrt{2}+1)}{(2\sqrt{2}-1)(2\sqrt{2}+1)-(\sqrt{2}-1)(\sqrt{2}+1)}\right) = \tan^{-1}\left(\frac{6}{6}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$x = -\frac{1}{\sqrt{2}}$$

$$= \tan^{-1}\left(\frac{-\frac{1}{\sqrt{2}}-1}{-\frac{1}{\sqrt{2}}-2}\right) + \tan^{-1}\left(\frac{-\frac{1}{\sqrt{2}}+1}{-\frac{1}{\sqrt{2}}+2}\right) = \tan^{-1}\left(\frac{\sqrt{2}+1}{2\sqrt{2}+1}\right) + \tan^{-1}\left(\frac{\sqrt{2}-1}{2\sqrt{2}-1}\right) \{\text{same as above}\}$$

$$= \tan^{-1}(1) = \frac{\pi}{4}$$

$\therefore \quad x = \pm\dfrac{1}{\sqrt{2}}$ are solutions **Ans.**

Illustration 21 : Solve the equation : $2\tan^{-1}(2x + 1) = \cos^{-1}x$.

Solution : Here, $2\tan^{-1}(2x + 1) = \cos^{-1}x$

or $\cos(2\tan^{-1}(2x+1)) = x$ $\qquad \left\{\text{We know } \cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}\right\}$

$\therefore \quad \dfrac{1-(2x+1)^2}{1+(2x+1)^2} = x \Rightarrow (1 - 2x - 1)(1 + 2x + 1) = x(4x^2 + 4x + 2)$

$\Rightarrow \quad -2x \,.\, 2(x + 1) = 2x(2x^2 + 2x + 1) \Rightarrow 2x(2x^2 + 2x + 1 + 2x + 2) = 0$

$\Rightarrow \quad 2x(2x^2 + 4x + 3) = 0$

$\Rightarrow \quad x = 0$ or $2x^2 + 4x + 3 = 0$ {No solution}

Verify $x = 0$

$2\tan^{-1}(1) = \cos^{-1}(1) \Rightarrow \dfrac{\pi}{2} = \dfrac{\pi}{2}$

$\therefore \quad x = 0$ is only the solution **Ans.**

7. INEQUATIONS INVOLVING INVERSE TRIGONOMETRIC FUNCTION :

Illustration 22 : Find the complete solution set of $\sin^{-1}(\sin 5) > x^2 - 4x$.

Solution : $\sin^{-1}(\sin 5) > x^2 - 4x \Rightarrow \sin^{-1}[\sin(5 - 2\pi)] > x^2 - 4x$

$\Rightarrow \quad x^2 - 4x < 5 - 2\pi \Rightarrow x^2 - 4x + (2\pi - 5) < 0$

$\Rightarrow \quad 2-\sqrt{9-2\pi} < x < 2+\sqrt{9-2\pi} \Rightarrow x \in (2-\sqrt{9-2\pi},\ 2+\sqrt{9-2\pi})$

Ans.

Illustration 23 : Find the complete solution set of $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \le 0$, where [.] denotes the greatest integer function.

Solution : $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \le 0$

$\Rightarrow \quad ([\cot^{-1}x] - 3)^2 \le 0 \;\Rightarrow\; [\cot^{-1}x] = 3 \;\Rightarrow\; 3 \le \cot^{-1}x < 4 \;\Rightarrow\; x \in (-\infty, \cot 3]$

Illustration 24 : If $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$, $n \in N$, then the maximum value of n is -

(A) 1 (B) 5 (C) 9 (D) none of these

Solution : $\cot^{-1}\frac{n}{\pi} > \frac{\pi}{6}$

$\Rightarrow \quad \cot\left(\cot^{-1}\left(\frac{n}{\pi}\right)\right) < \cot\left(\frac{\pi}{6}\right) \Rightarrow \frac{n}{\pi} < \sqrt{3}$

$\Rightarrow \quad n < \pi\sqrt{3} \quad \Rightarrow \quad n < 5.5$ (approx)

$\Rightarrow \quad n = 5 \qquad \because (n \in N)$ **Ans. (B)**

Do yourself - 7 :

1. Solve the following equation for x :

(i) $\sin\left[\sin^{-1}\left(\frac{1}{5}\right) + \cos^{-1}x\right] = 1$ (ii) $\cos^{-1}x + \sin^{-1}\frac{x}{2} = \frac{\pi}{6}$

2. (i) Solve the inequality $\tan^{-1}x > \cot^{-1}x$.

(ii) Complete solution set of inequation $(\cos^{-1}x)^2 - (\sin^{-1}x)^2 > 0$, is

(A) $\left[0, \frac{1}{\sqrt{2}}\right)$ (B) $\left[-1, \frac{1}{\sqrt{2}}\right)$ (C) $(-1, \sqrt{2})$ (D) none of these

3. (i) Solve the inequality : $(\text{arc sec}x)^2 - 6(\text{arc sec}x) + 8 > 0$

(ii) If $\sin^2x + \sin^2y < 1$; x, y $\in$ R then prove that $\sin^{-1}(\tan x \cdot \tan y) \in (-\pi/2, \pi/2)$.

4. Solve the following :

(i) $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$

(ii) $\tan^{-1}\frac{1}{1+2x} + \tan^{-1}\frac{1}{1+4x} = \tan^{-1}\frac{2}{x^2}$

(iii) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$

(iv) $3\cos^{-1}x = \sin^{-1}\left(\sqrt{1-x^2}(4x^2-1)\right)$

(v) $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$ and $\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$

5. Find all the positive integral solutions of, $\tan^{-1}x + \cos^{-1}\frac{y}{\sqrt{1+y^2}} = \sin^{-1}\frac{3}{\sqrt{10}}$.

6. Solve the following inequalities :

(i) $\text{arc cot}^2x - 5\,\text{arc cot}\,x + 6 > 0$ (ii) arc sin x > arc cos x

(iii) $\tan^2(\text{arc sin}x) > 1$

7. Solve the following system of inequalities :

$4\,\text{arc}\tan^2 x - 8\,\text{arc}\tan x + 3 < 0$ & $4\,\text{arc}\cot x - \text{arc}\cot^2 x - 3 \ge 0$

8. All x satisfying the inequality $(\cot^{-1} x)^2 - 7(\cot^{-1} x) + 10 > 0$, lie in the interval :-

(A) $(-\infty, \cot 5) \cup (\cot 4, \cot 2)$ (B) $(\cot 5, \cot 4)$

(C) $(\cot 2, \infty)$ (D) $(-\infty, \cot 5) \cup (\cot 2, \infty)$

9. Solve for x : $\sin^{-1}\left(\sin\left(\dfrac{2x^2+4}{1+x^2}\right)\right) < \pi - 3$.

10. Solve the following inequalities:

(i) $\cos^{-1} x > \cos^{-1} x^2$ (ii) $\text{arccot}^2\, x - 5\,\text{arccot}\, x + 6 > 0$ (iii) $\sin^{-1} x > -1$

(iv) $\cos^{-1} x < 2$ (v) $\cot^{-1} x < -\sqrt{3}$

11. Solve for x

(i) $\cos(2\sin^{-1}x) = \dfrac{1}{3}$ (ii) $\cot^{-1} x + \tan^{-1} 3 = \dfrac{\pi}{2}$

(iii) $\tan^{-1}\left(\dfrac{x-1}{x-2}\right) + \tan^{-1}\left(\dfrac{x+1}{x+2}\right) = \dfrac{\pi}{4}$ (iv) $\sin^{-1}x + \sin^{-1}2x = \dfrac{2\pi}{3}$

12. The solution of the equation $\sin^{-1}\left(\tan\dfrac{\pi}{4}\right) - \sin^{-1}\left(\sqrt{\dfrac{3}{x}}\right) - \dfrac{\pi}{6} = 0$ is

(A) x = 2 (B) x = –4 (C) x = 4 (D) x = 3

13. Number of solutions of the equation $\cot^{-1}\sqrt{4-x^2} + \cos^{-1}(x^2-5) = \dfrac{3\pi}{2}$ is :

(A) 2 (B) 4 (C) 6 (D) 8

14. If $x \ge 0$ and $\theta = \sin^{-1}x + \cos^{-1}x - \tan^{-1}x$, then

(A) $\dfrac{\pi}{2} \le \theta \le \dfrac{3\pi}{4}$ (B) $0 \le \theta \le \dfrac{\pi}{4}$ (C) $0 \le \theta < \dfrac{\pi}{2}$ (D) $\dfrac{\pi}{4} \le \theta \le \dfrac{\pi}{2}$

15. If α is a real root of the equation $x^3 + 3x - \tan 2 = 0$, then $\cot^{-1}\alpha + \cot^{-1}\dfrac{1}{\alpha} - \dfrac{\pi}{2}$ can be equal to

(A) 0 (B) $\dfrac{\pi}{2}$ (C) π (D) $\dfrac{3\pi}{2}$

16. If $\sin^{-1}x + \cot^{-1}\left(\dfrac{1}{2}\right) = \dfrac{\pi}{2}$, then x is equal to

(A) 0 (B) $\dfrac{1}{\sqrt{5}}$ (C) $\dfrac{2}{\sqrt{5}}$ (D) $\dfrac{\sqrt{3}}{2}$

17. Solve the following :

(i) $\tan^{-1}\dfrac{x-1}{x+1} + \tan^{-1}\dfrac{2x-1}{2x+1} = \tan^{-1}\dfrac{23}{36}$

(ii) $2\tan^{-1}x = \cos^{-1}\dfrac{1-a^2}{1+a^2} - \cos^{-1}\dfrac{1-b^2}{1+b^2}$ (a > 0, b > 0)

8. SUMMATION OF SERIES :

Illustration 25 : Prove that :

$$\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right)+\tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right)+\tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right)+\ldots+\tan^{-1}\left(\frac{c_n-c_{n-1}}{1+c_nc_{n-1}}\right)+\tan^{-1}\left(\frac{1}{c_n}\right)=\tan^{-1}\left(\frac{x}{y}\right)$$

Solution : L.H.S.

$$\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right)+\tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right)+\tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right)+\ldots+\tan^{-1}\left(\frac{c_n-c_{n-1}}{1+c_nc_{n-1}}\right)+\tan^{-1}\left(\frac{1}{c_n}\right)$$

$$=\tan^{-1}\left(\frac{\frac{x}{y}-\frac{1}{c_1}}{1+\frac{x}{y}.\frac{1}{c_1}}\right)+\left(\tan^{-1}c_2-\tan^{-1}c_1\right)+\left(\tan^{-1}c_3-\tan^{-1}c_2\right)+\ldots.+\left(\tan^{-1}c_n-\tan^{-1}c_{n-1}\right)+\tan^{-1}\left(\frac{1}{c_n}\right)$$

$$=\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left(\frac{1}{c_1}\right)-\tan^{-1}c_1+\tan^{-1}c_n+\tan^{-1}\left(\frac{1}{c_n}\right)$$

$$=\tan^{-1}\left(\frac{x}{y}\right)-\left(\cot^{-1}c_1+\tan^{-1}c_1\right)+\left(\tan^{-1}c_n+\cot^{-1}c_n\right)$$

$$=\tan^{-1}\left(\frac{x}{y}\right)-\frac{\pi}{2}+\frac{\pi}{2}=\tan^{-1}\left(\frac{x}{y}\right)=\text{R.H.S.}$$

Do yourself - 8 :

1. Find the sum of each of the following series :

(i) $\tan^{-1}\dfrac{1}{x^2+x+1}+\tan^{-1}\dfrac{1}{x^2+3x+3}+\tan^{-1}\dfrac{1}{x^2+5x+7}+\tan^{-1}\dfrac{1}{x^2+7x+13}$ .... upto n terms.

(ii) $\tan^{-1}\dfrac{1}{3}+\tan^{-1}\dfrac{2}{9}+\ldots..+\tan^{-1}\dfrac{2^{n-1}}{1+2^{2n-1}}+$ ....upto infinite terms

(iii) $\sin^{-1}\dfrac{1}{\sqrt{2}}+\sin^{-1}\dfrac{\sqrt{2}-1}{\sqrt{6}}+\ldots..+\sin^{-1}\dfrac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}+$ .... upto infinite terms

2. Find the sum of the series :

(i) $\cot^{-1}7+\cot^{-1}13+\cot^{-1}21+\cot^{-1}31+$ ..... to n terms.

(ii) $\sin^{-1}\dfrac{1}{\sqrt{5}}+\sin^{-1}\dfrac{1}{\sqrt{65}}+\sin^{-1}\dfrac{1}{\sqrt{325}}+\ldots..+\sin^{-1}\dfrac{1}{\sqrt{4n^4+1}}+\ldots...\infty$ terms.

3. If $x\in(0,1)$ and $f(x)=\sec\left\{\tan^{-1}\left(\dfrac{\sin(\cos^{-1}x)+\cos(\sin^{-1}x)}{\cos(\cos^{-1}x)+\sin(\sin^{-1}x)}\right)\right\}$, then $\sum\limits_{r=2}^{10}f\left(\dfrac{1}{r}\right)$ is

4. Find the value of $3\sum\limits_{n=1}^{\infty}\left\{\dfrac{1}{\pi}\sum\limits_{k=1}^{\infty}\cot^{-1}\left(1+2\sqrt{\sum\limits_{r=1}^{k}r^3}\right)\right\}^n$